



Snow and Snowmelt Processes

C. David Moeser

United States Geological Survey, New Mexico Water Science Center

Physical Hydrology [Snow Processes]

08th September 2025



Snow and Snowmelt Processes

C. David Moeser

Hydrologist

Snow and Watershed Modeler

Physical Hydrology

09th September 2025

Interested in:

Forest – Snow Interactions

‘Change’

Contrarian Viewpoints

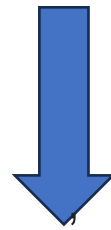
Mountains



Snow Accumulation and Snowmelt Processes

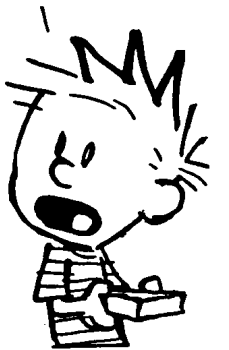


Contrarian Viewpoints

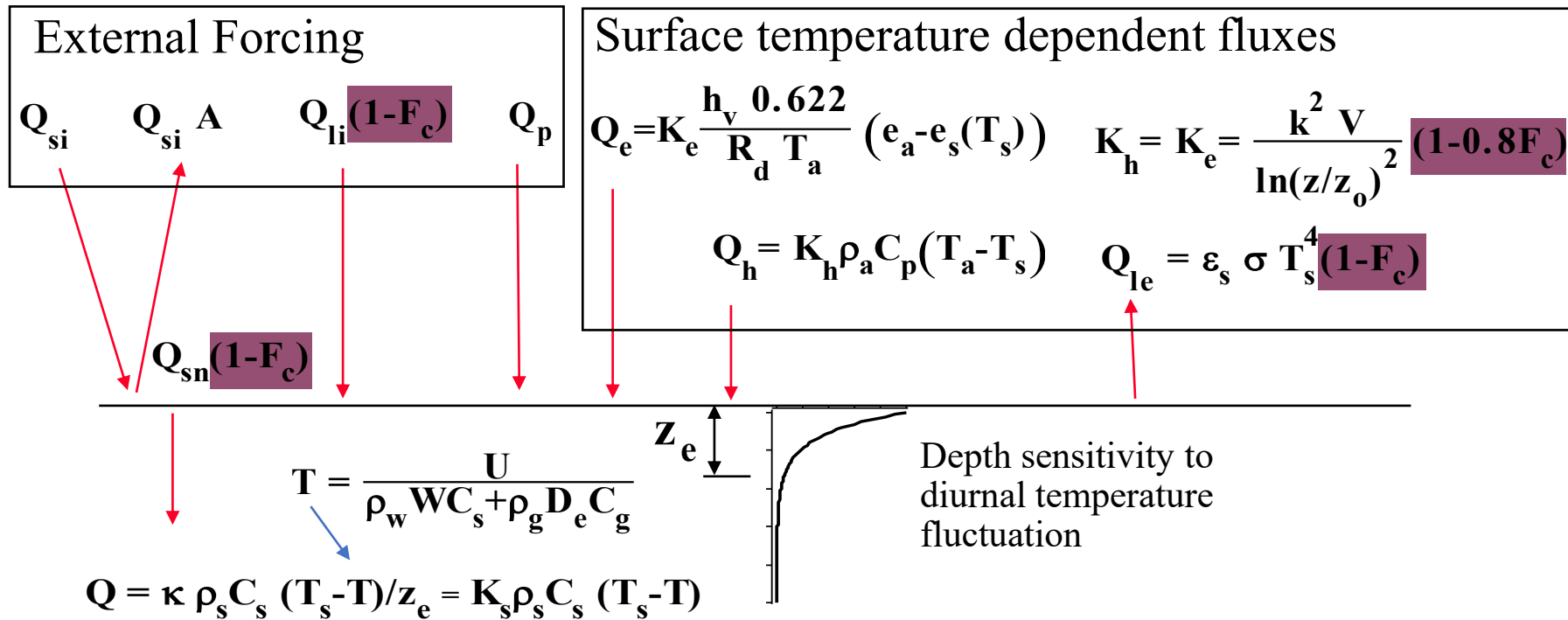


VERY SIMPLE !!!!

I THINK WE MIGHT ALL
LEARN A LESSON FROM
HOW THIS UTTERLY
UNIQUE AND EXQUISITE
CRYSTAL ...



Surface Energy Balance – With Forest Canopy



Energy Balance /Equilibrium at surface

$$Q_{si}(1-A) + Q_{li} + Q_p + Q_h(T_s) + Q_e(T_s) - Q_{le}(T_s) = Q(T_s, U, W)$$

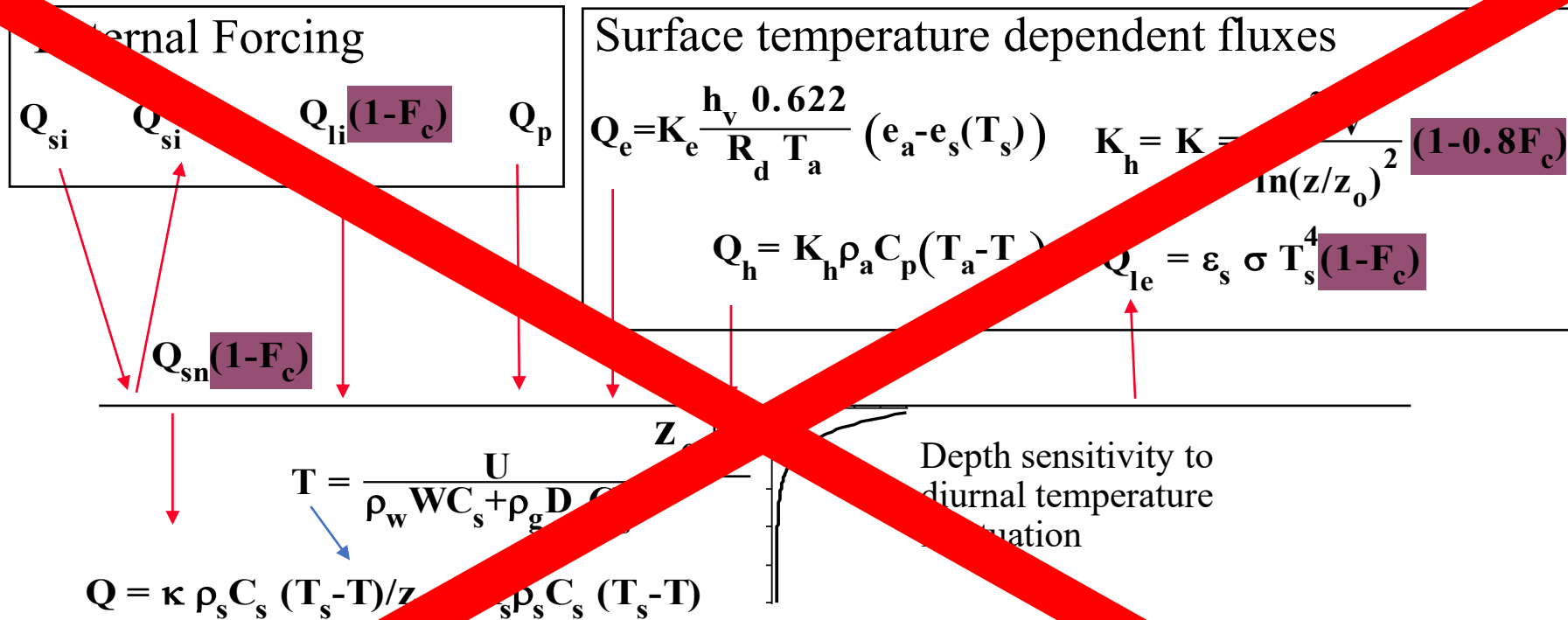
Solve for T_s : Linearize and iterate

If $T_s > 0^\circ\text{C} \Rightarrow$ Conduction cannot accommodate surface energy inputs

$\Rightarrow$ Surface melt is generated. Set $T_s = 0^\circ\text{C}$

$\Rightarrow$ Additional energy is advected downwards by surface melt inflow

Surface Energy Balance – With Forest Canopy



Energy Balance equilibrium at surface

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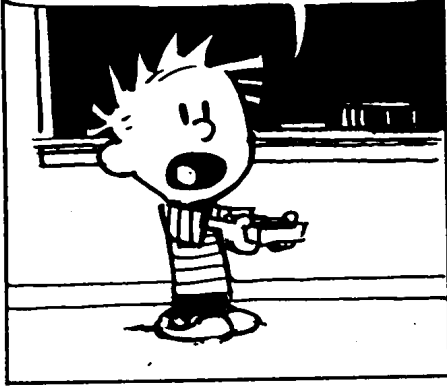
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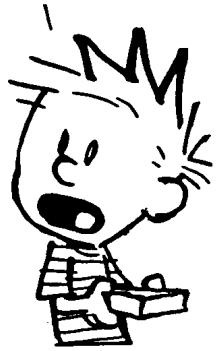
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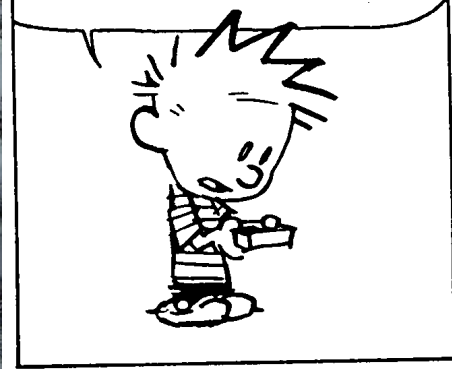
TODAY FOR SHOW AND TELL,
I'VE BROUGHT A TINY
MARVEL OF NATURE :
A SINGLE SNOWFLAKE.



I THINK WE MIGHT ALL
LEARN A LESSON FROM
HOW THIS UTTERLY
UNIQUE AND EXQUISITE
CRYSTAL ...



...TURNS INTO AN ORDINARY,
BORING MOLECULE OF WATER,
JUST LIKE EVERY OTHER
ONE, WHEN YOU BRING
IT IN THE CLASSROOM.



AND NOW, WHILE
THE ANALOGY
SINKS IN, I'LL
BE LEAVING YOU
DRIPS AND GOING
OUTSIDE.



CALVIN!

Snow Accumulation and Snowmelt Processes

(VERY SIMPLE !!!!)

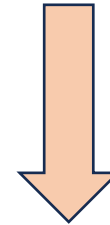


Mass Balance

Precipitation

Affected by things like wind, terrain, and
landscape

$$\text{Total Snow} = \text{Snow}_{\text{in}} - \text{Snow}_{\text{out}}$$



Energy Balance

Melt Energy = Total Radiation + Total Heat

Affected by terrain, and landscape

$$\text{SWR}_{\text{in}} + \text{-SWR}_{\text{out}} + \text{LWR}_{\text{in}} + \text{-LWR}_{\text{out}} + \text{Heat}_{\text{latent}} + \text{Heat}_{\text{sensible}}$$



I. Significance of Snow



Global:



Northern Hemisphere:



Regional: Western U.S.



Watershed: Rio Grande



Local: Headwaters



II. Manual Measurement Techniques

III. Automated Measurements

IV. Remote Sensing

V. Modeling

VI. Modeling forest snow example*

RIO GRANDE FOCUS

Visualize and understand basic publicly available
data

Global Significance of snow

- One sixth of the world's population relies on snowmelt water for agriculture and human consumption
- Climate services



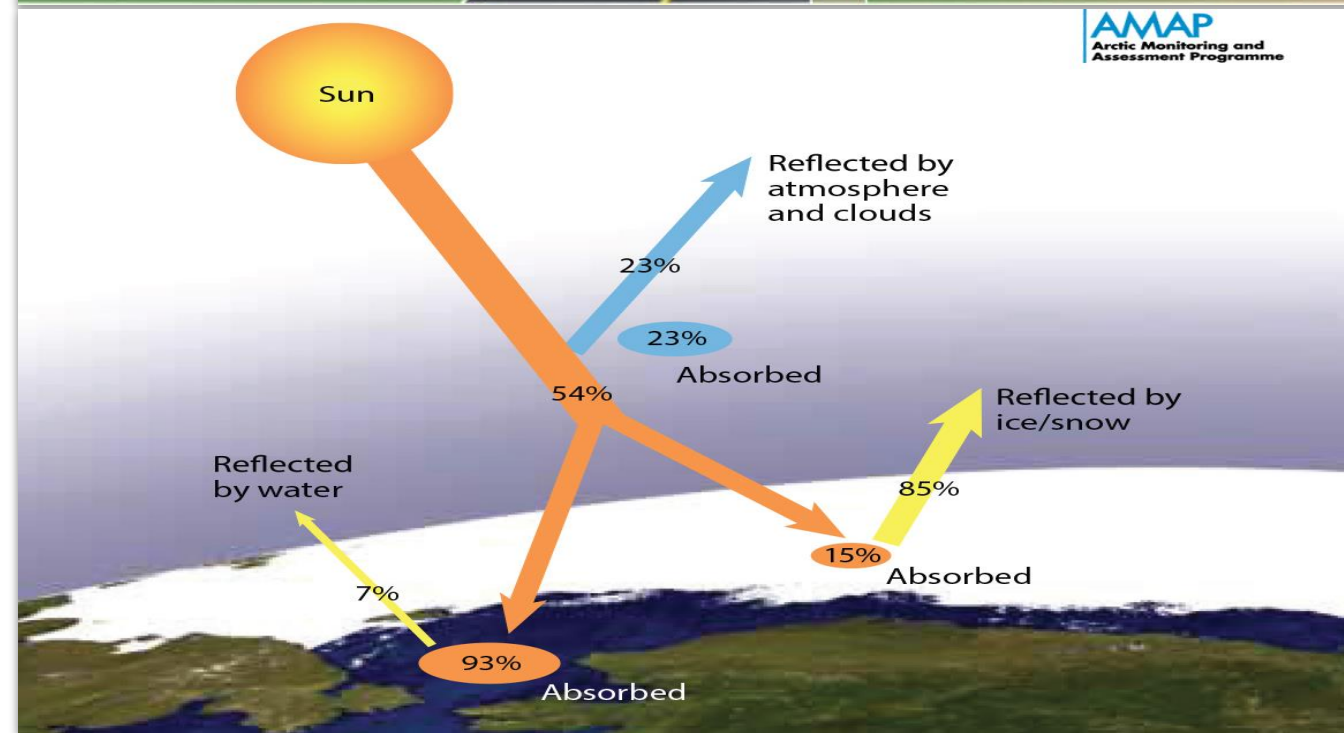
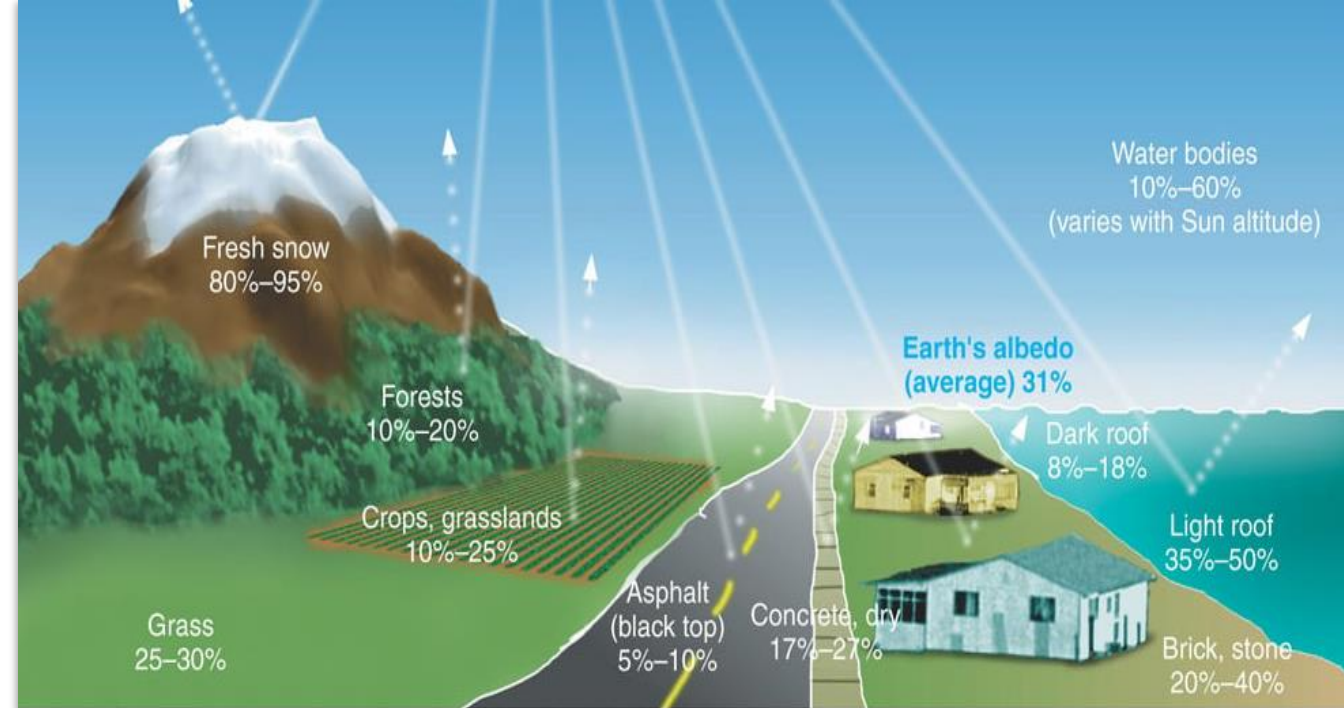
Global Significance of snow

- **Climate services:**

Albedo

Is a measure of Reflectivity.

New Snow has an albedo of up to 95%

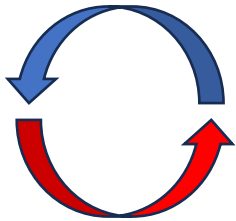


Global Significance of snow

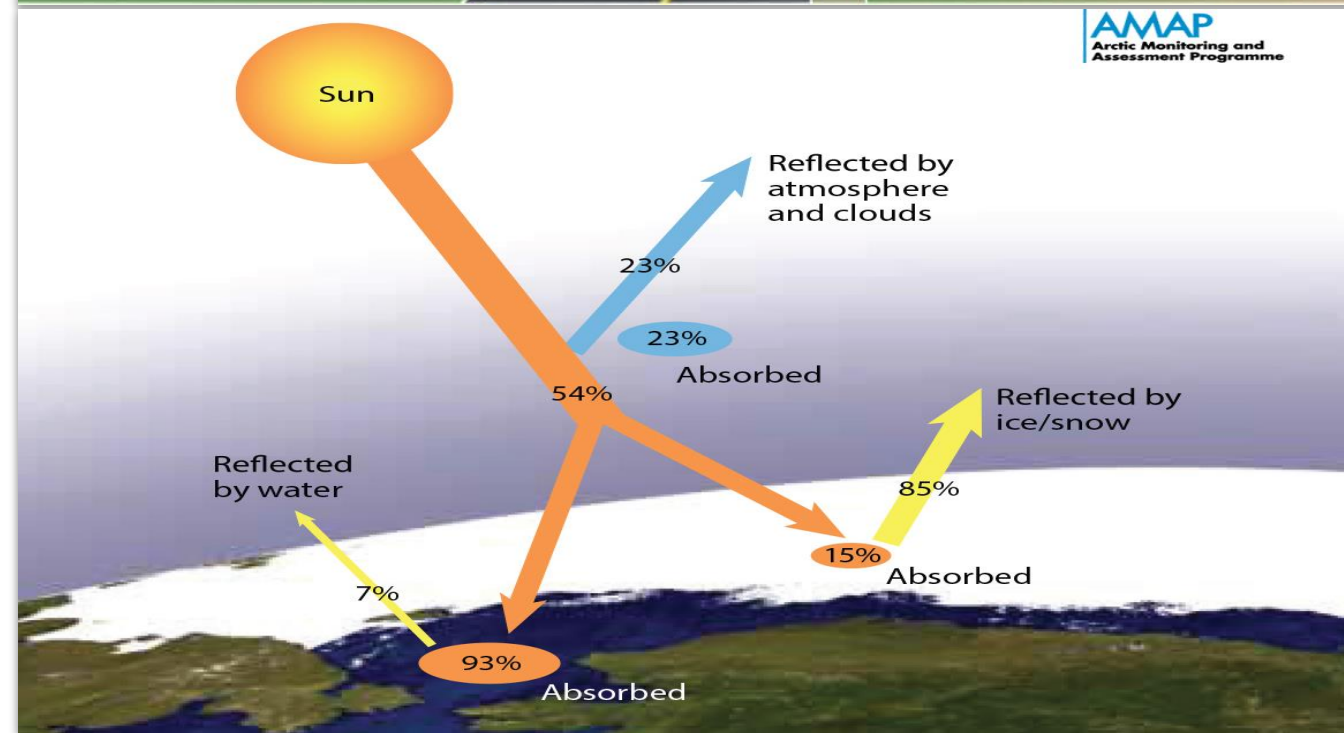
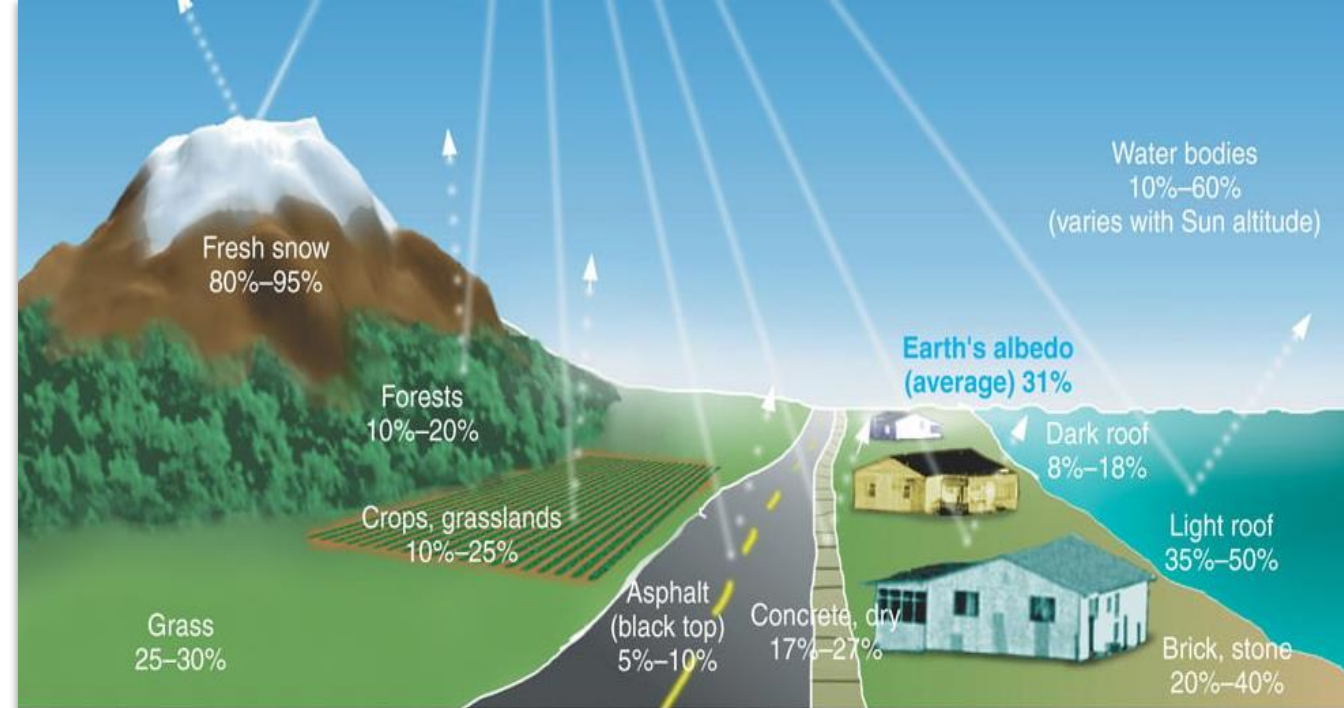
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Responsible for a variety of climactic Feedback loops.

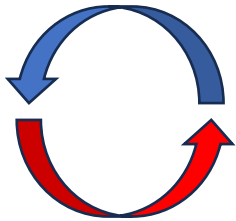


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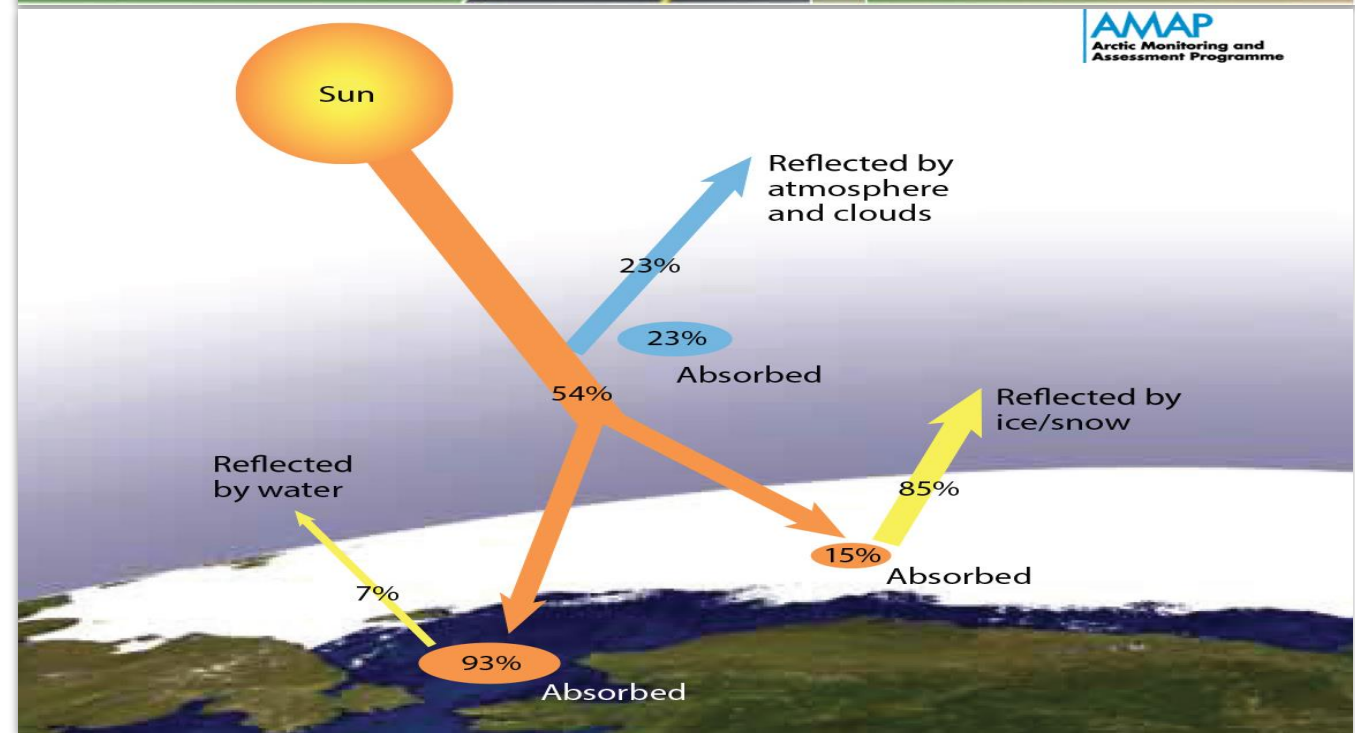
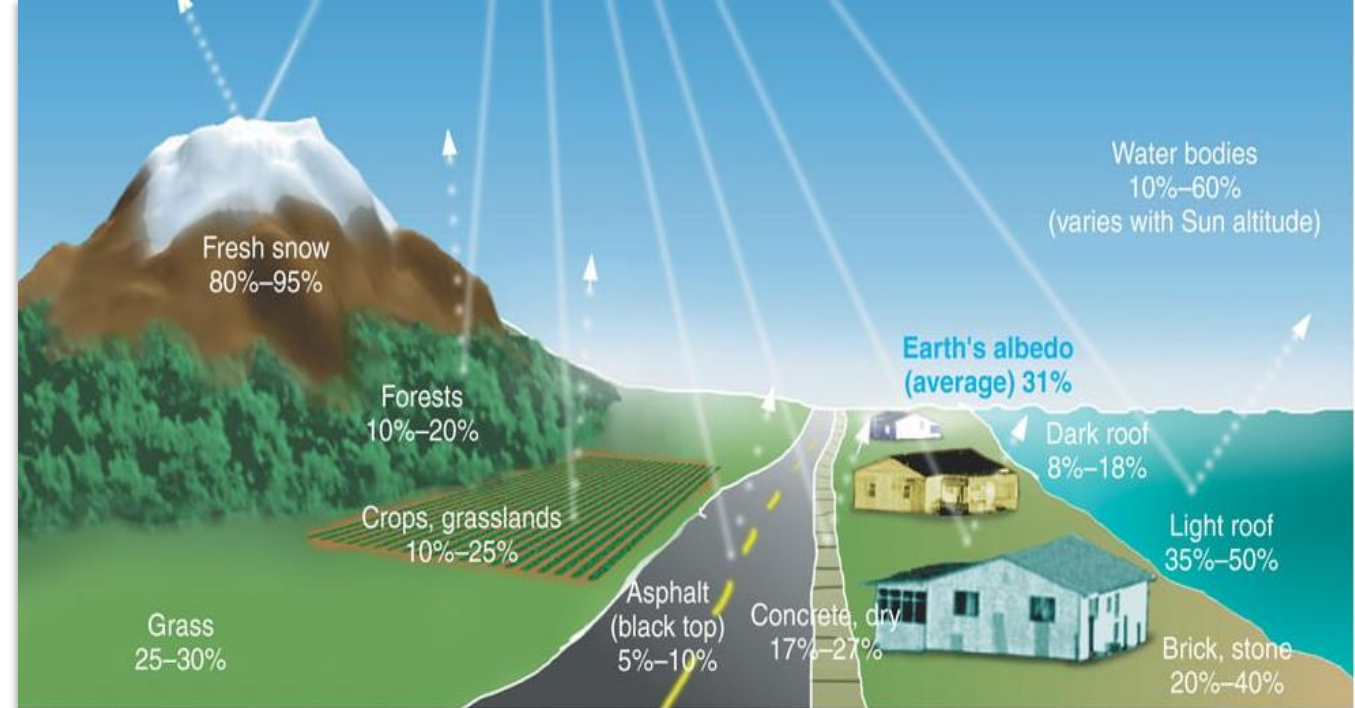
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What happens if Snow Covered Area (SCA) is reduced?



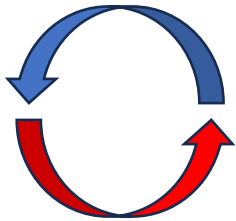
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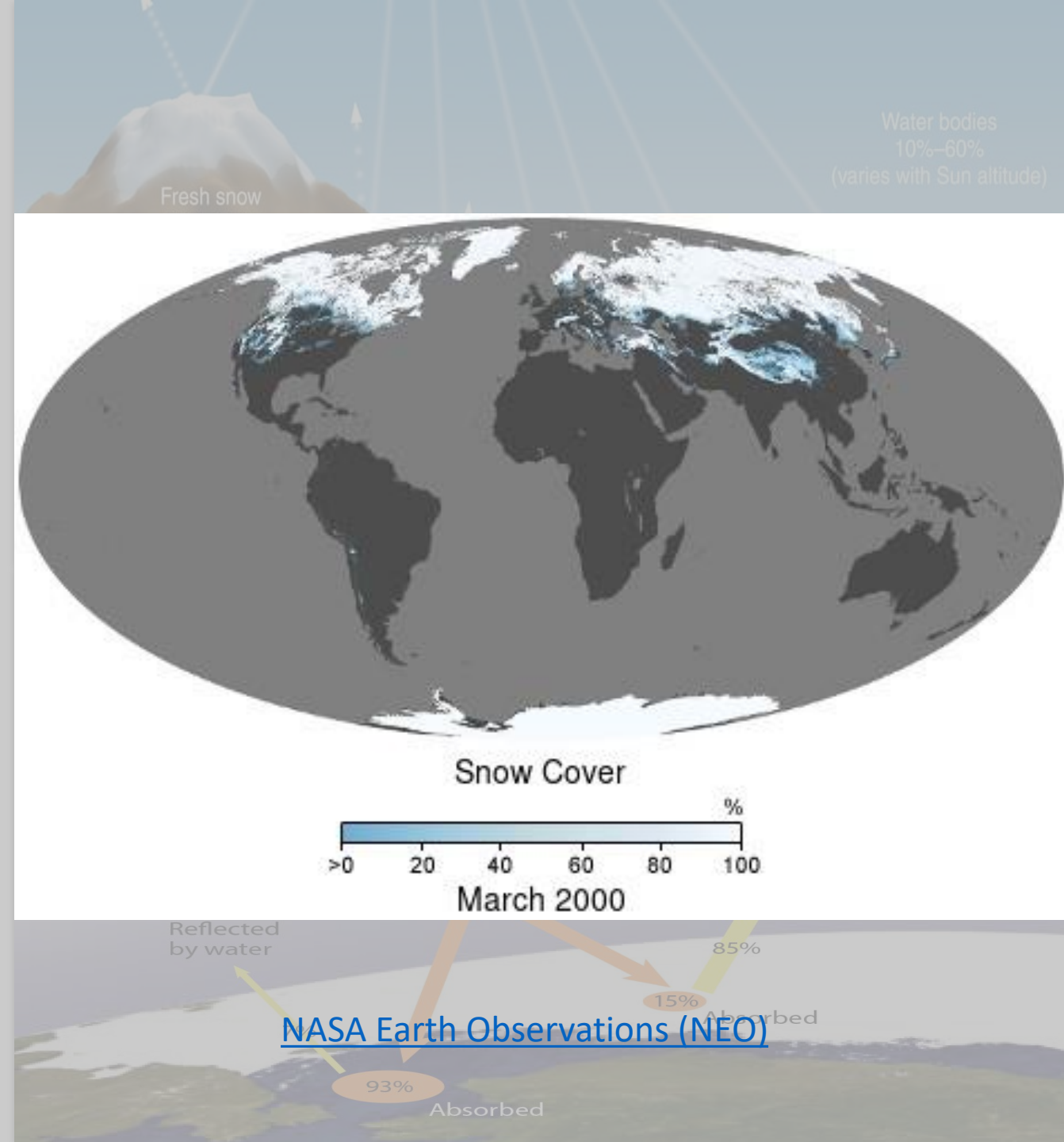
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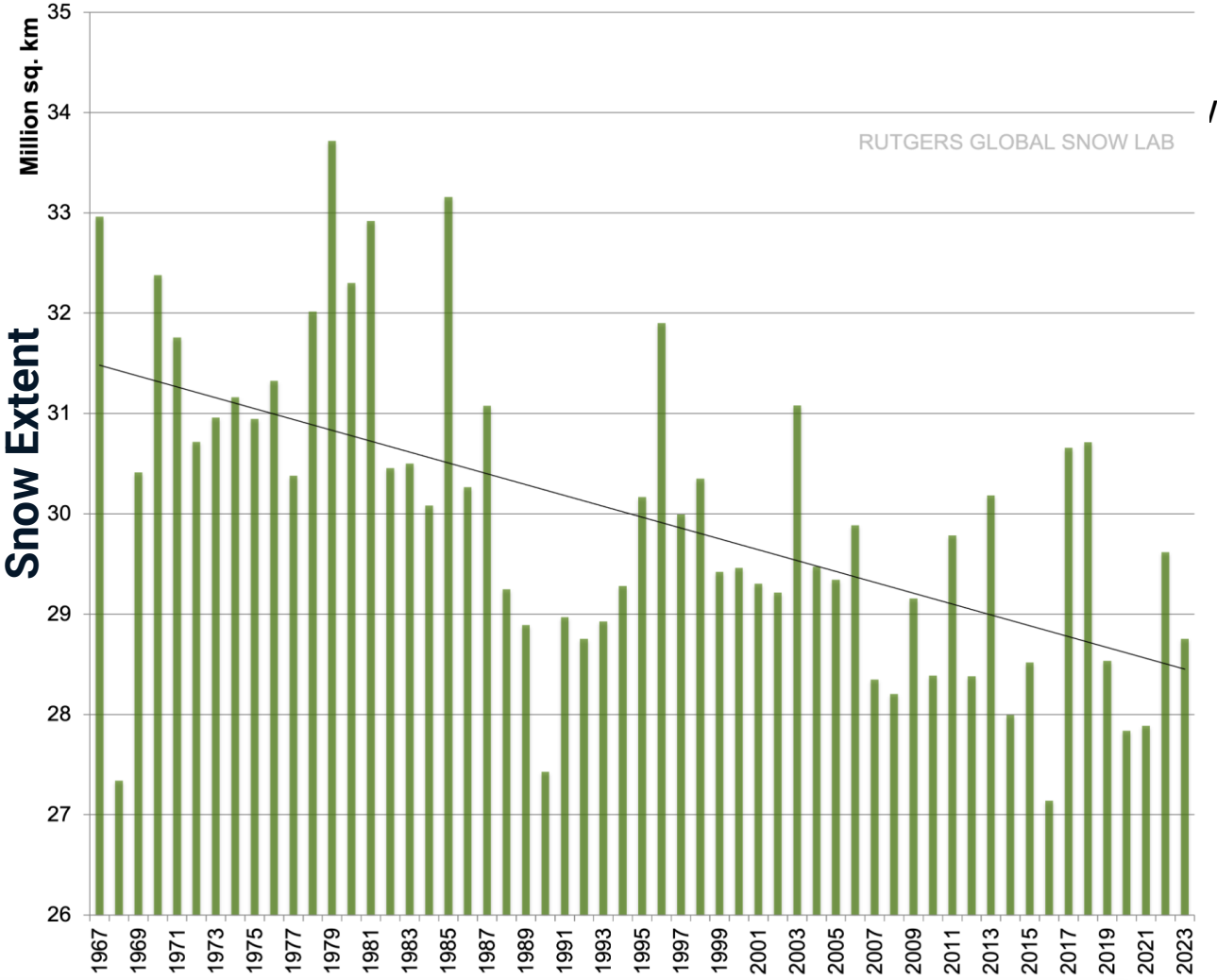


Global Significance of Snow

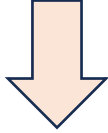


CLIMATE FEEDBACK LOOPS:

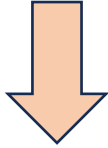
Spring Northern Hemisphere Snow Extent



Snow Cover is
disappearing earlier
(melt-out date)



Albedo is reduced



More energy is retained



Snow Cover disappears
earlier



Albedo is reduced



More energy is retained

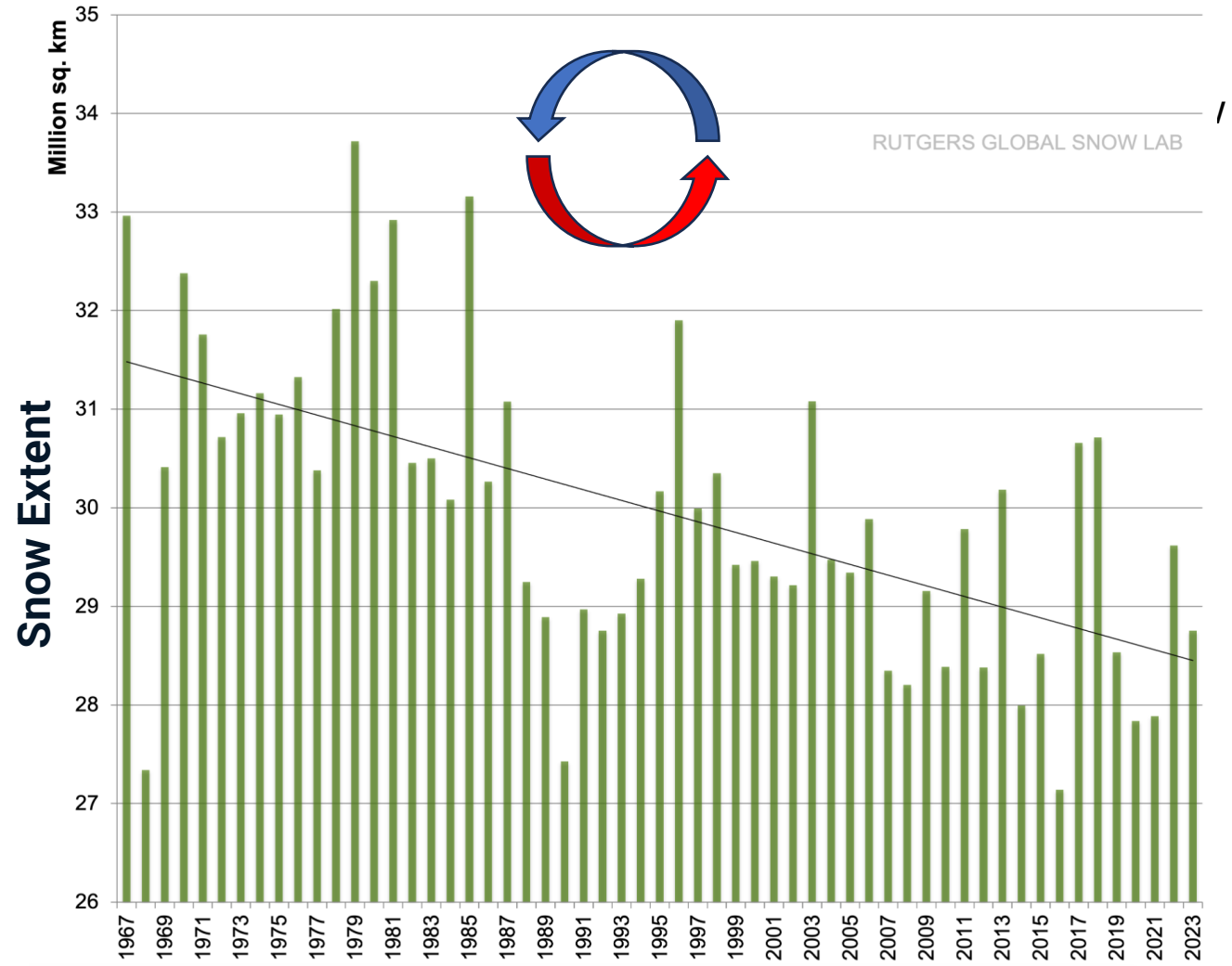


Global Significance of Snow



CLIMATE FEEDBACK LOOPS:

Spring Northern Hemisphere Snow Extent

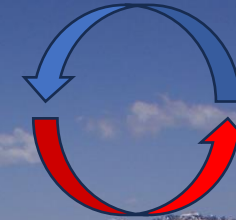


Global Significance of Snow

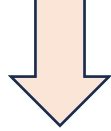


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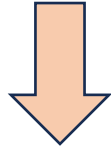
San Juan Mountains



Aridification



Increased Dust on Snow
Events



Albedo is Reduced



Snow Cover disappears
earlier



More Energy is Retained



Changing Climate
Patterns



Global Significance of Snow



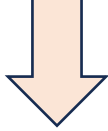
CLIMATE FEEDBACK LOOPS:

Spring Northern Hemisphere Snow Extent

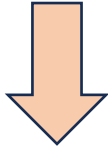
Pyrenees, Spain



Aridification



Increased Dust on Snow
Events



Albedo is Reduced



Snow Cover disappears
earlier



More Energy is Retained



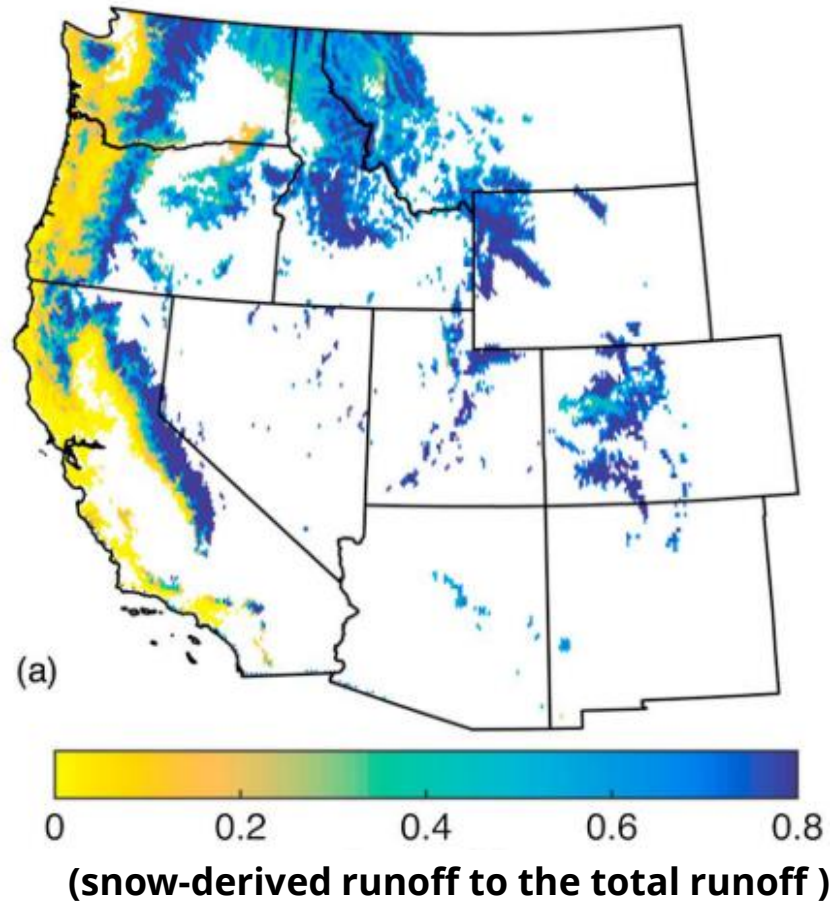
Changing Climate
Patterns



Regional Significance of Snow



- 25% of the area of the Western United States produces 90% of the total runoff.

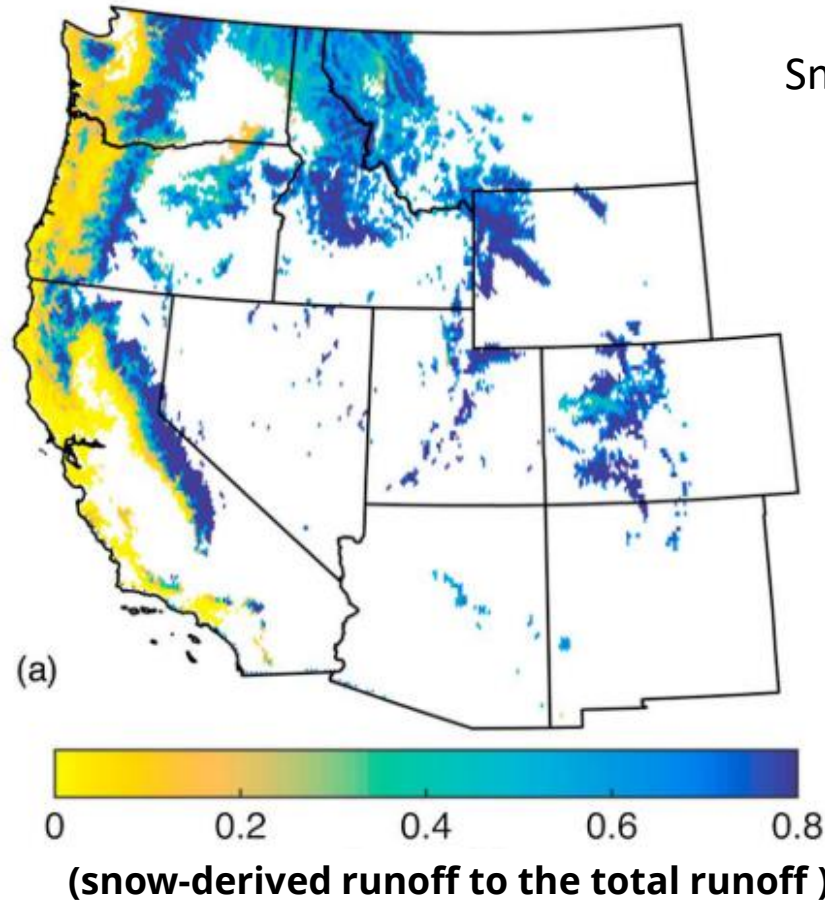


Regional Significance of Snow



- 25% of the area of the Western United States produces 90% of the total runoff.

blue = snowmelt dominated
Yellow = rain dominated



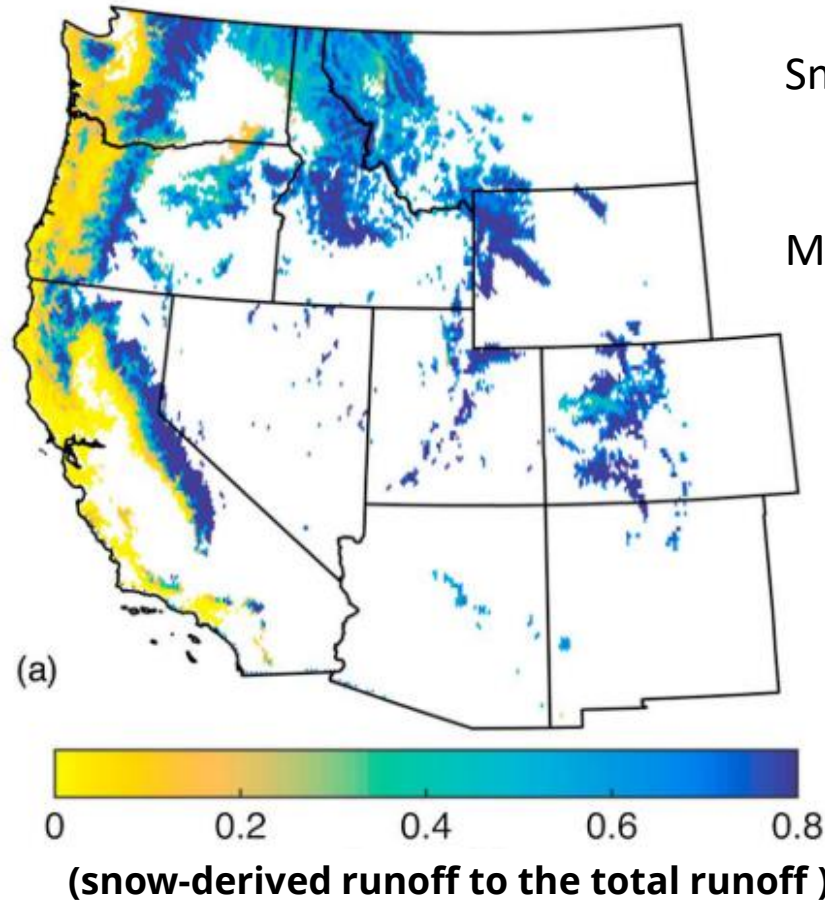
Snow accounts for 50-99% of total runoff

Regional Significance of Snow



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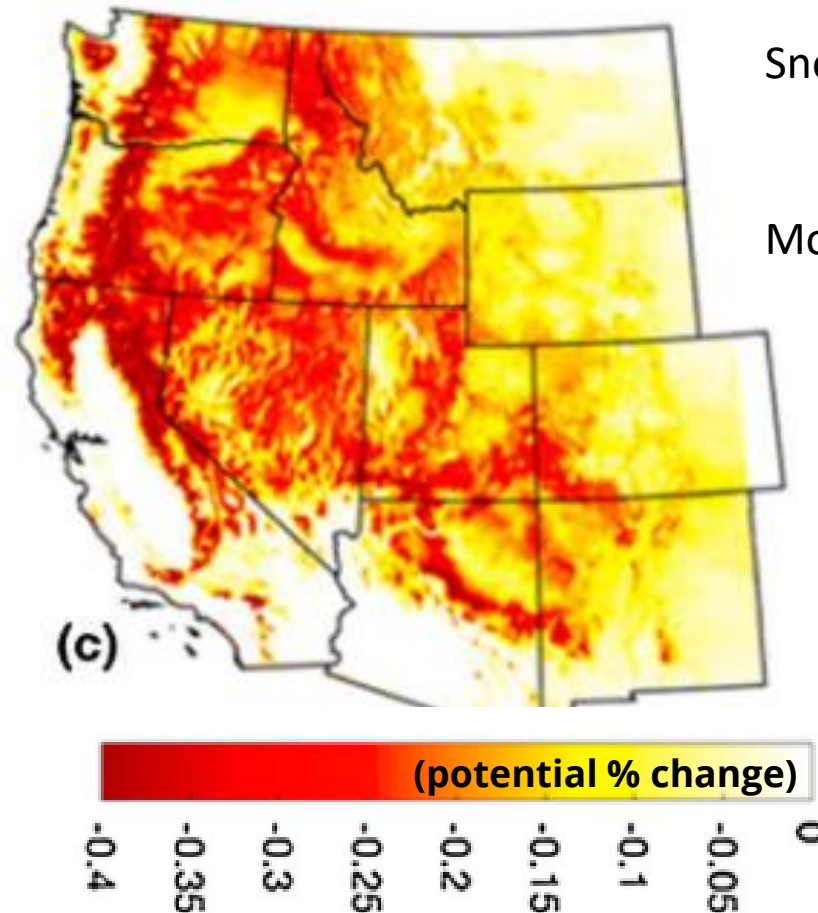
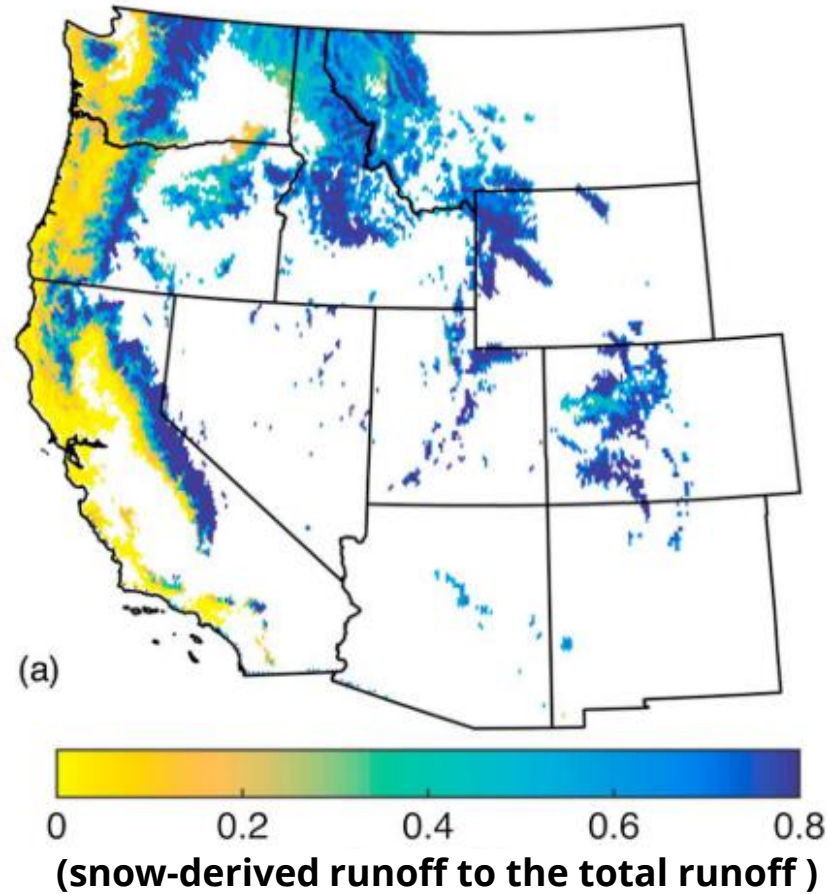


Mountains: snow is 70% of the total runoff.

Regional Significance of Snow



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Snow accounts for 50-99% of total runoff



Mountains: snow is 70% of the total runoff.



Some projections show this being reduced to 46%

The Upper Rio Grande Watershed:



General Facts For Rio Grande

75% of streamflow originates from snowpack

- primarily from forested headwaters
- in a small area relative to the size of the total basin

90% of the spruce forest in the Rio Grande National Forest has been affected by bark beetles

‘in the headwater regions of the Rio Grande and Colorado River Basins, burn areas are projected to increase 300% to 700% with a 1°C increase in mean global temperature’

{ What will happen to the snow if the forest changes? }



(river)



Significance of Forest Snow



Global: 17% of total terrestrial winter water storage



Northern Hemisphere: 20% seasonal snow cover occurs in forested areas



Regional: Western U.S. reliant on forest snow



Watershed: Primary generator of downstream water in URGB.



Local: Agricultural and Municipal Water Managers: struggling to keep up with the 'change'

(Think of snow as a giant reservoir)
The reservoir has been changing (timing, volume)

Snow Measurement Techniques

Snow Accumulation and Snowmelt Processes

have a high degree of
Spatial and Temporal Heterogeneity

giving no one right measurement method

there is not yet one technique that is spatially and
temporally robust

Snow Accumulation and Snowmelt Processes: Spatial and Temporal Heterogeneity

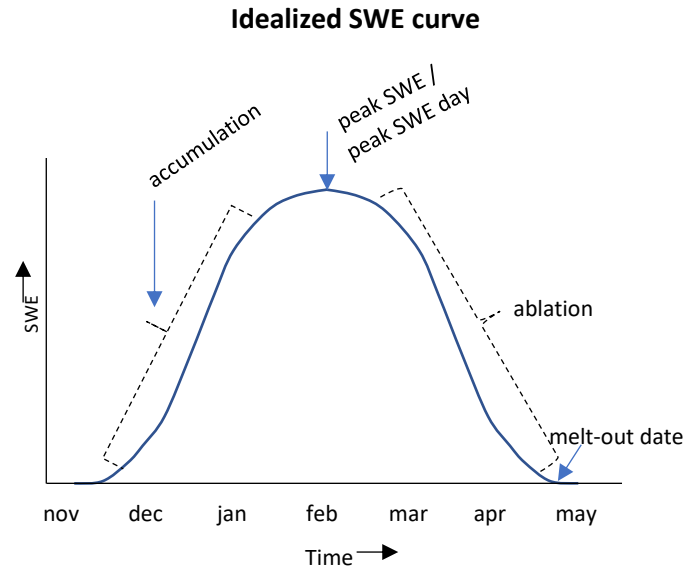


Reynolds Creek. Photograph Keith Cooley

Snow Processes in a Forested Environment

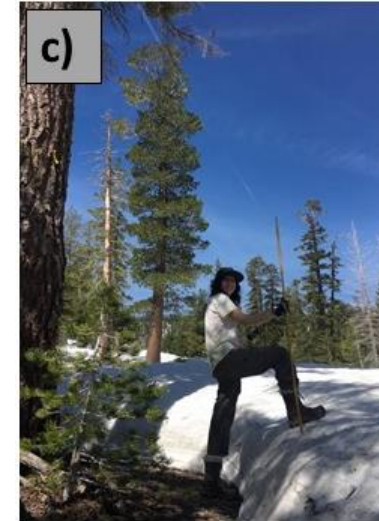
Accumulation processes
Governed by the mass balance

(interception, unloading, wind distribution)



Ablation processes
Governed by the Energy Balance

(~short and longwave radiation)



Snow Pro

ironment

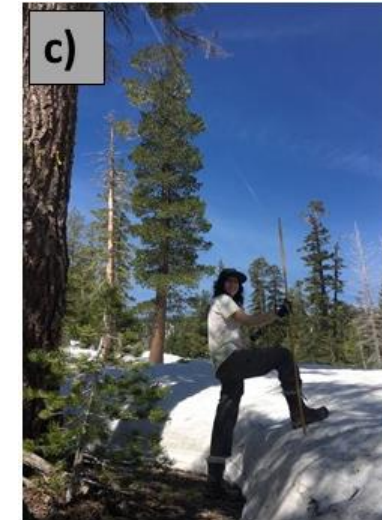
Accumulation processes
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Ablation processes
Governed by the Energy Balance

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Manual Snow Measurements

Point measurements haven't really changed since the early 1900's

The Mt. Rose Federal Snow Sample still used today to measure SWE

Consists of a hollow tube with a cutter.

The tube is used to collect a snow sample and the weight of the sample is then taken



James Church first hired in 1892 by University of Nevada to teach Latin and Greek. Ended up pioneering snow surveying on Mt. Rose.



Snow Measurements and Calculations

Initial calculations



The interior diameter of the snow tube: 3.756 cm

Radius: 1.878 cm

Interior surface area: πr^2 $[3.14(1.878)^2] = 11.08 \text{ cm}^2$

Symbols

P_{snow} = **Density of snow** (units: g/cm³)

P_{water} = **Density of water** (units: g/cm³)

SD = **Snow Depth** (cm)

SWE = **Snow Water Equivalence** (cm)

Density is always a weight over a volume: $\frac{\text{weight (g)}}{\text{volume (cm}^3\text{)}}$

Math

$$SWE = \frac{P_{\text{snow}}}{P_{\text{water}}} * SD \quad \text{The density of water is 1 (well close enough to call it one....) so we can then say: } SWE = P_{\text{snow}} * SD$$

$$P_{\text{snow}} = \frac{\text{weight (g)}}{\text{volume (cm}^3\text{)}} \text{ or } \frac{\text{weight (g)}}{\text{surface area (cm}^2\text{)} * \text{depth (cm)}} \quad \text{In this case, depth is the snow depth, and the surface area is the surface area of the interior of the tube so:}$$

$$SWE = \frac{\text{weight (g)}}{\text{surface area (cm}^2\text{)} * \cancel{SD \text{ (cm)}}} * \cancel{SD \text{ (cm)}} \quad \text{SD on the numerator and denominator cancel each other so:}$$

$$SWE = \frac{\text{weight (g)}}{\text{surface area (cm}^2\text{)}} \quad \text{Weight is the weight of snow in the tube in grams and surface area is 11.08 so: } SWE(\text{cm}) = \frac{\text{weight (g)}}{11.08}$$

(If you want inches divide by 2.54)

Automated Snow Measurements

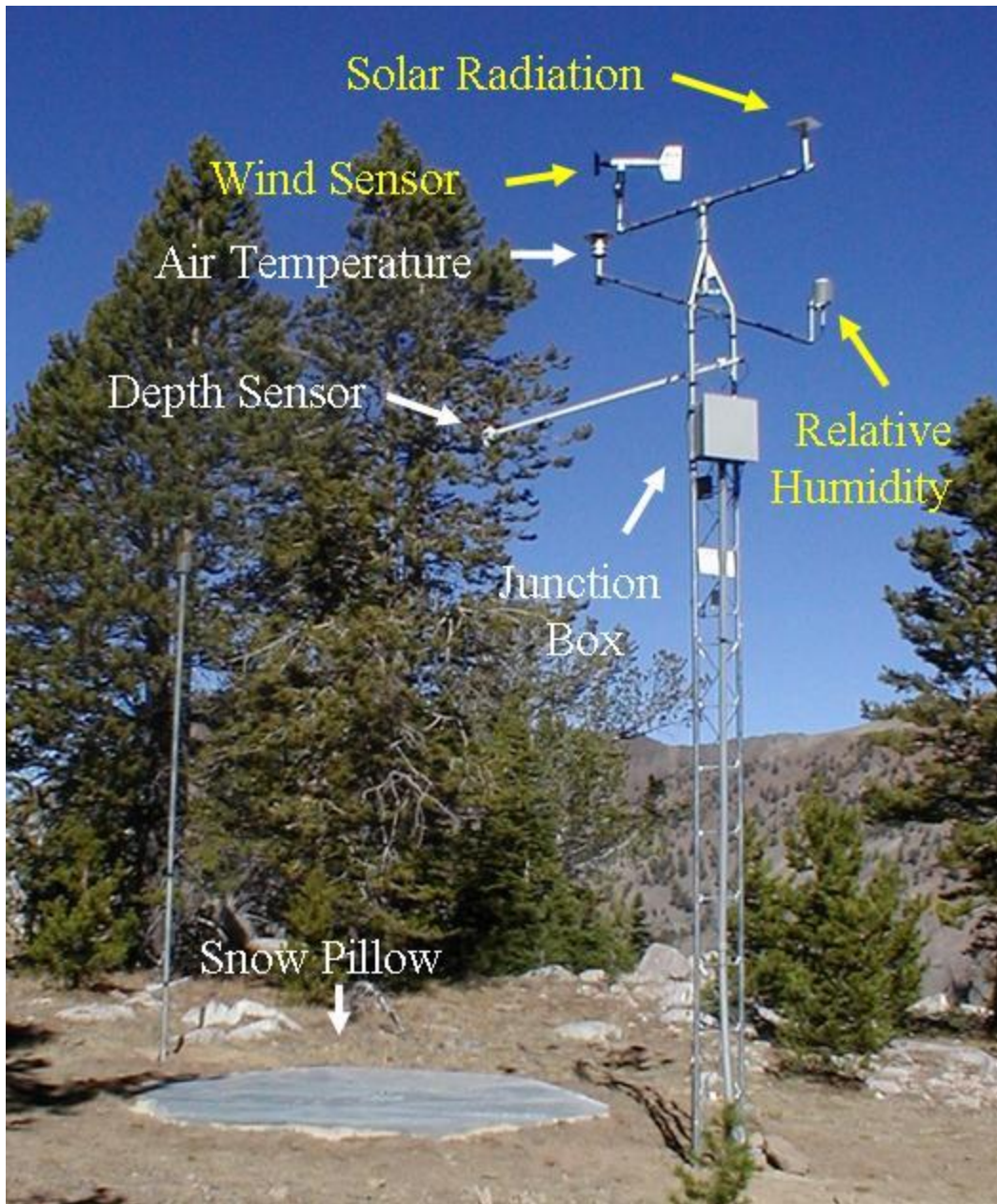
SNOTEL SITES:

Snow Pillow:

large 'pillow' of rubber containing antifreeze.

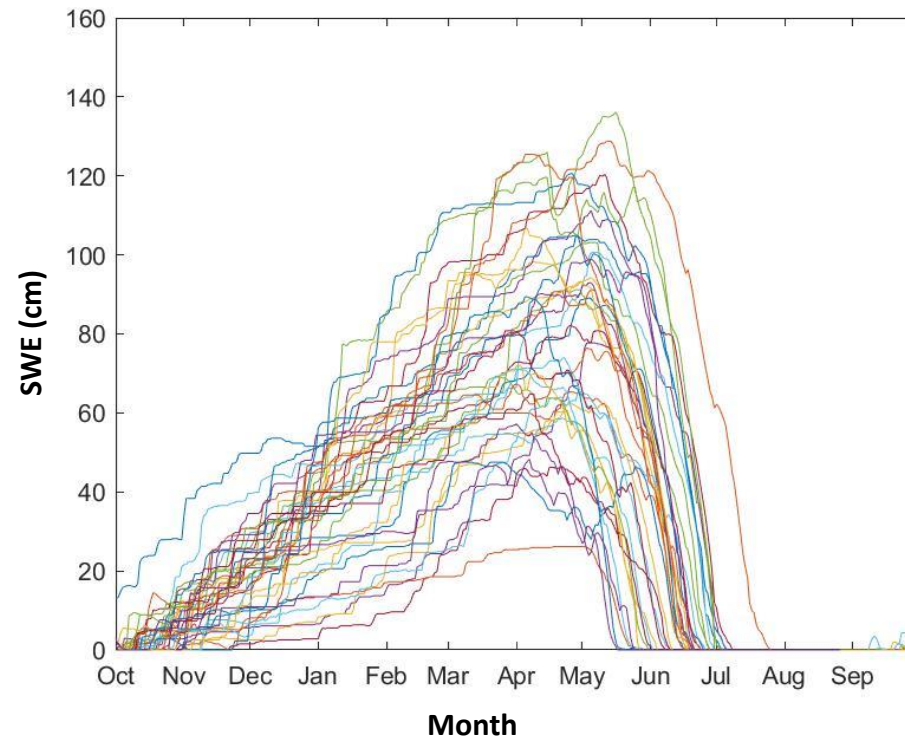
As snow accumulates it increases pressure on the pillow

Pressure is measured with a transducer and converted into a SWE value



SWE CURVE

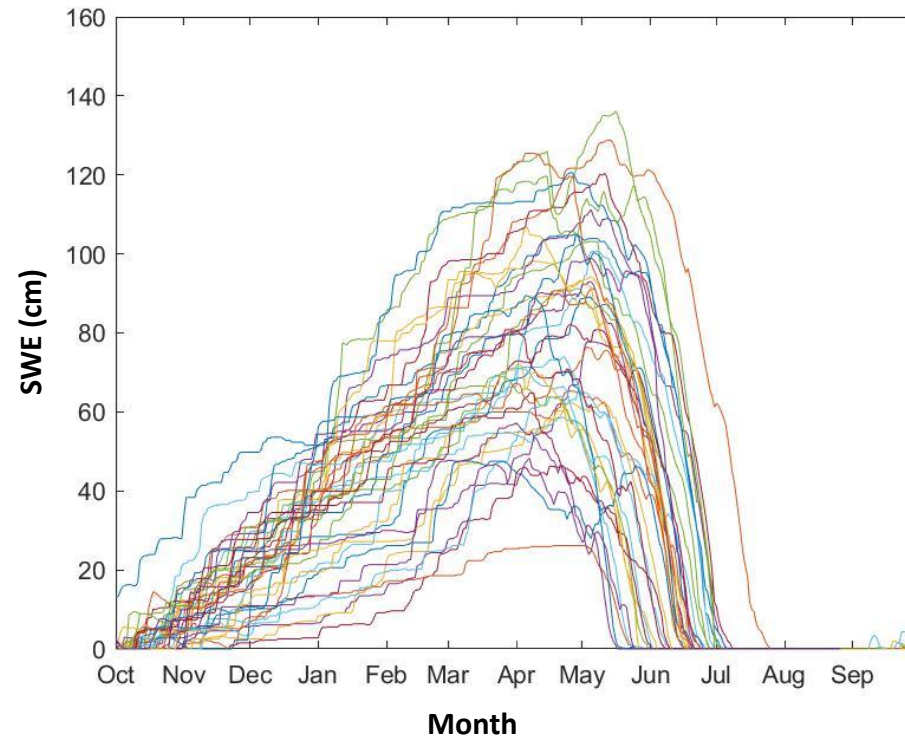
Wolf Creek Snotel (1980 – Present)



What Can you See?

SWE CURVE

Wolf Creek Snotel (1980 – Present)



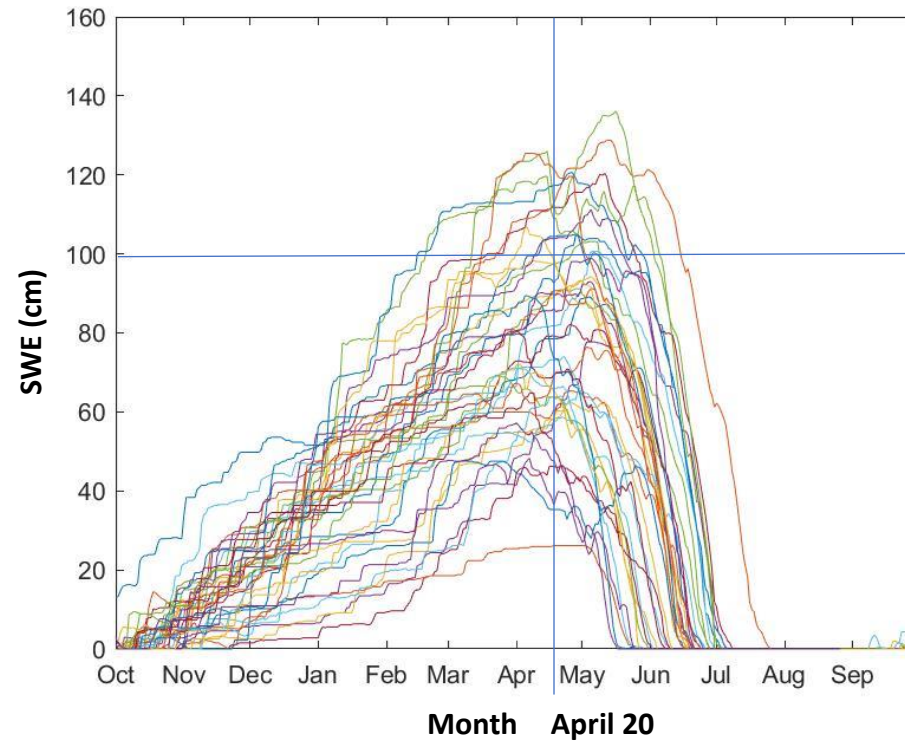
[Average]
Snow Cover Duration:
Accumulation Rate:
Ablation Rate:
Peak SWE:
Peak SWE Date:
Melt-out Date

?

What Can you See?

SWE CURVE

Wolf Creek Snotel (1980 – Present)



[Average]

Snow Cover Duration: ~285 days

Accumulation Rate: ~5.5 mm/ day

Ablation Rate: ~16 mm/day

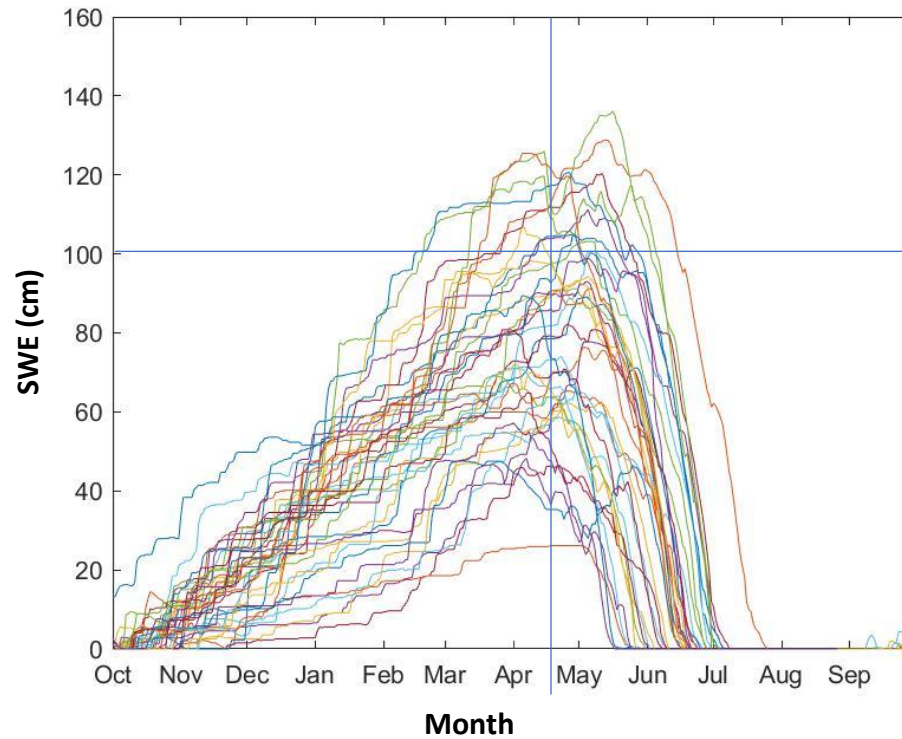
Peak SWE: ~1 meter

Peak SWE Date: ~April 20

Melt-out Date: ~June 18

SWE CURVE

Wolf Creek Snotel (1980 – Present)



[Average]

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Accumulation Rate: ~5.5 mm/ day

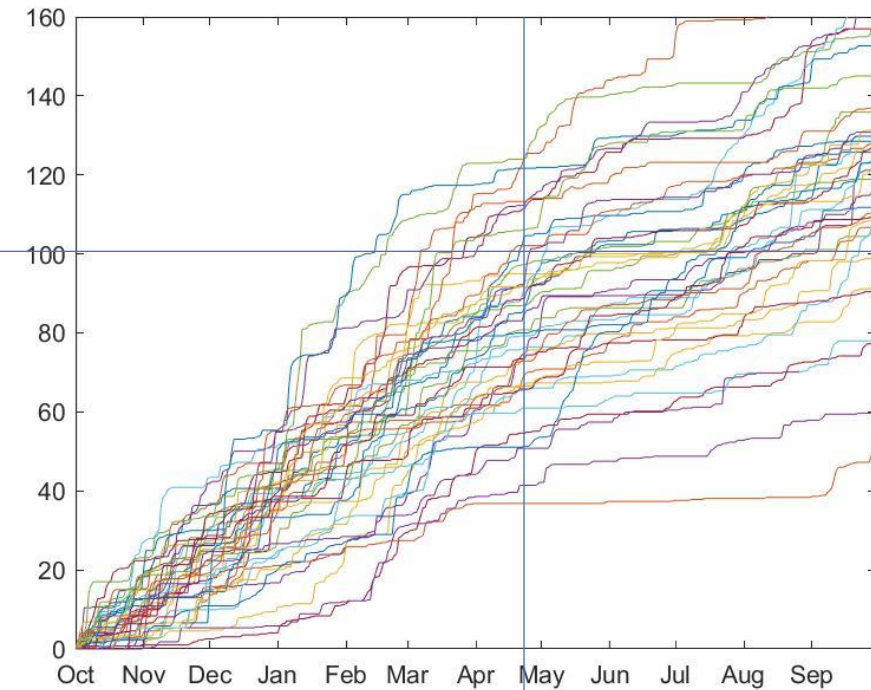
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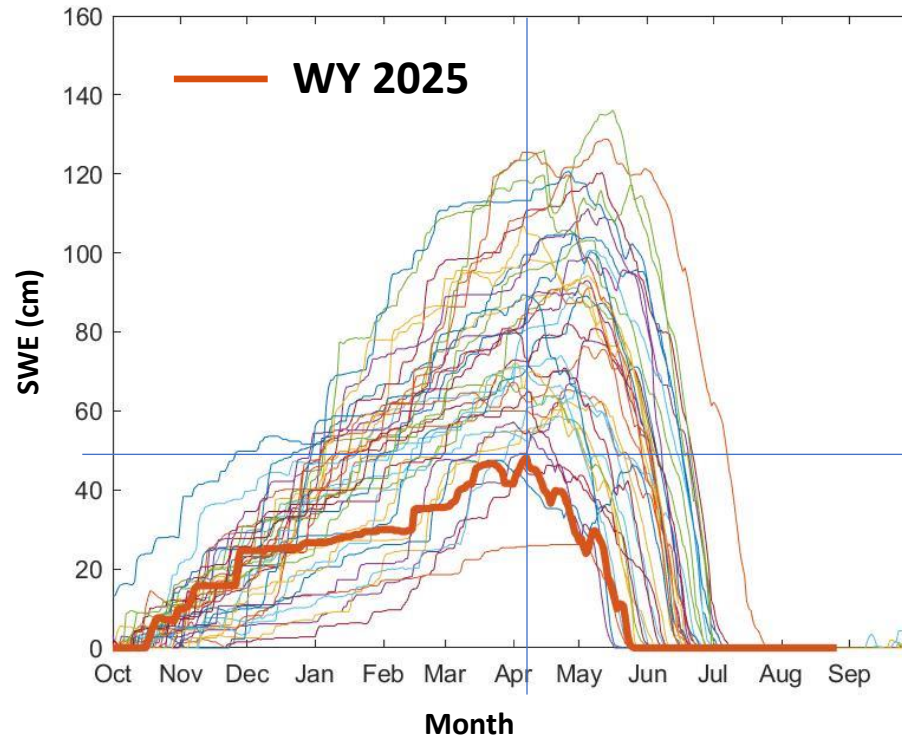
Cumulative Hydrograph



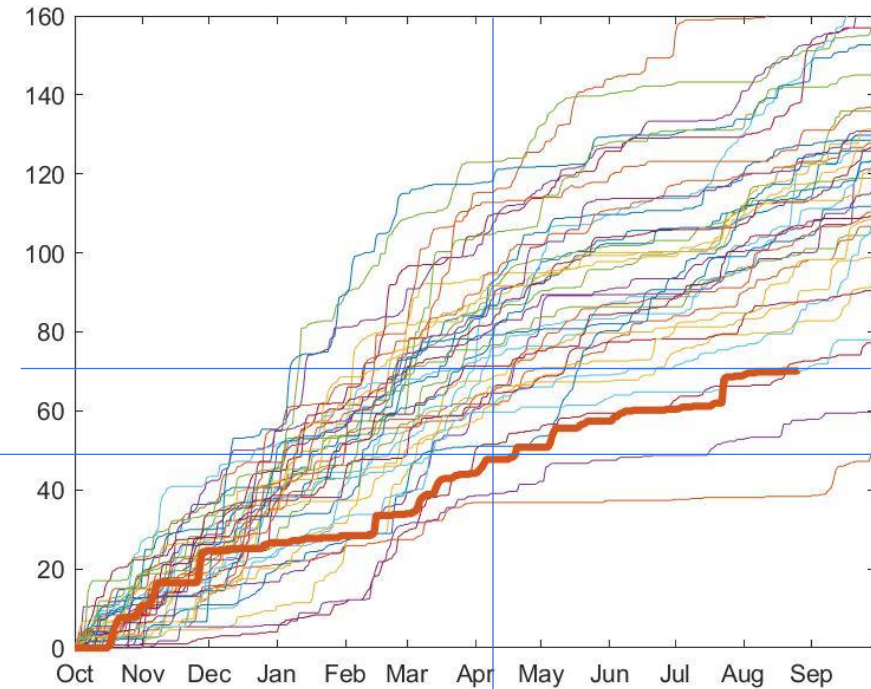
What else can we see?

SWE CURVE

Wolf Creek Snotel (1980 – Present)



Cumulative Hydrograph



[Average]

Snow Cover Duration: ~285 days
Accumulation Rate: ~5.5 mm/ day
Ablation Rate: ~16 mm/day
Peak SWE: ~100 cm
Peak SWE Date: ~April 20
Melt-out Date: ~June 18

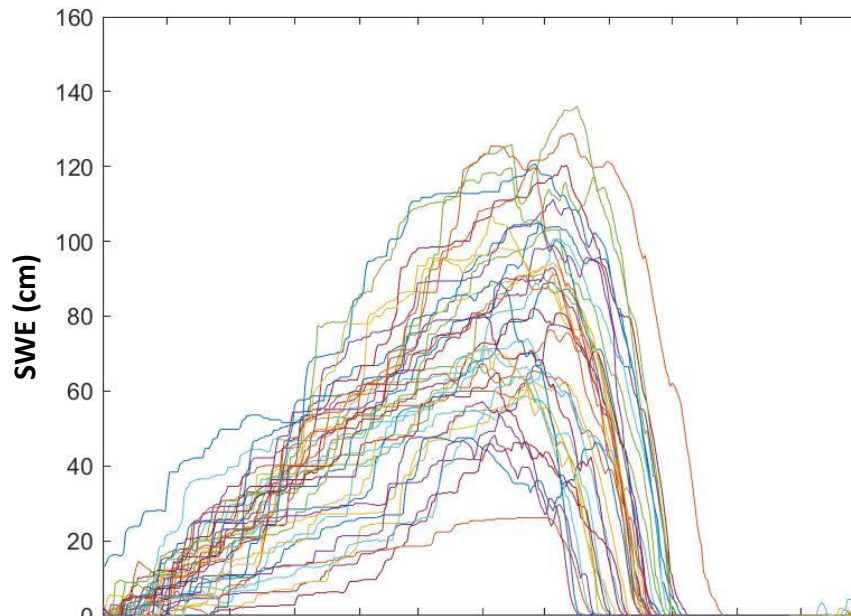
[— WY 2025]

Snow Cover Duration: ~210 days
Accumulation Rate: ~2.6 mm/ day
Ablation Rate: ~12 mm/day
Peak SWE: ~47 cm
Peak SWE Date: ~April 08
Melt-out Date: ~May 18

Total Precipitation: 71 cm
Precipitation Ratio [SWE/Total]: 0.66



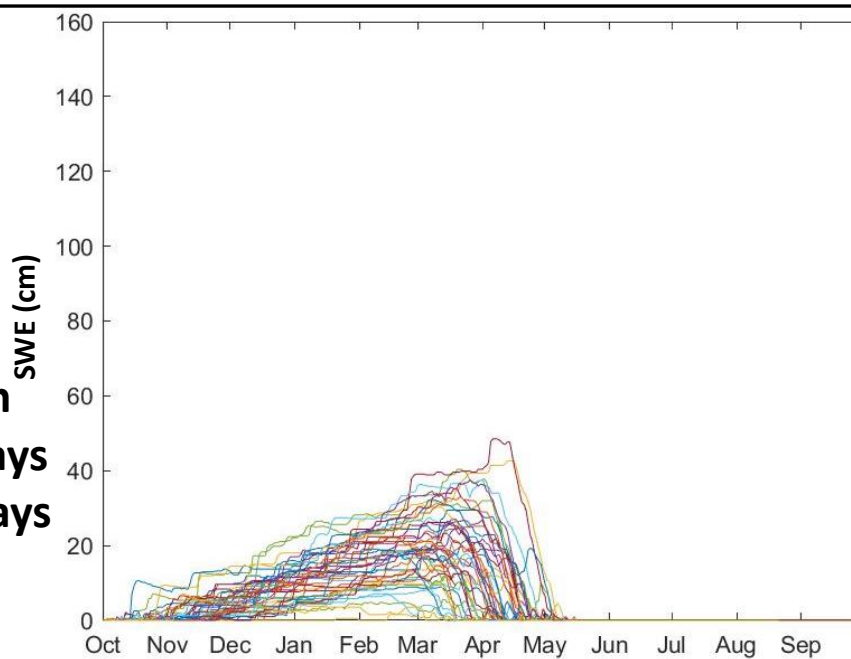
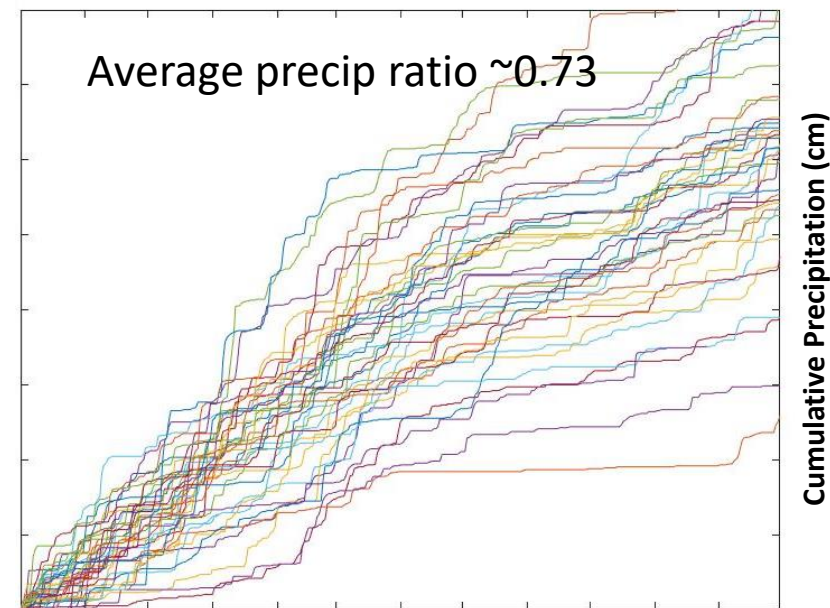
SWE CURVE



Cumulative Hydrograph

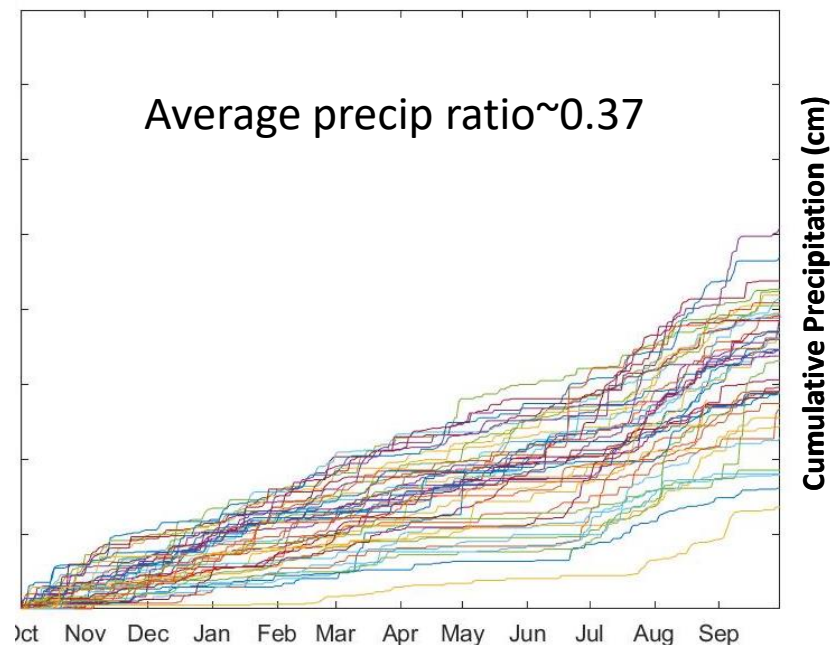
Average precip ratio ~ 0.73

Wolf Creek



Average precip ratio ~ 0.37

Quemazon

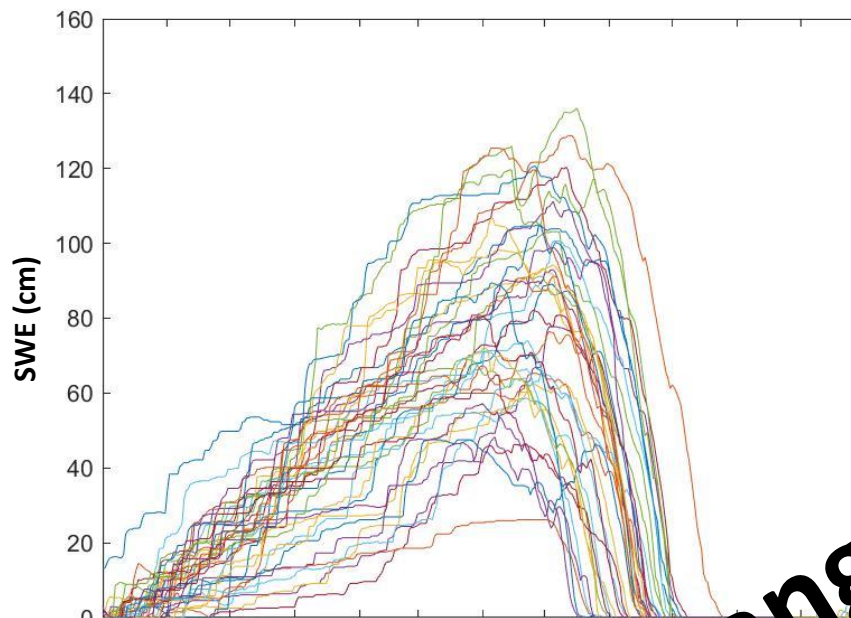


[Average Differences]

- Δ Peak SWE: ~ 75 cm
- Δ Peak SWE Date: ~ 20 days
- Δ Melt-out Date: ~ 50 days
- Δ Precipitation Ratio: ~ 0.36



SWE CURVE

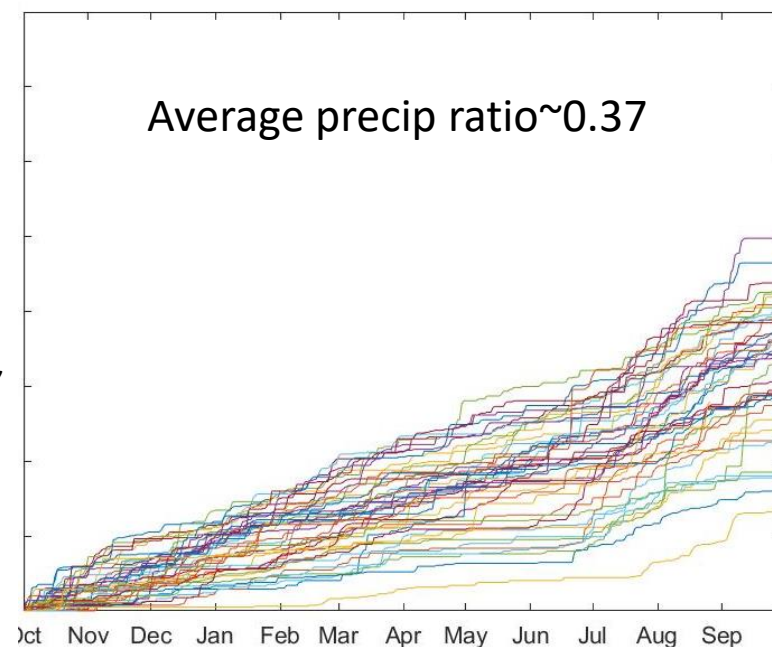


Cumulative Hydrograph

Average precip ratio ~ 0.73

Wolf Creek

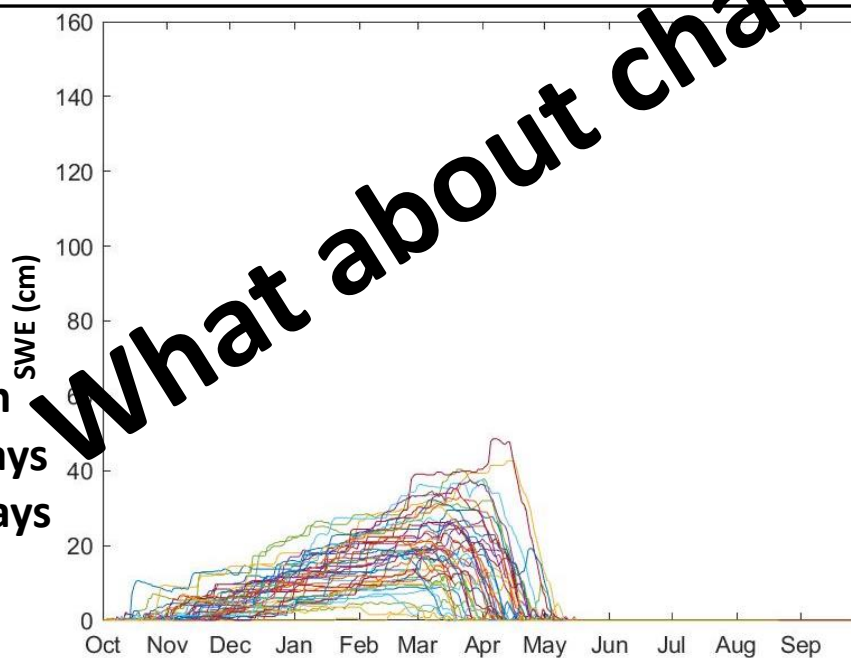
Cumulative Precipitation (cm)



Average precip ratio ~ 0.37

Quemazon

Cumulative Precipitation (cm)



[Average Differences]

Δ Peak SWE:

~ 75 cm

Δ Peak SWE Date:

~ 20 days

Δ Melt-out Date:

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Δ Precipitation Ratio:

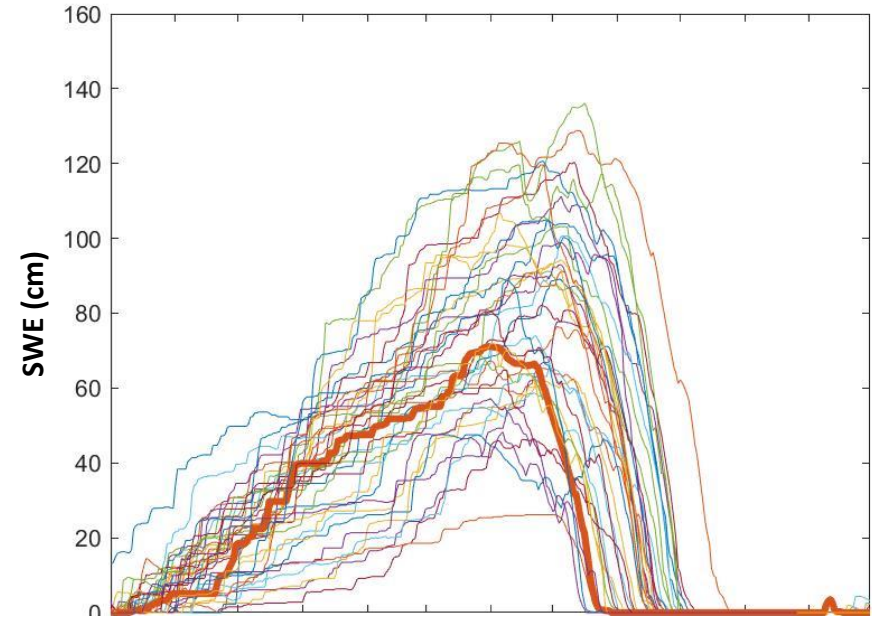
~ 0.36

What about changes through time?

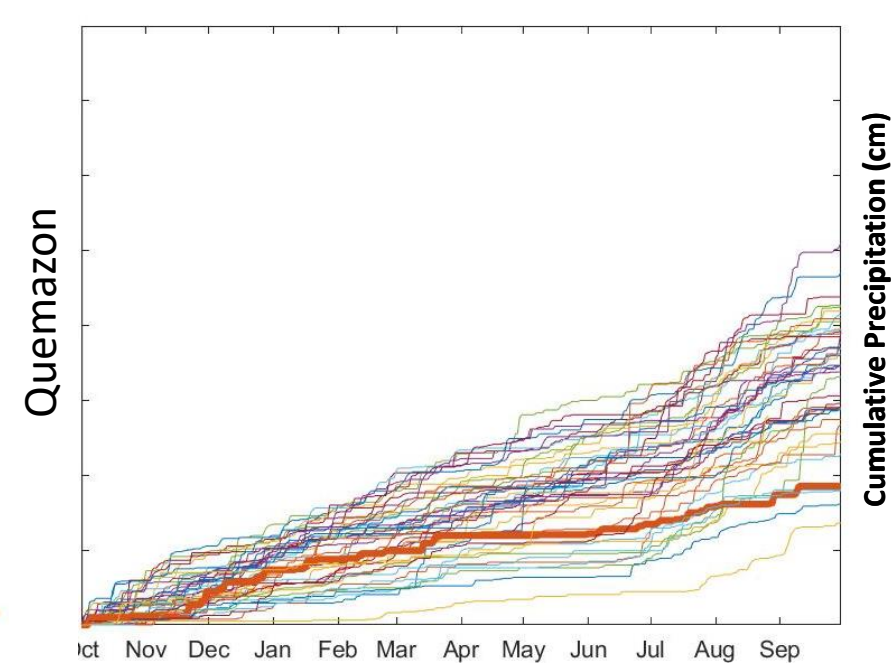
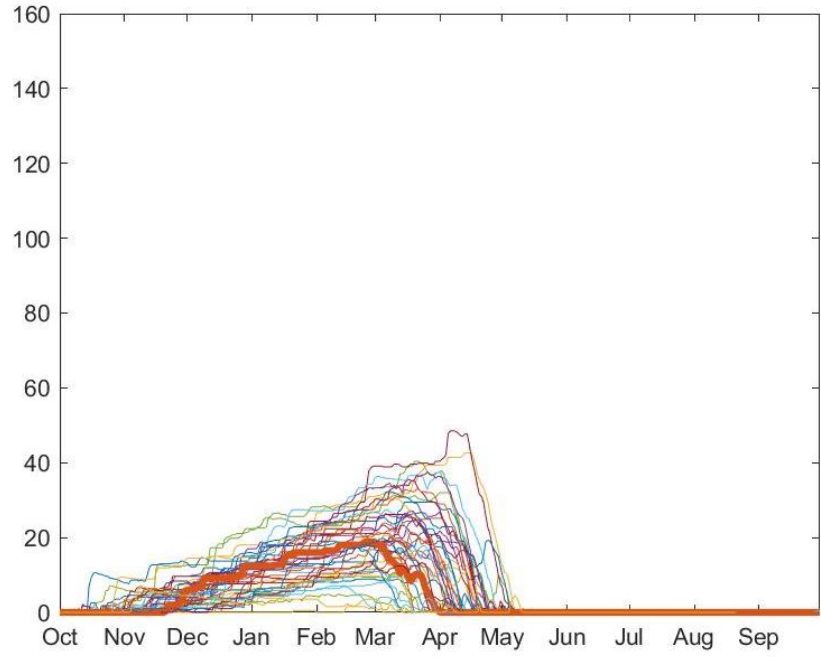
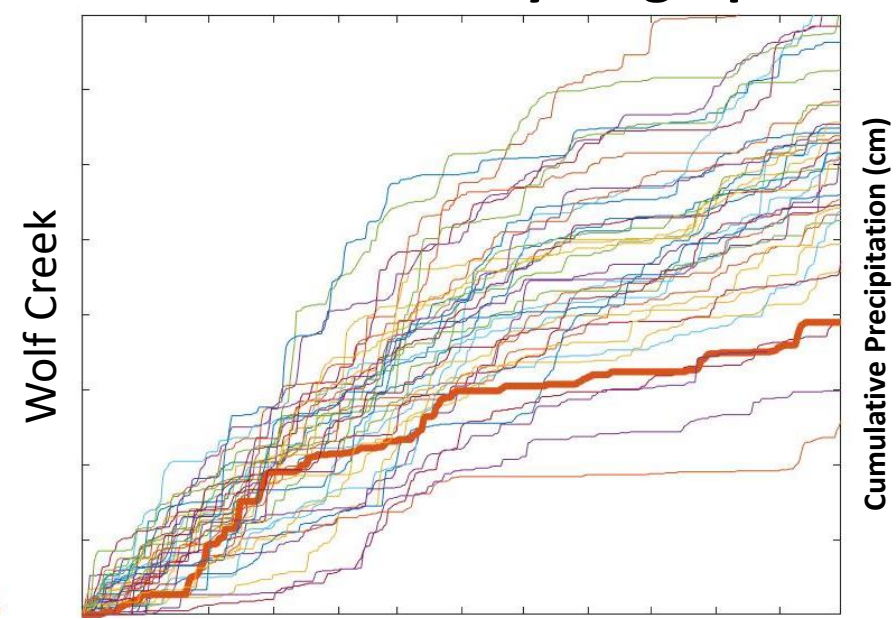


— WY 2020

SWE CURVE



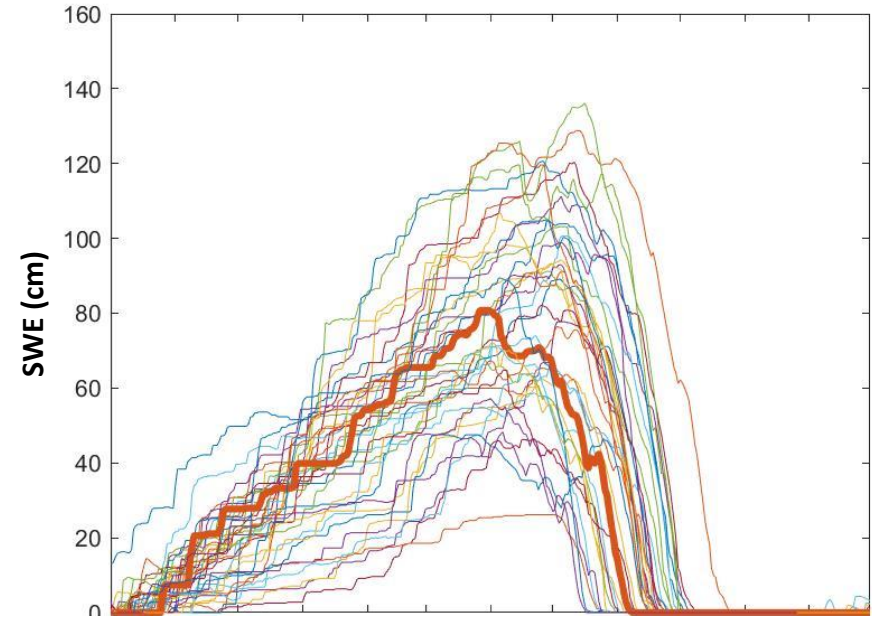
Cumulative Hydrograph



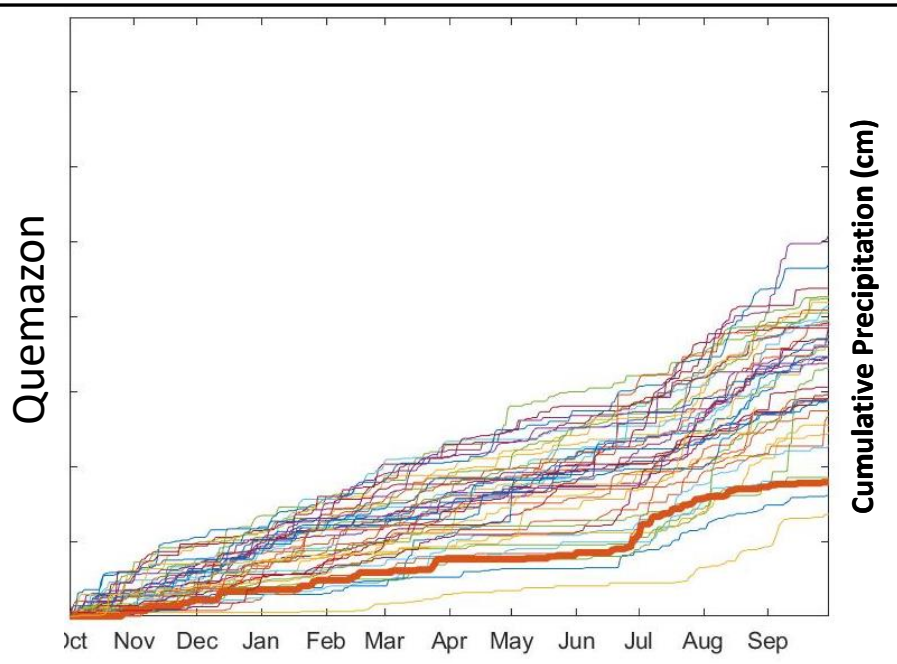
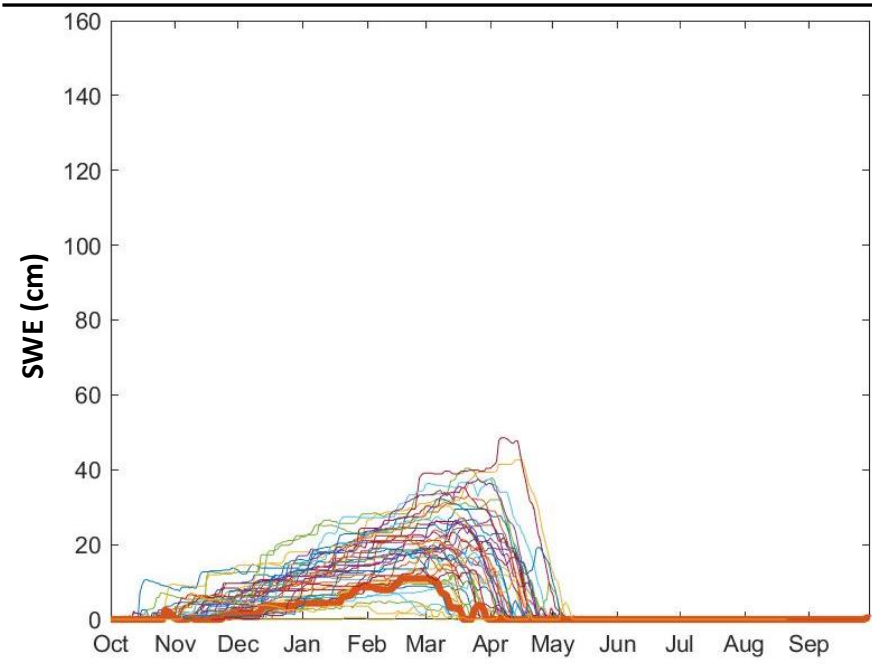
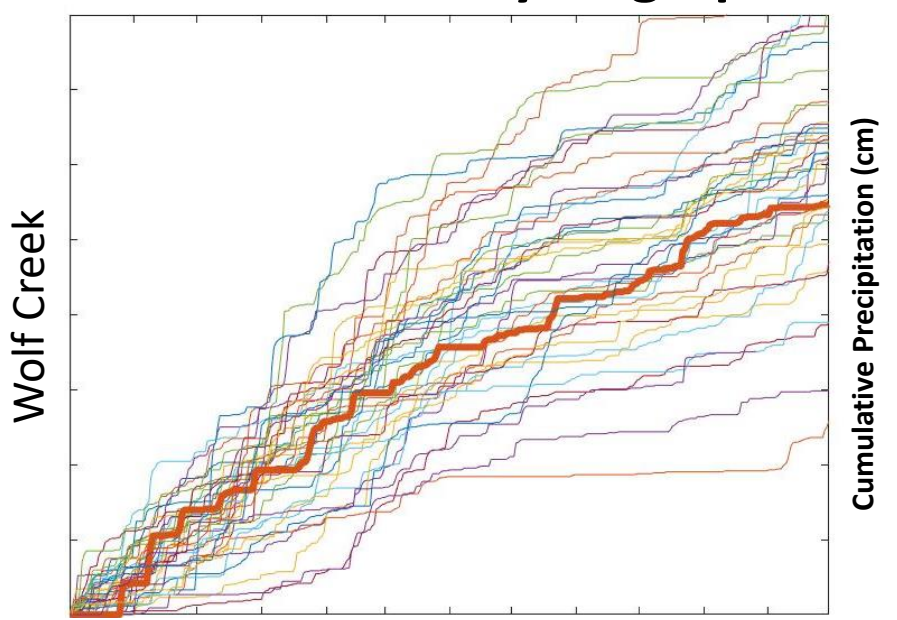


— WY 2021

SWE CURVE



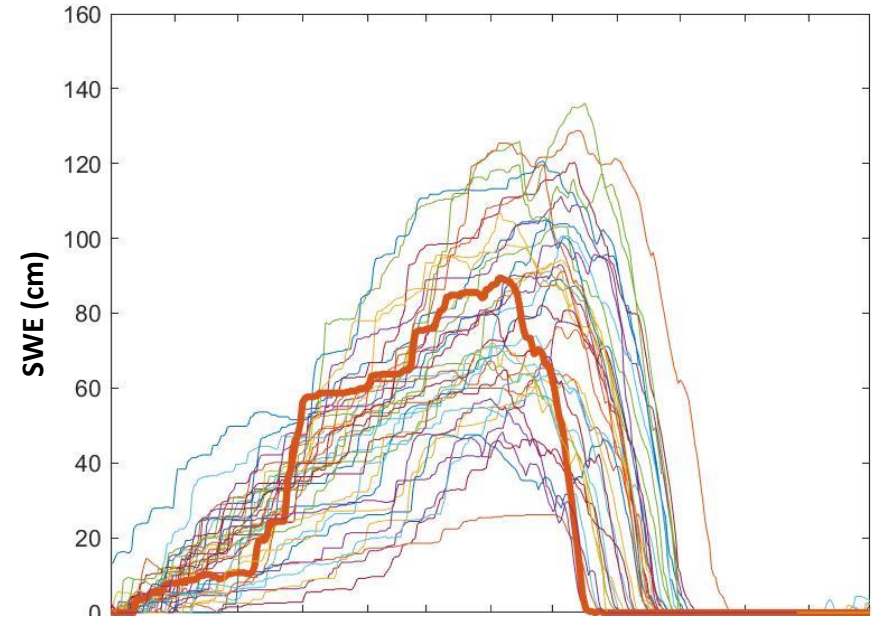
Cumulative Hydrograph



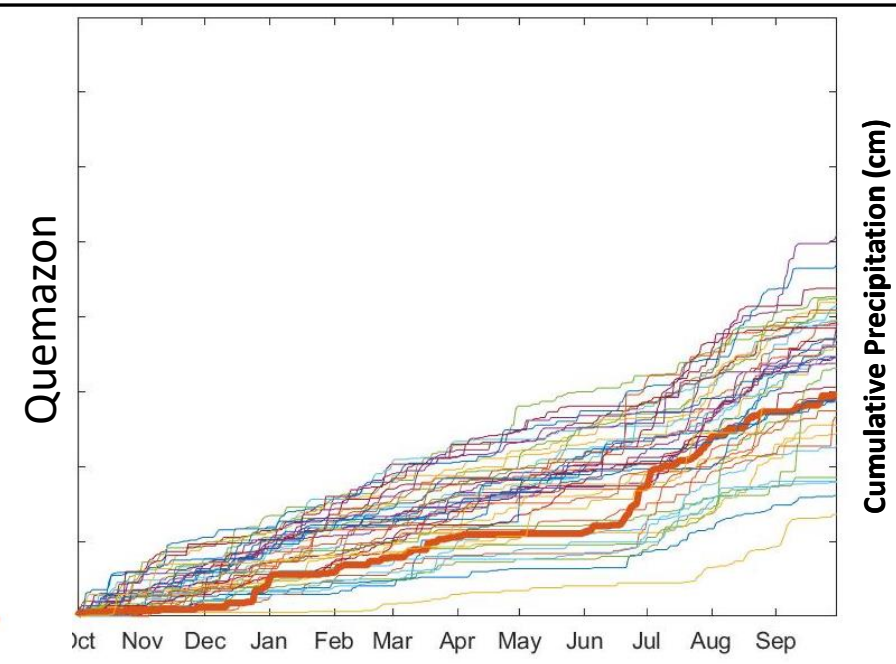
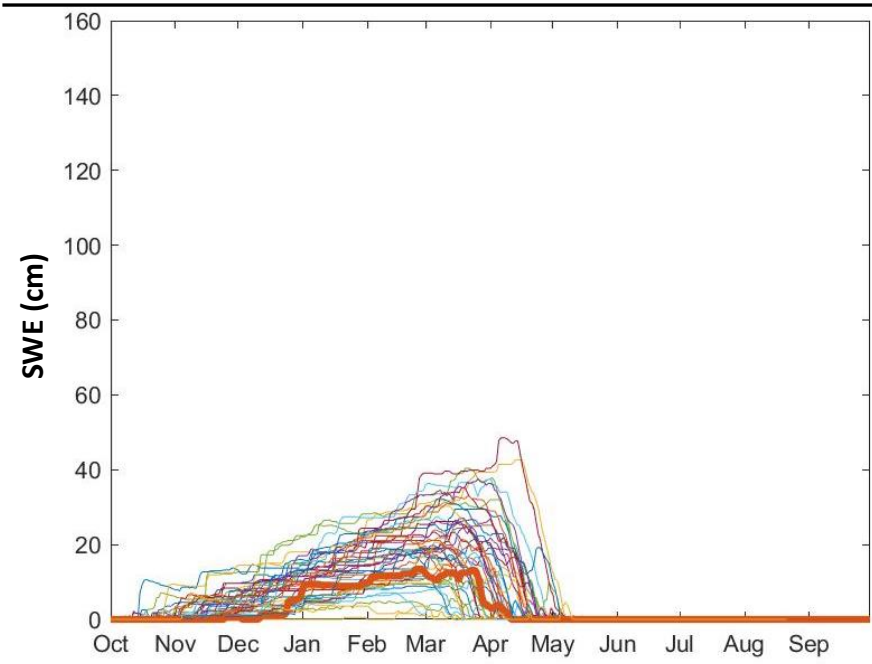
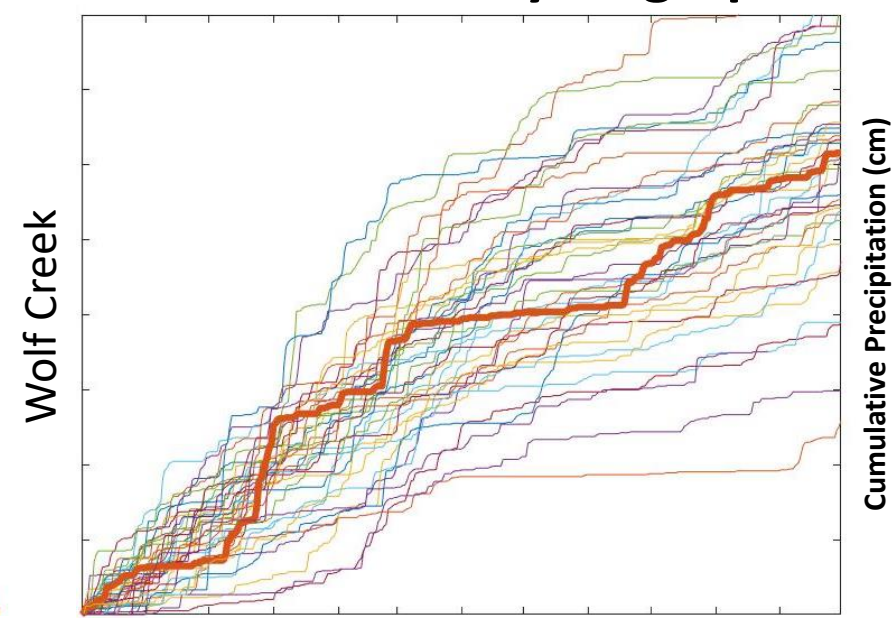


— WY 2022

SWE CURVE



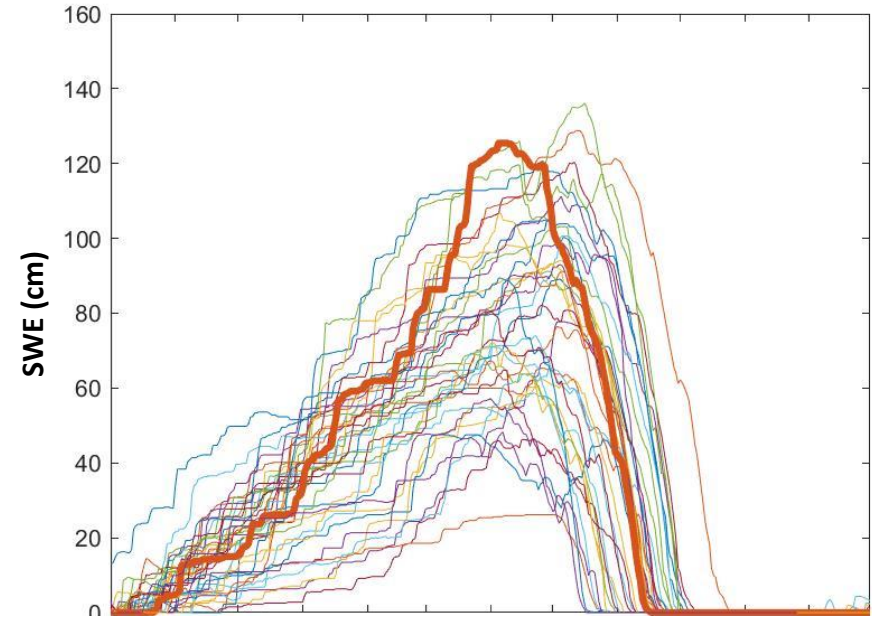
Cumulative Hydrograph



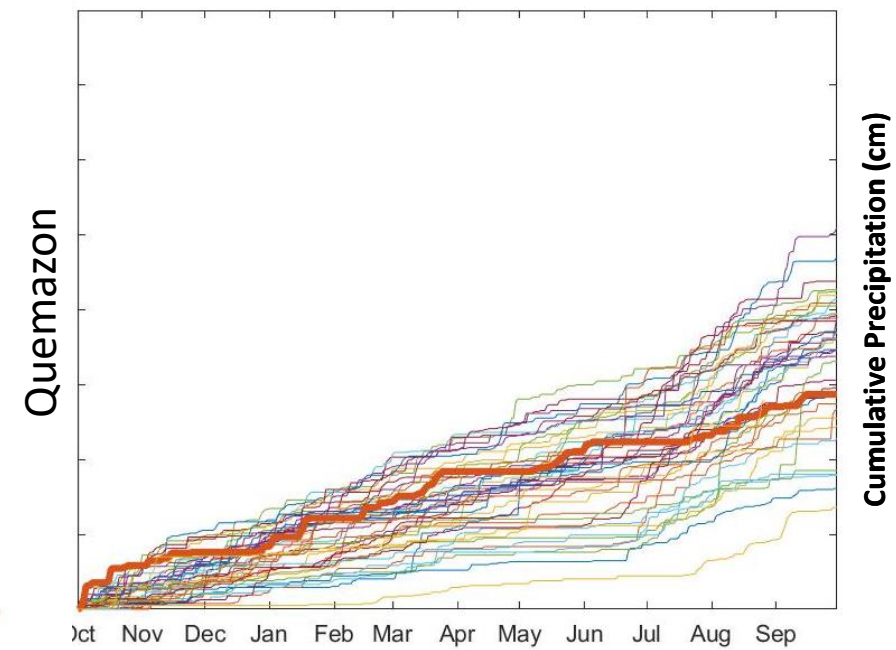
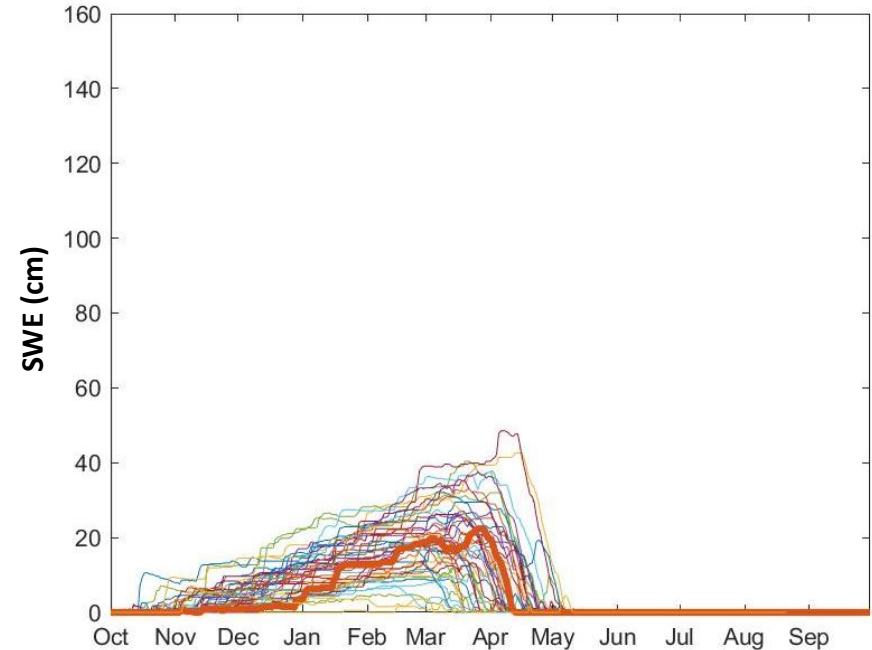
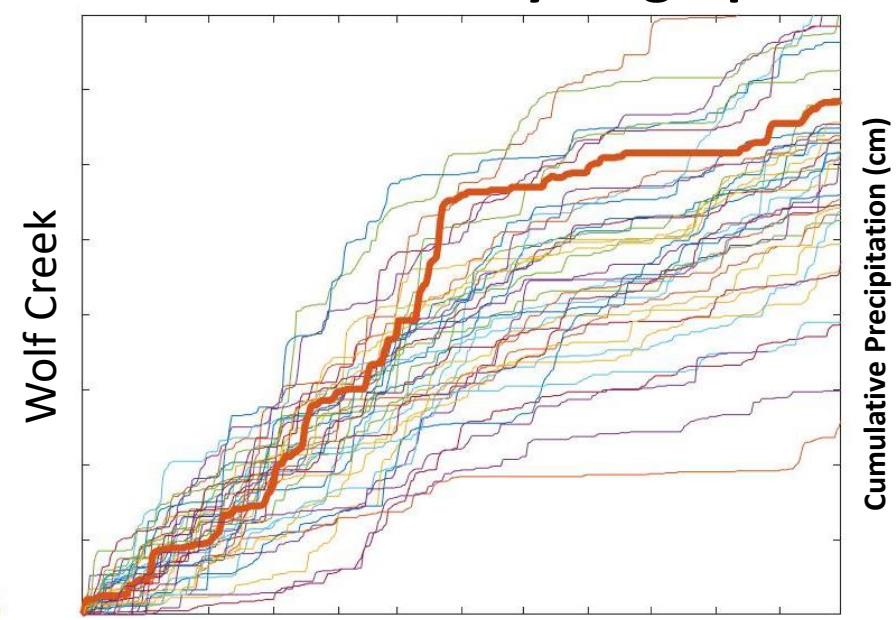


— WY 2023

SWE CURVE



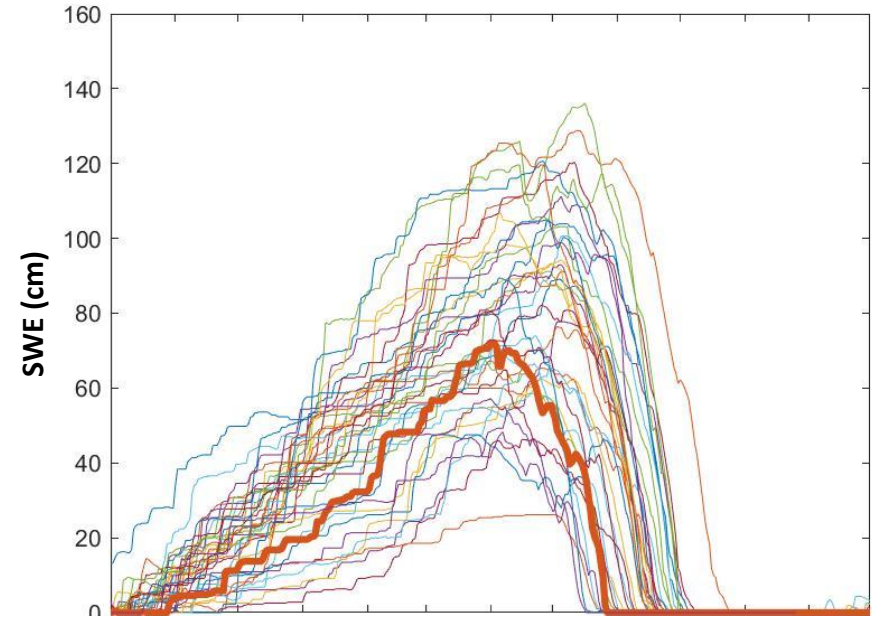
Cumulative Hydrograph



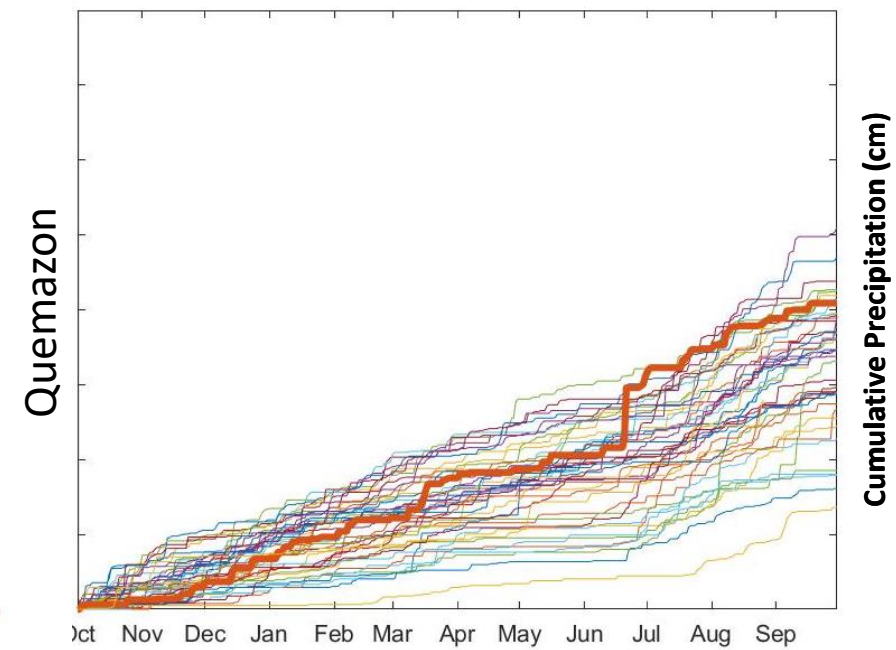
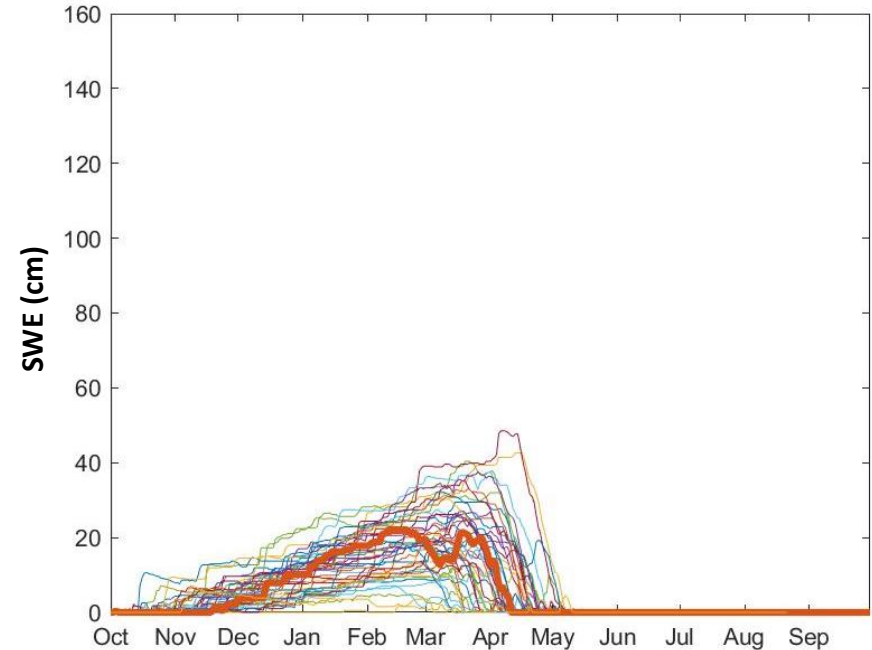
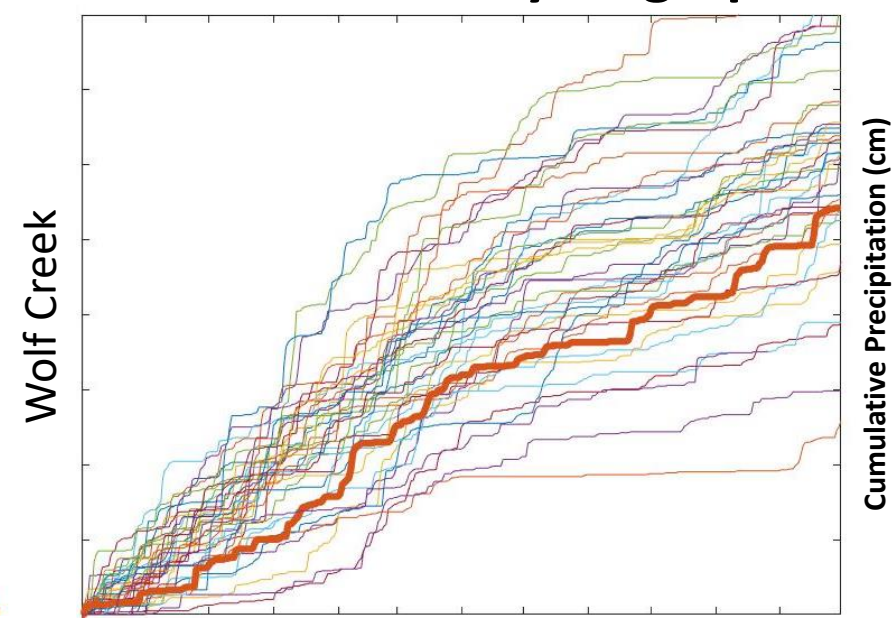


— WY 2024

SWE CURVE



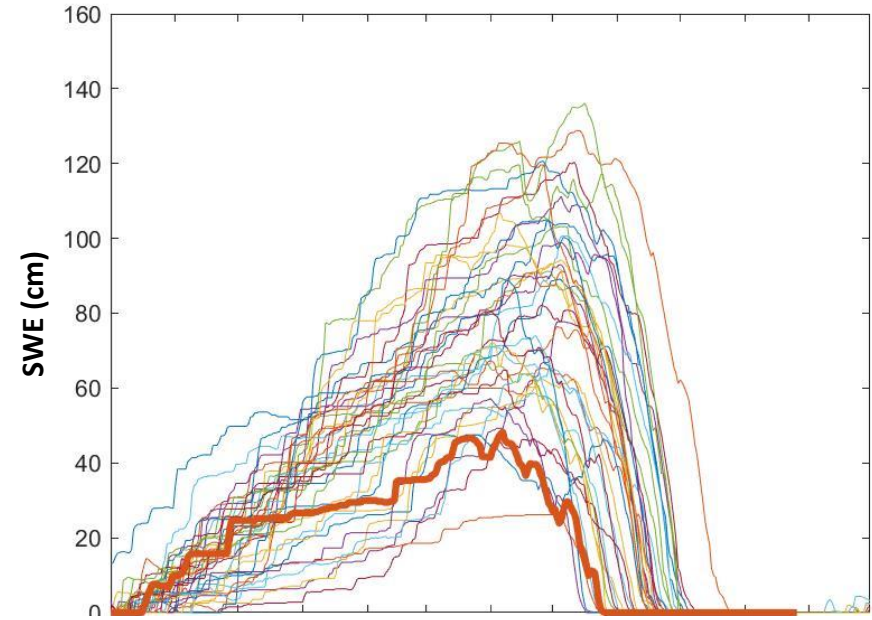
Cumulative Hydrograph



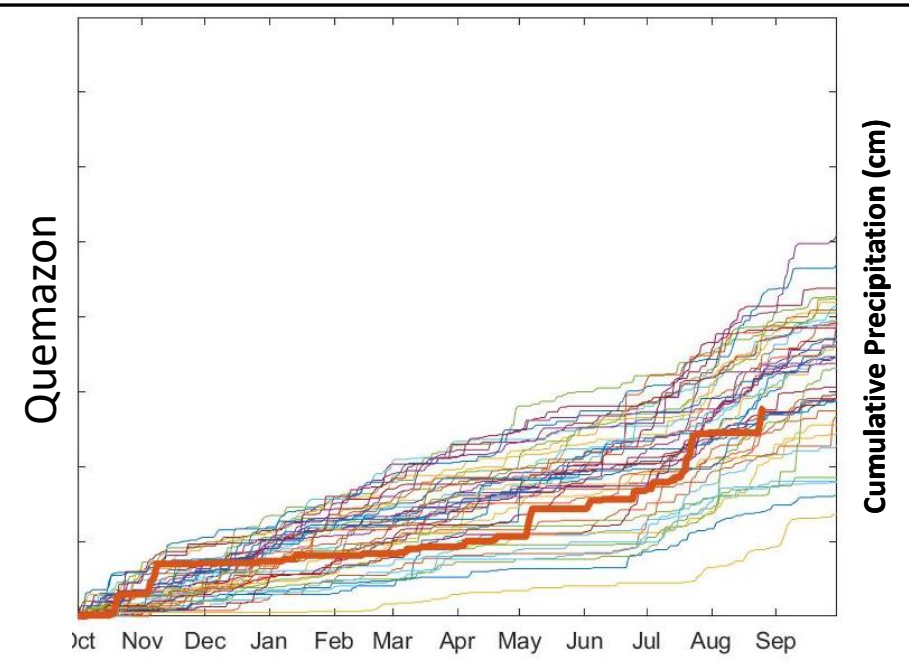
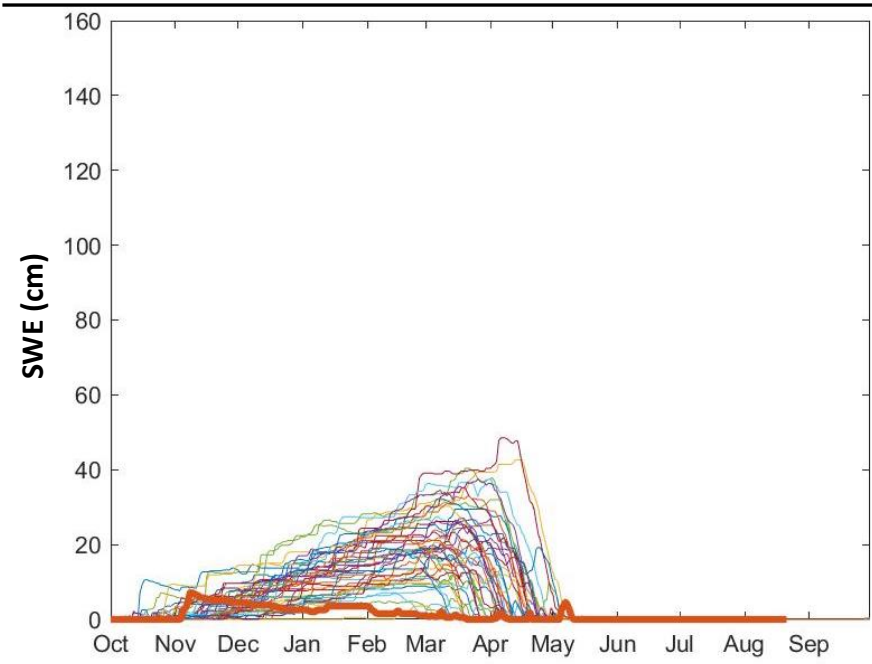
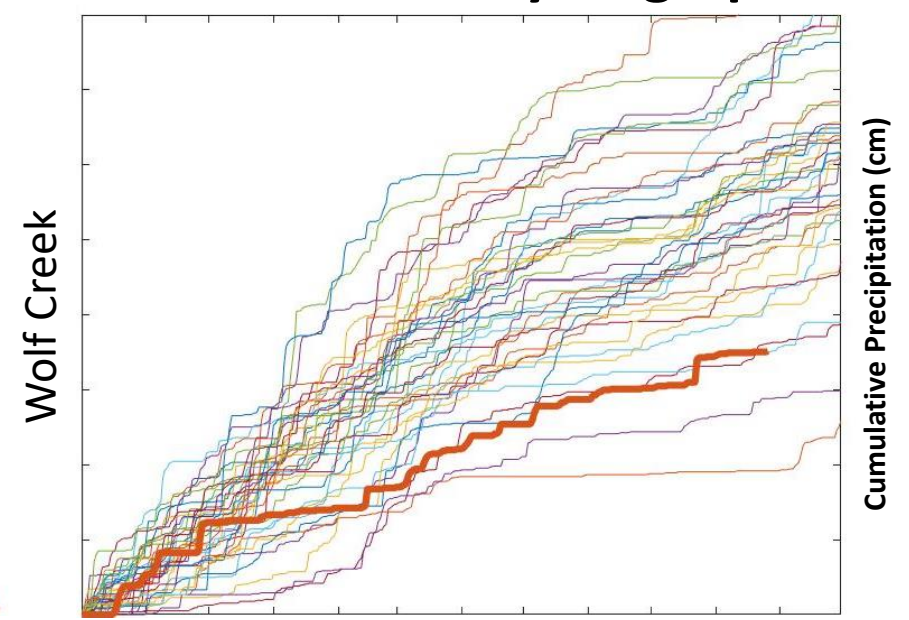


— WY 2025

SWE CURVE



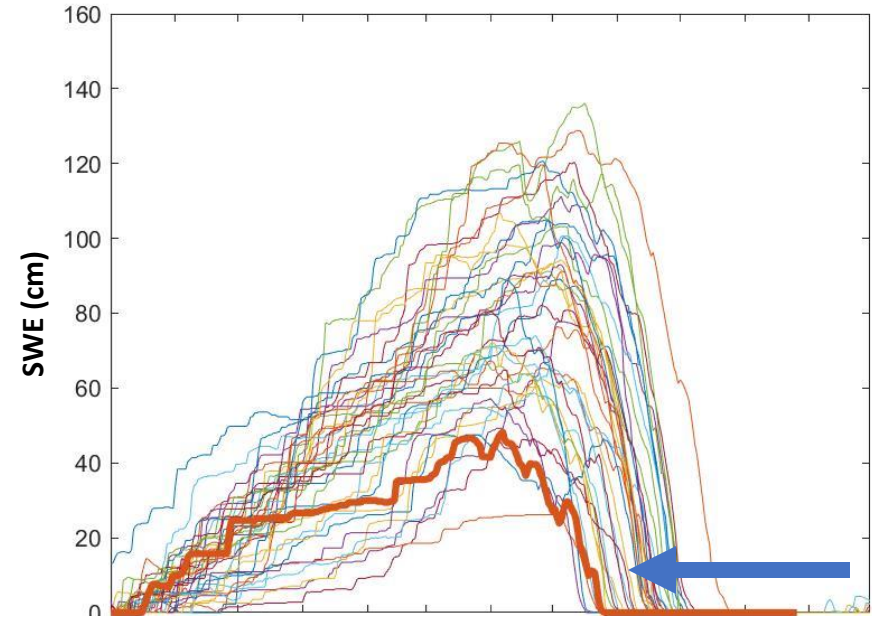
Cumulative Hydrograph



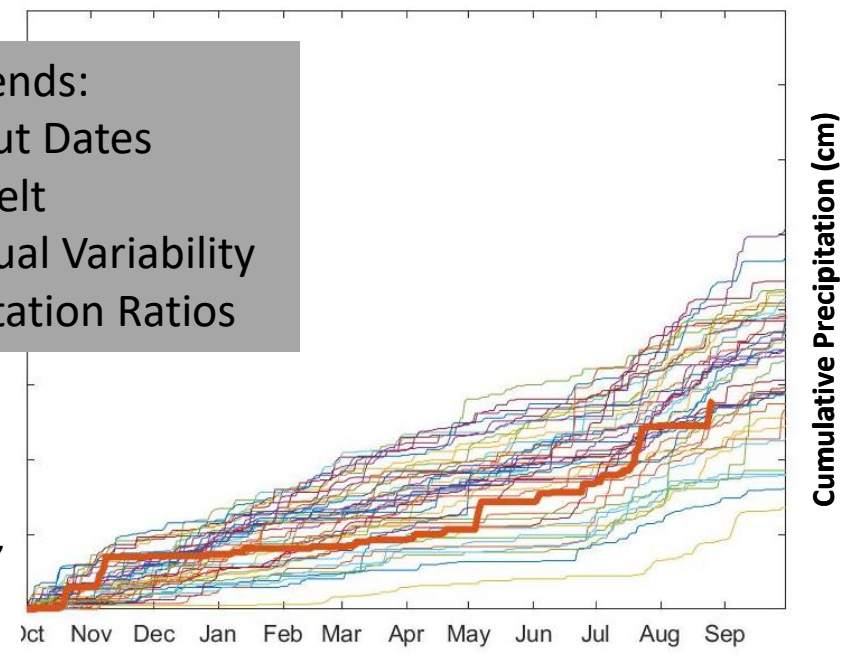
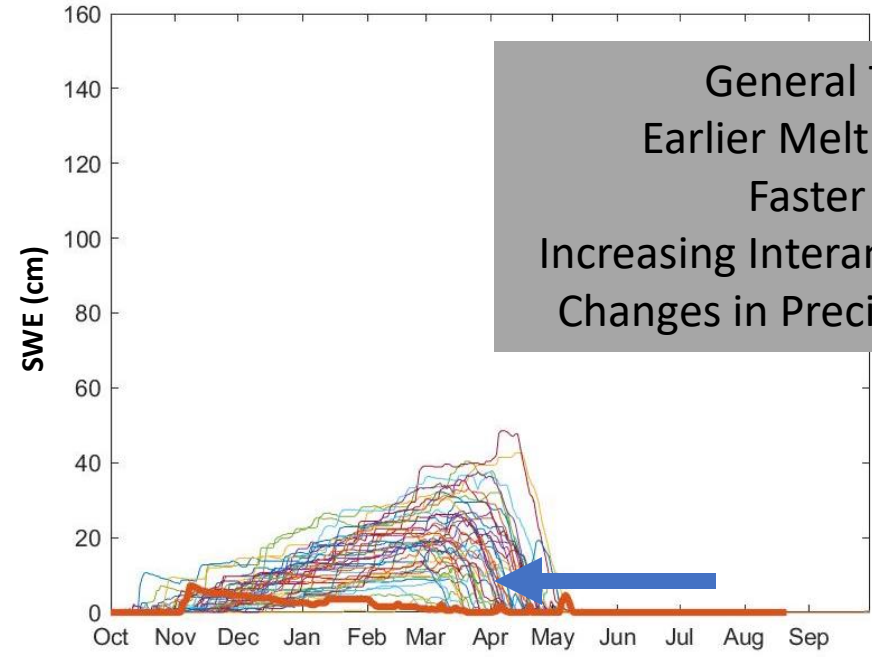
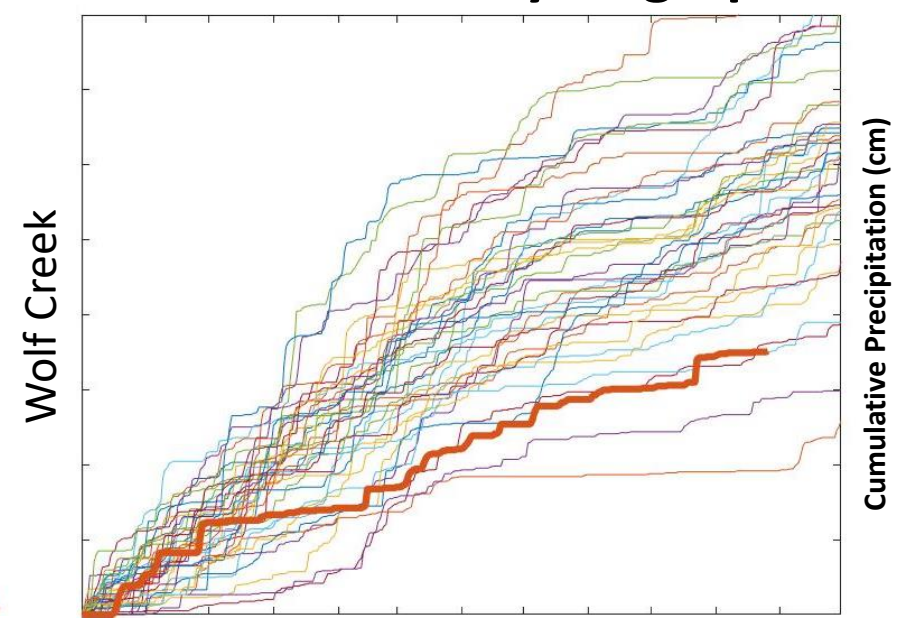


— WY 2025

SWE CURVE



Cumulative Hydrograph

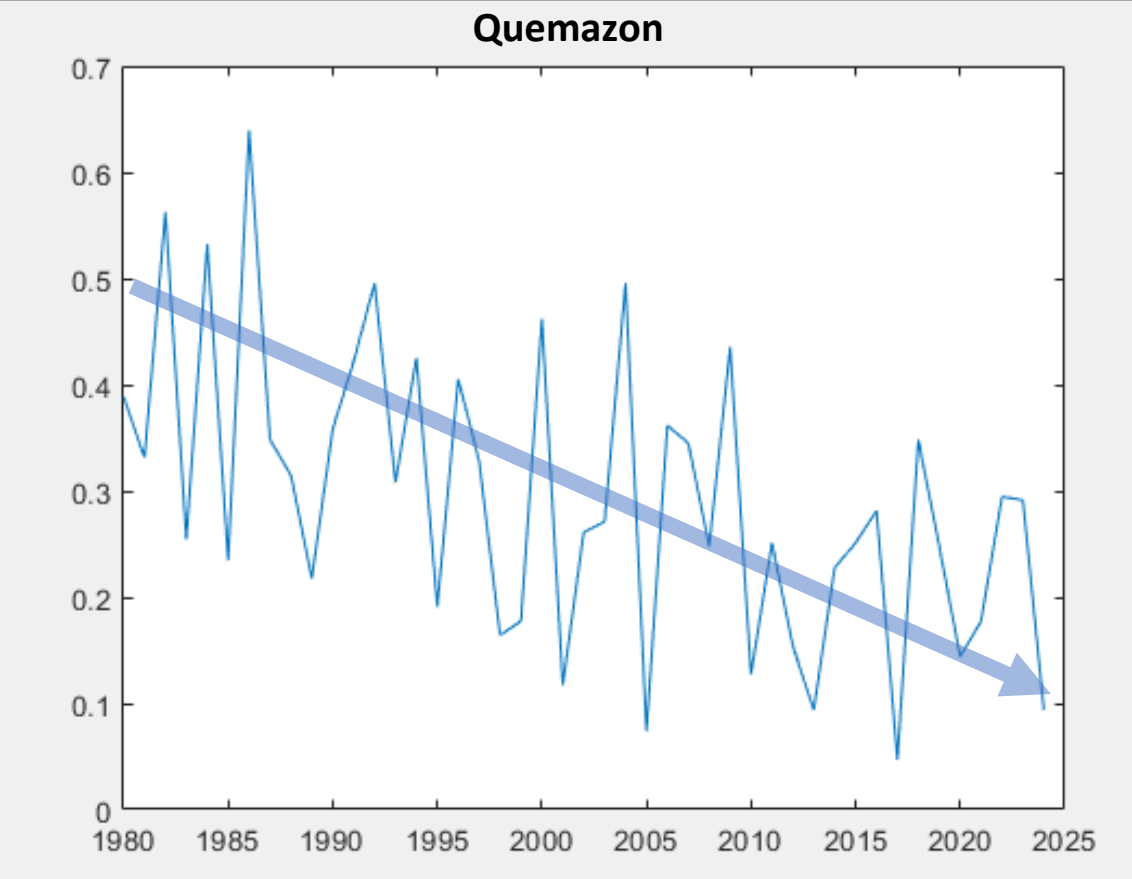


General Trends:
Earlier Melt Out Dates
Faster Melt
Increasing Interannual Variability
Changes in Precipitation Ratios

General Trends:
Changes in Precipitation Ratios



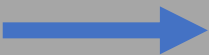
Snow to Total Precipitation Ratio



Total Precipitation is lowering through time



Primarily from Reduction in snow fall and snow permanence characteristic of low latitude sites

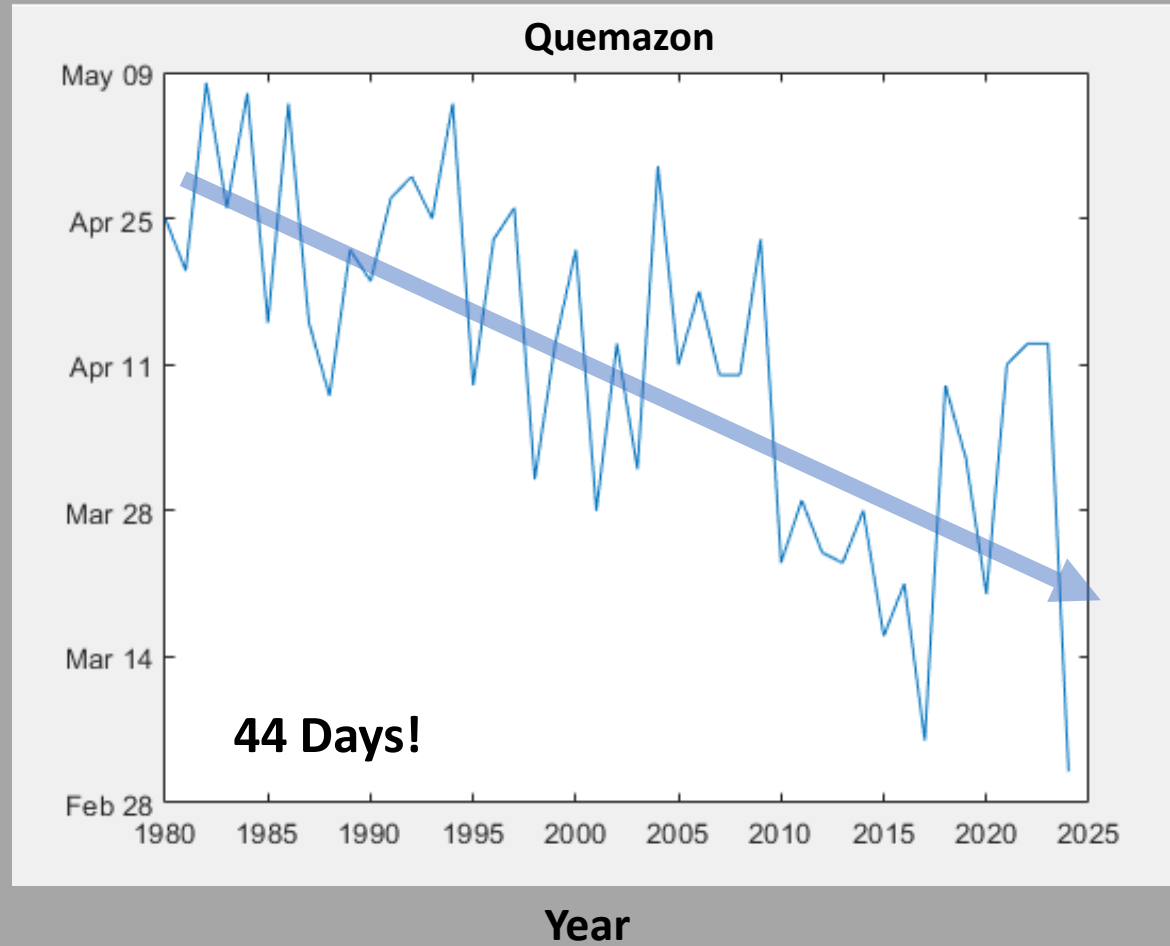


Increasing change in timing and type of precipitation

General Trends: Changes in Precipitation Ratios



Melt-Out Date



Total Precipitation is lowering through time

Primarily from Reduction in snow fall and snow permanence characteristic of low latitude sites

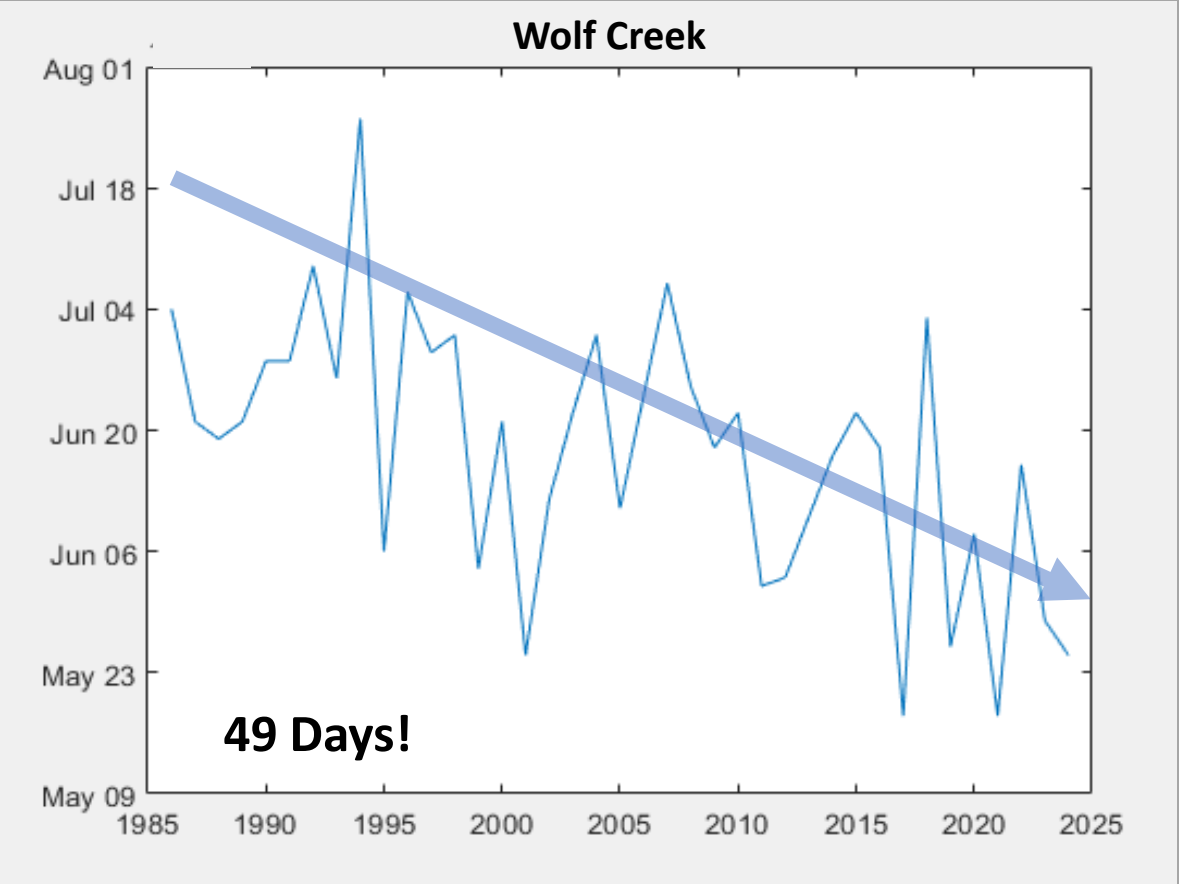
Snow has less time to infiltrate

Increasing change in timing and type of precipitation

General Trends:
Changes in Precipitation Ratios



Melt-Out Date



Total Precipitation is lowering through time

Primarily from Reduction in snow fall and snow permanence characteristic of low latitude sites

Snow has less time to infiltrate

Increasing change in timing and type of precipitation

Snow Accumulation and Snowmelt Processes:



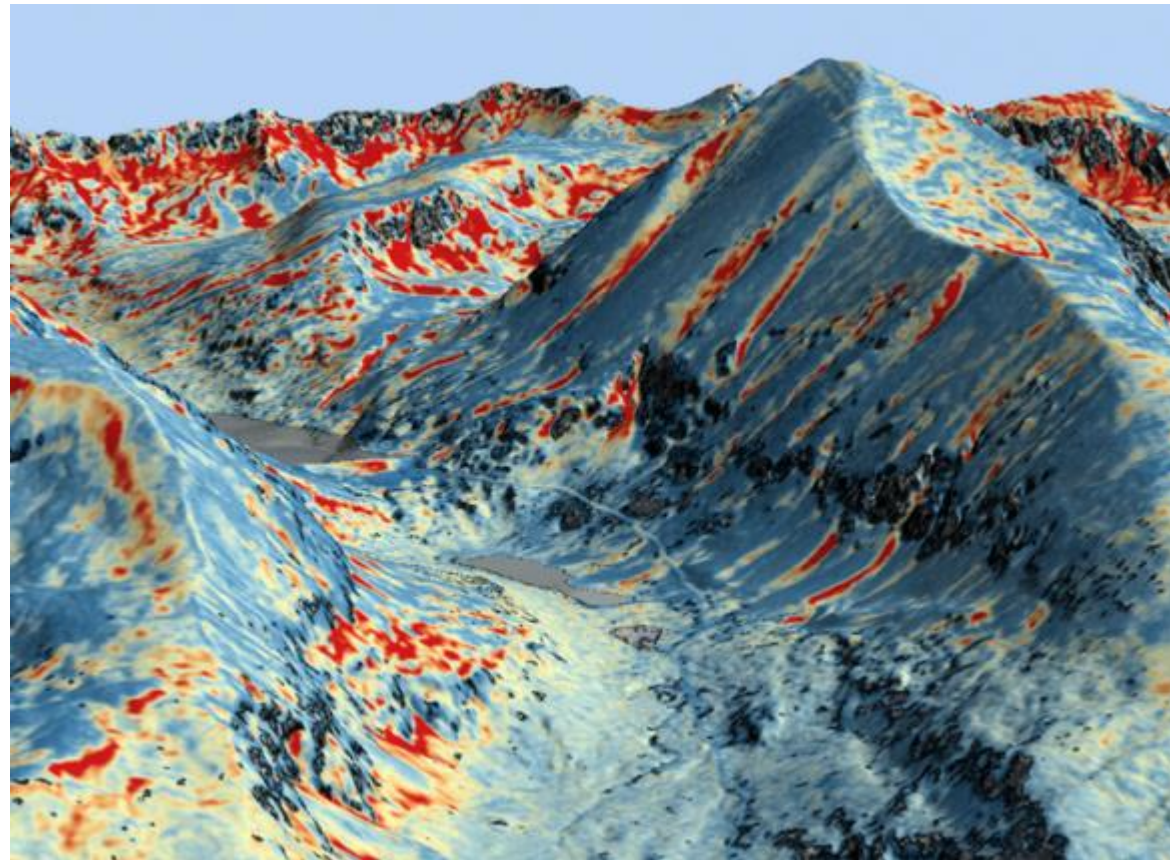
Ground
Measurements can
be Temporally
Robust

Remote Sensing:

What Happens
between the
measurement
points?

A Snapshot in Time:

Aerial and Satellite Derived Snow products



Airborne Snow Observatories- snow depth flyover map

Satellite derived:

NASA Moderate Resolution Imaging Spectroradiometer (MODIS) Products – resolution 1 km

European Space Agency Visible Infrared Imaging Radiometer Suite (VIIRS) – resolution ~750m

Snow Covered Area

Also a SWE** product

Limited by Cloud Cover, Forest Cover and Resolution

Aerial derived:

Airborne Snow Observatory

Light Detection and Ranging (LiDAR) Snow Depth derived from repeat flyovers at a 3-m resolution

Also a SWE** product

Limited by Weather and funds

However to quantify snowmelt runoff we need to know:



integration of observation and modeling



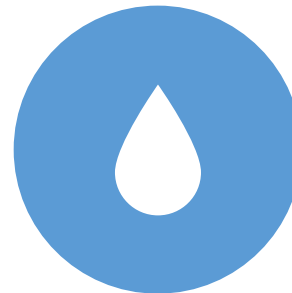
the quantity of water held in snow packs



the magnitude and rate of water lost to the atmosphere by sublimation

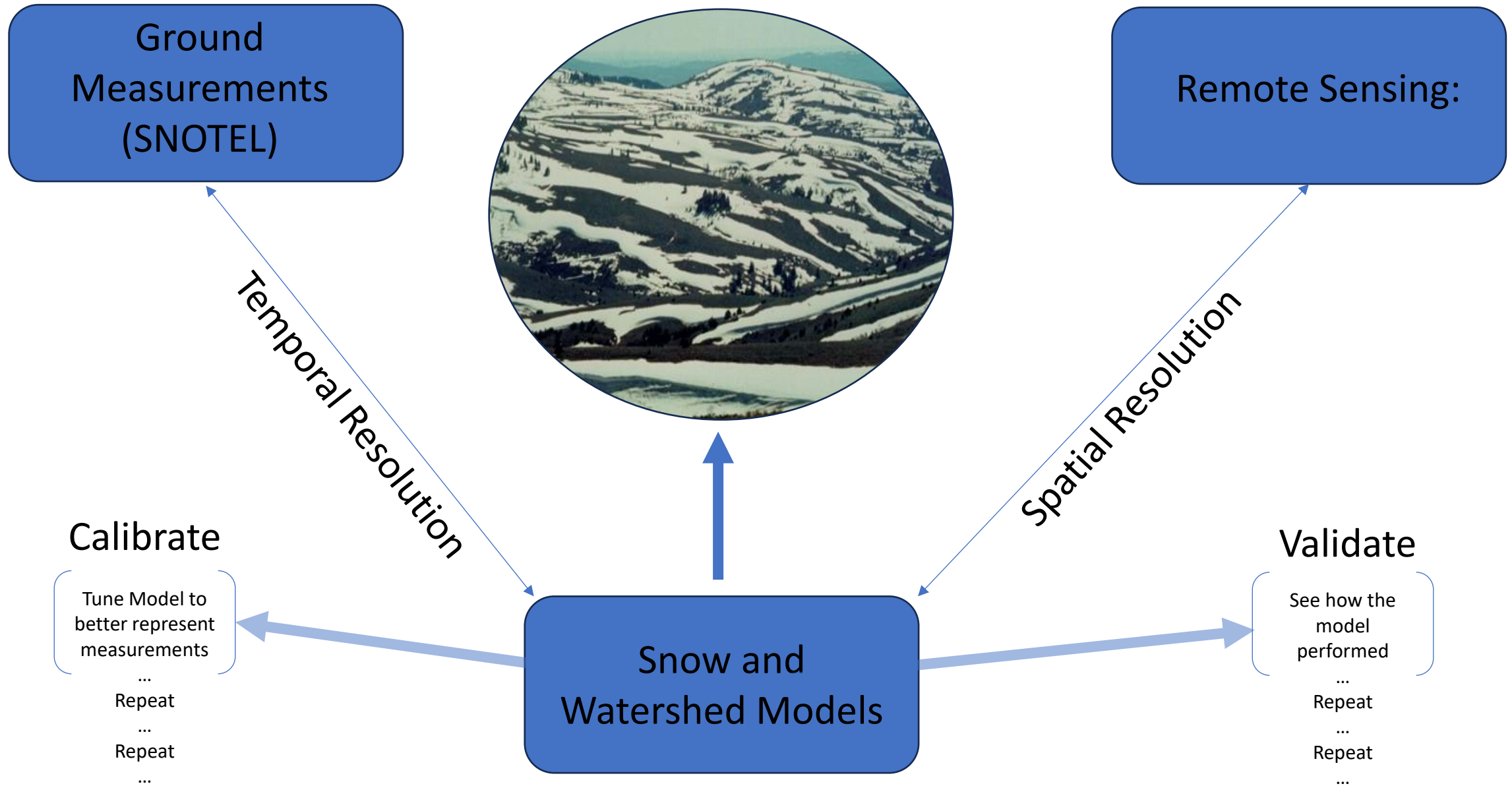


the timing, rate, and magnitude of snow melt

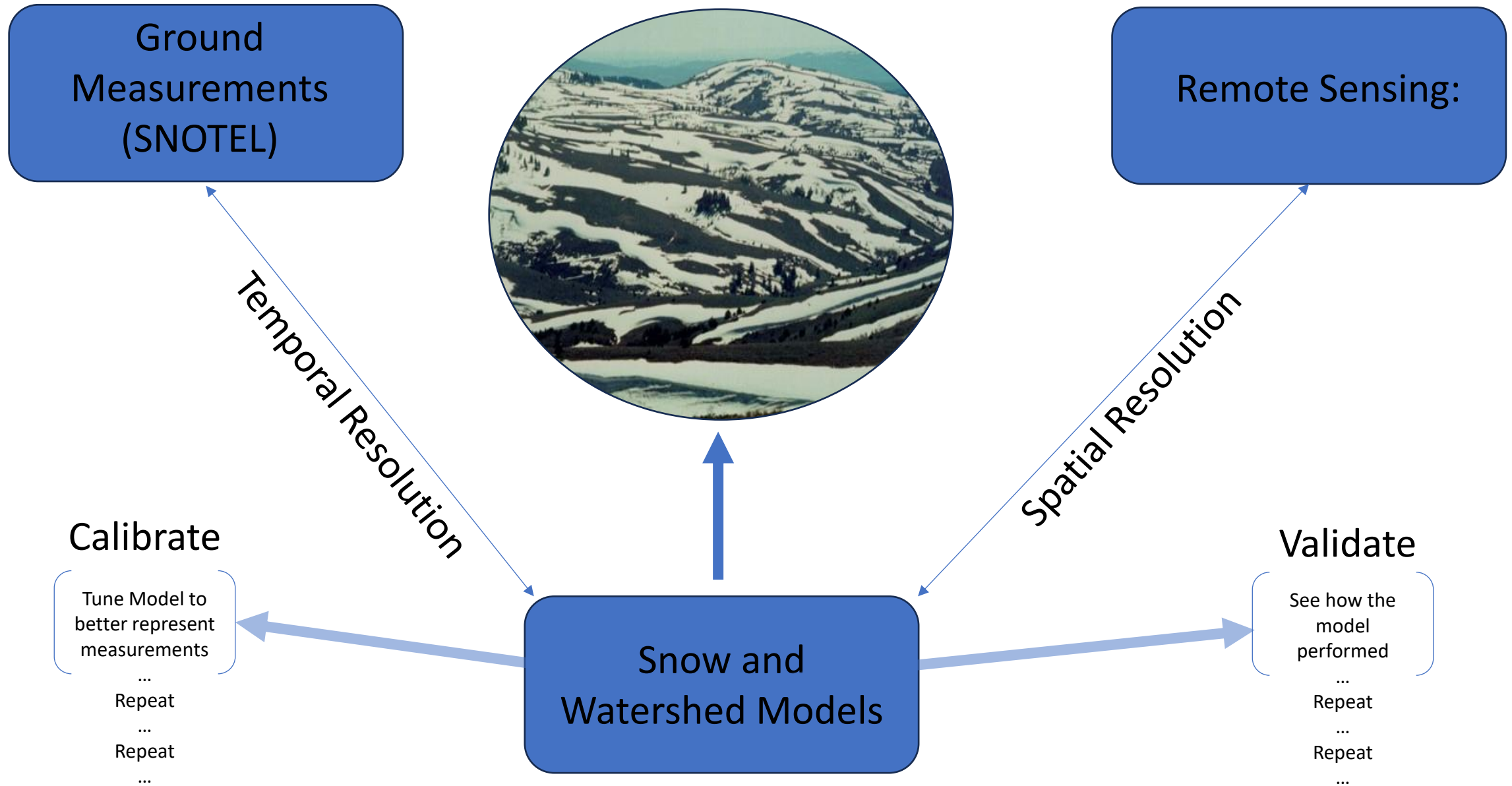


the fate of melt water

Models (can) integrate information through time and space



DON'T Believe the hype: THERE IS NOT ONE BEST MODEL





DON'T Believe the hype: THERE IS NOT ONE BEST MODEL



Despite many models claiming to have the best goodness of fit metrics, the best model comes down to several initial questions

Geography / Terrain

Climate / Meteorology

Available Data

Spatial Scale

Time Scale

Scientific Question

?



ALL Models benefit from
Calibration and Validation

All Models can be improved
and may have to change
depending on initial
questions





DON'T Believe the hype: THERE IS NOT ONE BEST MODEL

Most models and research that estimate canopy change impacts to snow and water only show increases in snow and water resources after a forest disturbance: This is incorrect.



All Models can be improved
and may have to change
depending on initial
questions



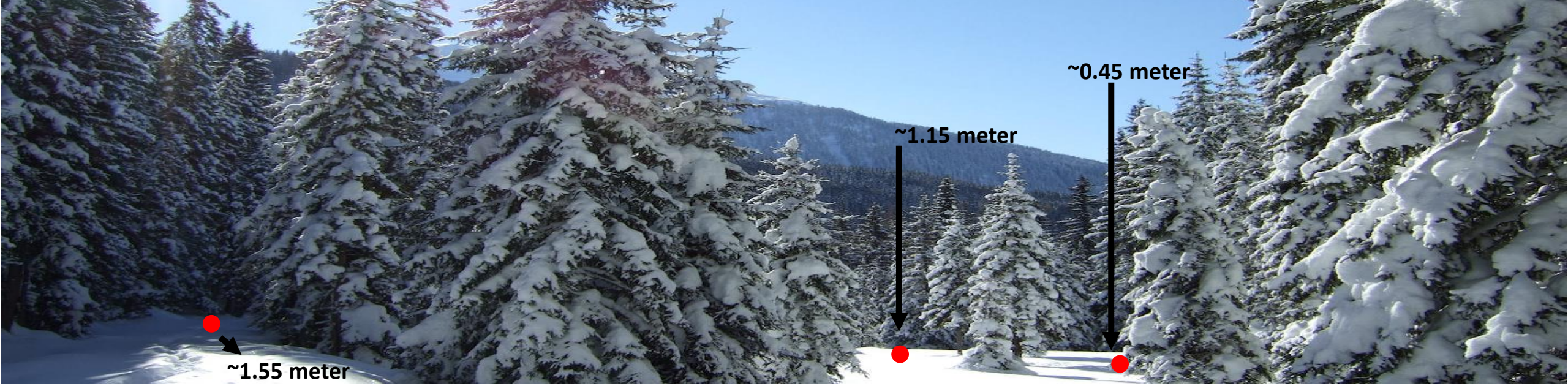


¿ Is the red dot in the forest ?



¿ Is the red dot in the forest ?

Models define the red dots in the same way



¿ Is the red dot in the forest ?

Models define the red dots in the same way

Snow Processes in a Forested Environment

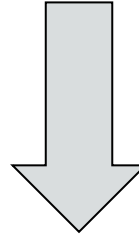
Accumulation processes

Governed by the mass balance

(interception, unloading, wind distribution)



***Historically : models only integrate canopy density



However, canopy structure has a large effect on the mass and energy balance

FOREST GEOMETRY!

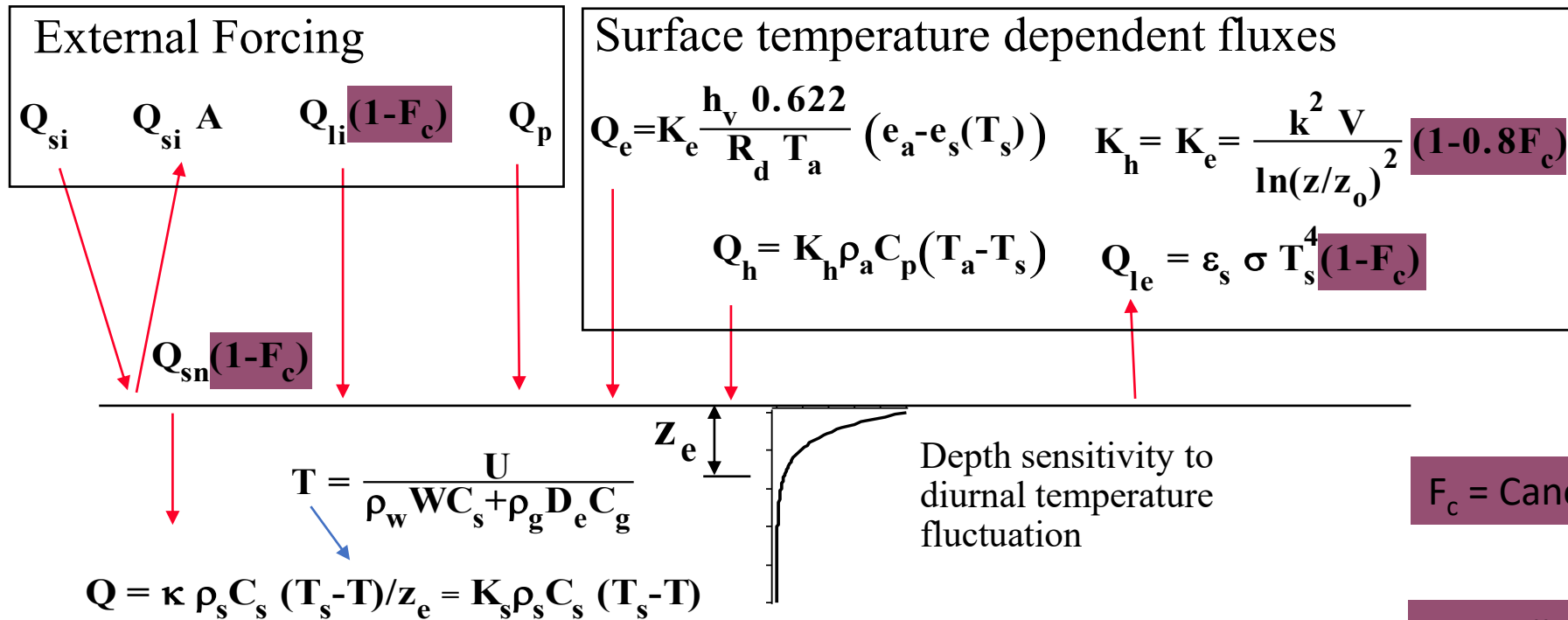
Ablation processes

Governed by the Energy Balance

(~short and longwave radiation)



Surface Energy Balance – With Forest Canopy



F_c = Canopy density

Basically, states denser forests have less sensitivity to energy

ONLY LOOKS AT VERTICAL INFLUENCES....

Energy Balance /Equilibrium at surface

$$Q_{si}(1-A) + Q_{li} + Q_p + Q_h(T_s) + Q_e(T_s) - Q_{le}(T_s) = Q(T_s, U, W)$$

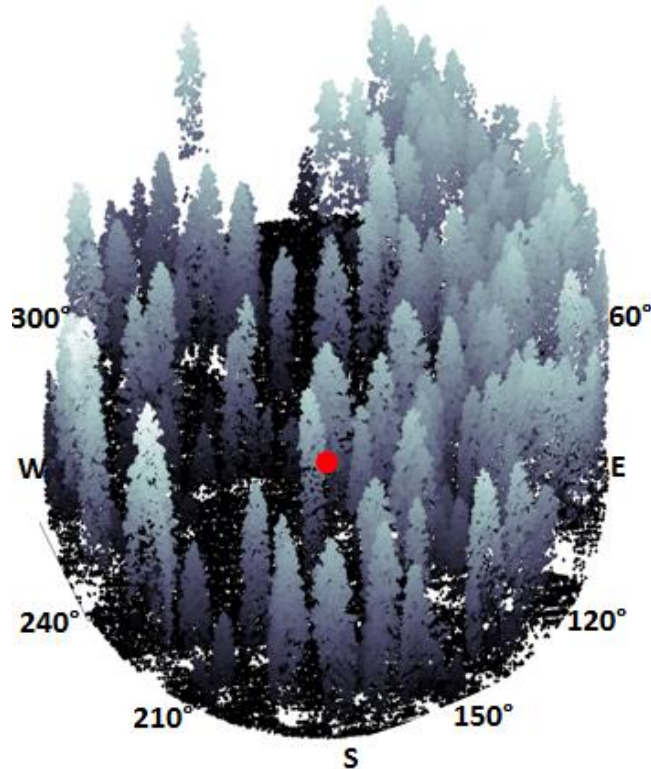
Solve for T_s : Linearize and iterate

If $T_s > 0^\circ\text{C} \Rightarrow$ Conduction cannot accommodate surface energy inputs

$\Rightarrow$ Surface melt is generated. Set $T_s = 0^\circ\text{C}$

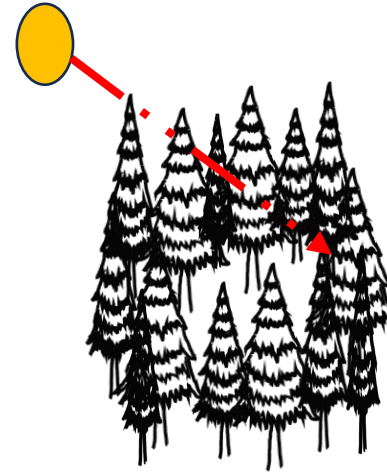
$\Rightarrow$ Additional energy is advected downwards by surface melt inflow

1. Integrate Aerial Lidar to define canopy geometry

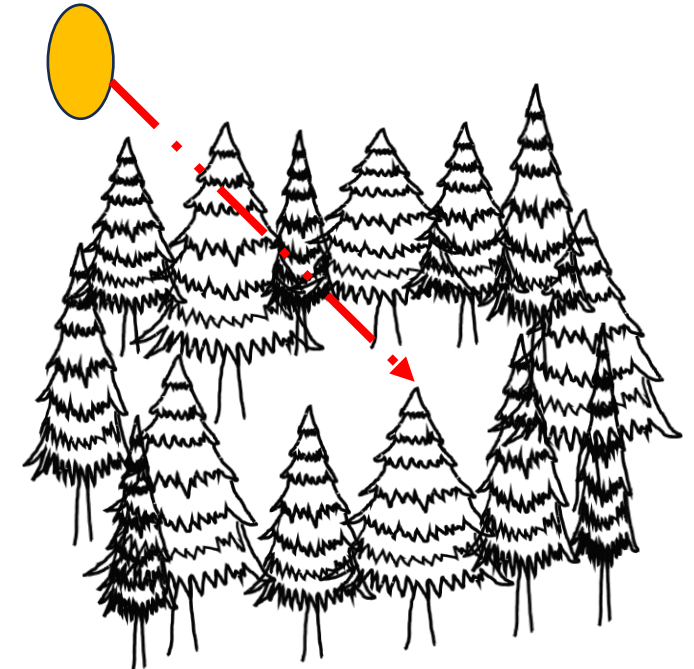


Define Canopy Shading Proxy

(Horizontal influences)



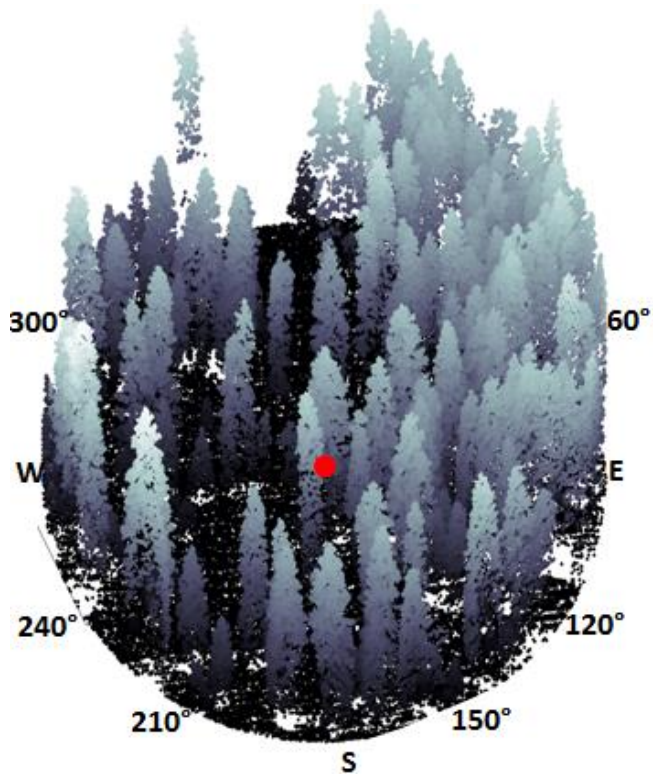
Small Gap and Edge :
less potential melt energy
more canopy shading



Large Gap and Edge:
More potential melt energy
Less canopy shading

1.
Aerial Lidar to
define canopy
geometry

2.
Integrate
with Manual
Measurements



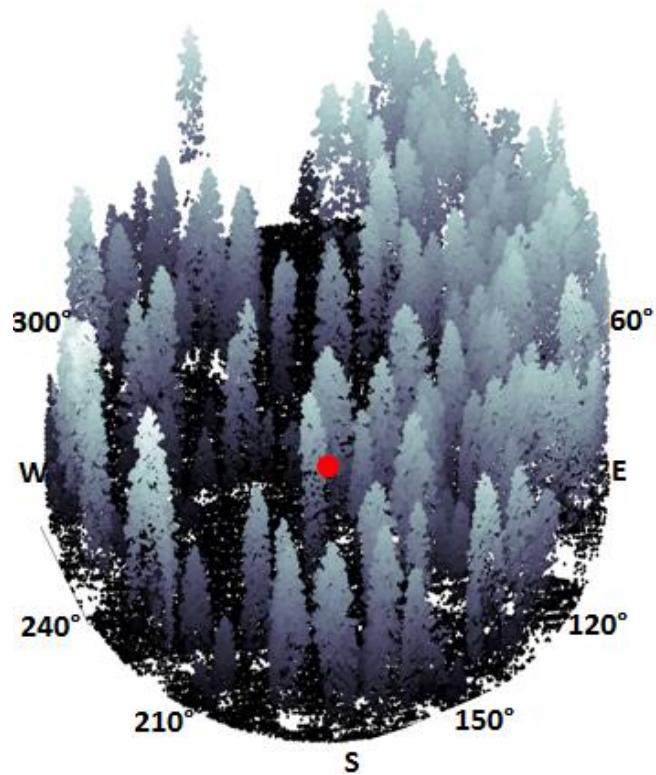
~5000 measurements of
SWE @

- △ North facing edge
- ▼ South facing edge
- Canopy Gap
- Interior Forest

1.
Aerial Lidar to
define canopy
geometry

2.
Integrate
with Manual
Measurements

3.
Quantify
relationships
between
geometry and
measurements



SWE @

- △ North facing edge
- ▼ South facing edge
- Canopy Gap
- Interior Forest

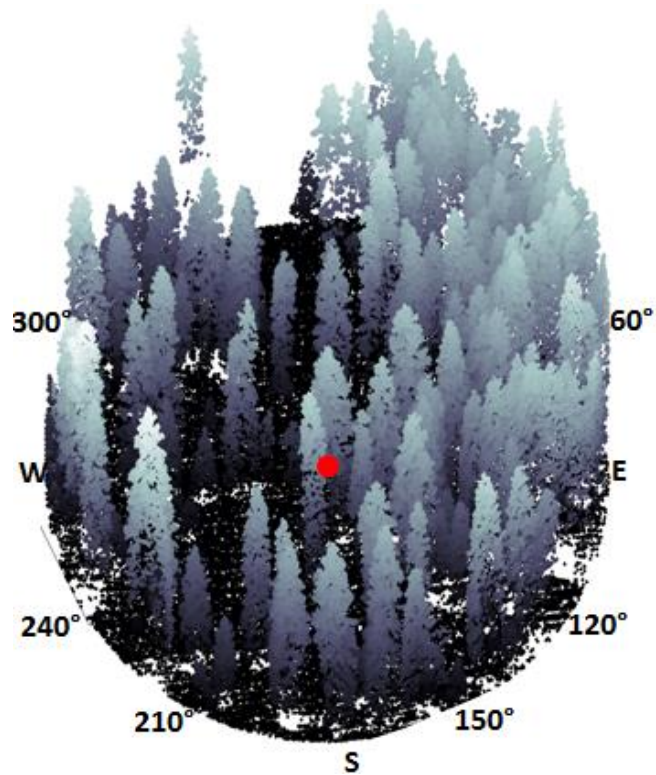
$$SWE^*(t) = mx + b$$

1.
Aerial Lidar to
define canopy
geometry

2.
Integrate
with Manual
Measurements

3.
Quantify
relationships
between
geometry and
measurements

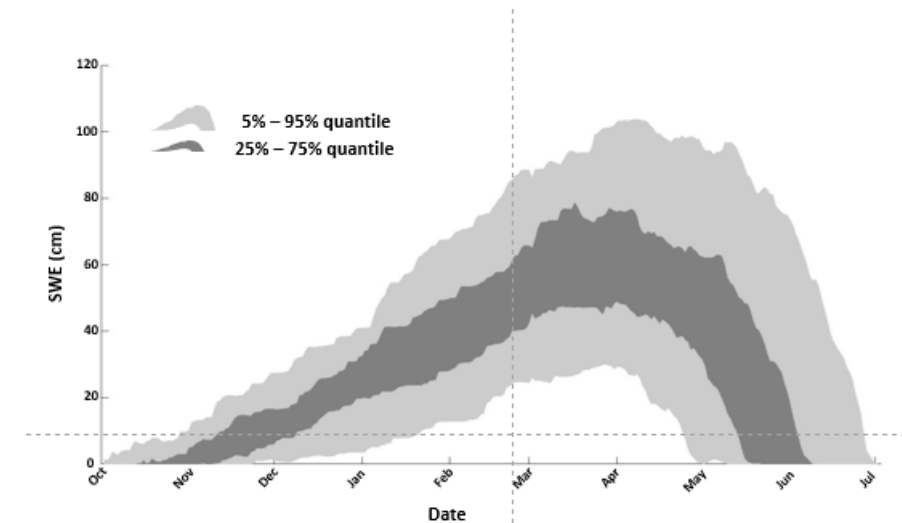
4.
Integrate with
model output



SWE @

- △ North facing edge
- ▼ South facing edge
- Canopy Gap
- Interior Forest

$$SWE^*(t) = mx + b$$

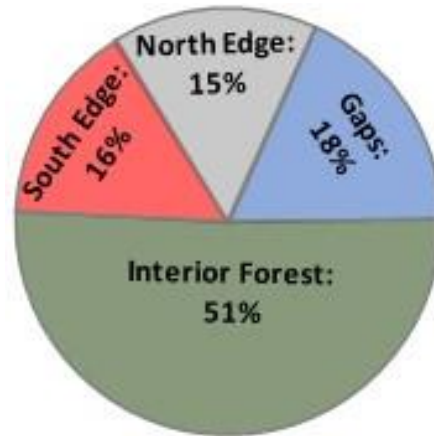


!Gives output range rather than one solution!

4. Integrate with model output

Example Location 'A'

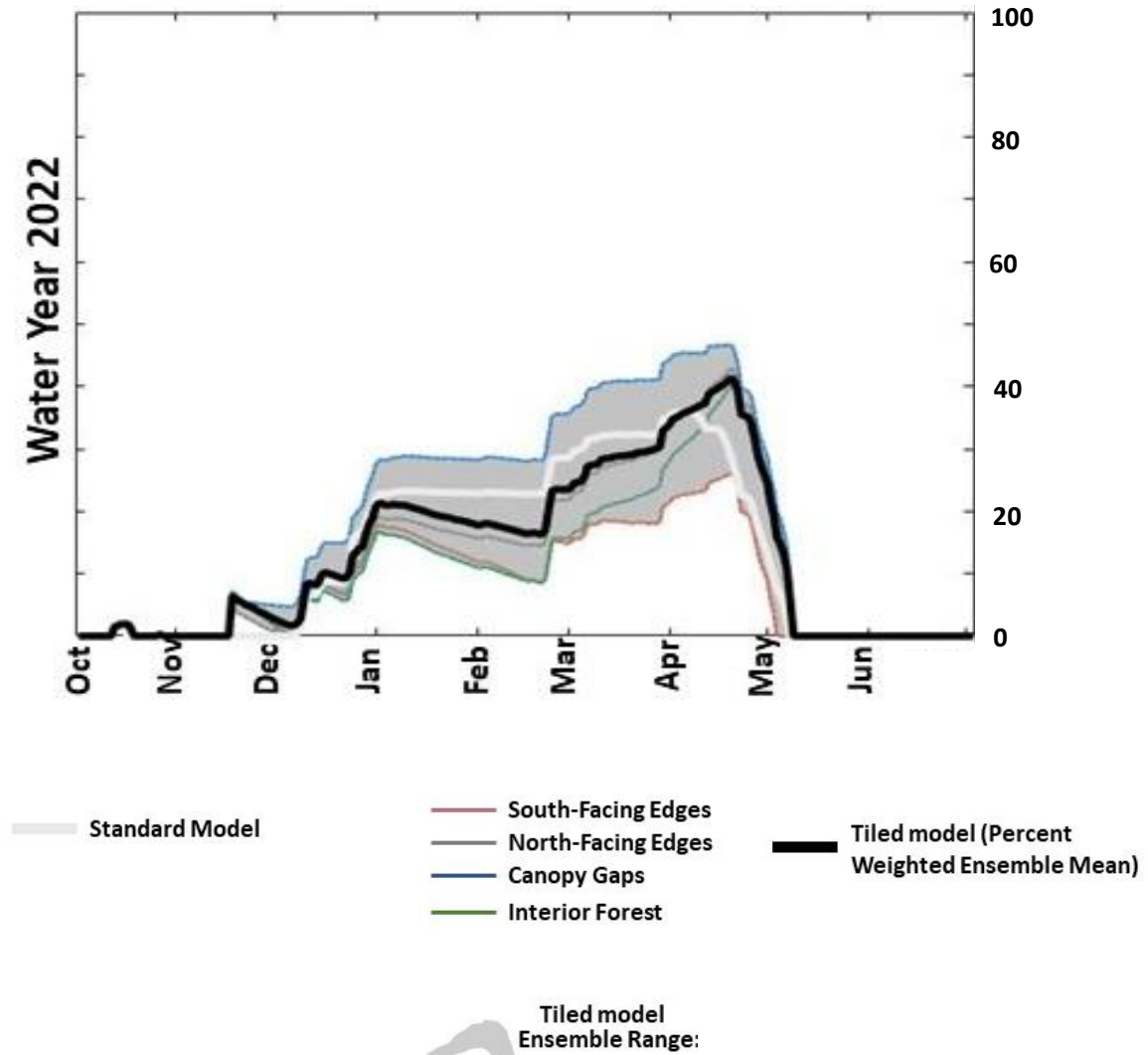
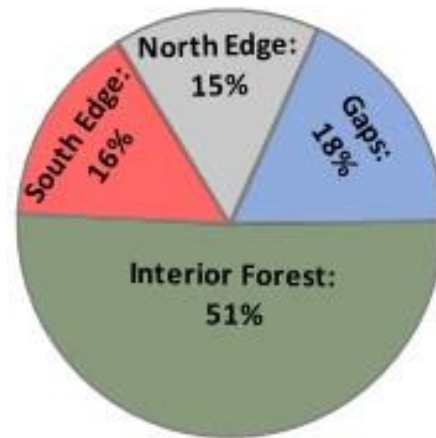
South-Facing Edges: 0.25
North-Facing Edges: 0.24
Relative Gap Area: 2.47
Canopy Closure: 0.60



4. Integrate with model output

Example Location 'A'

South-Facing Edges: 0.25
North-Facing Edges: 0.24
Relative Gap Area: 2.47
Canopy Closure: 0.60



Interactive Model Webpage

STEP ONE:

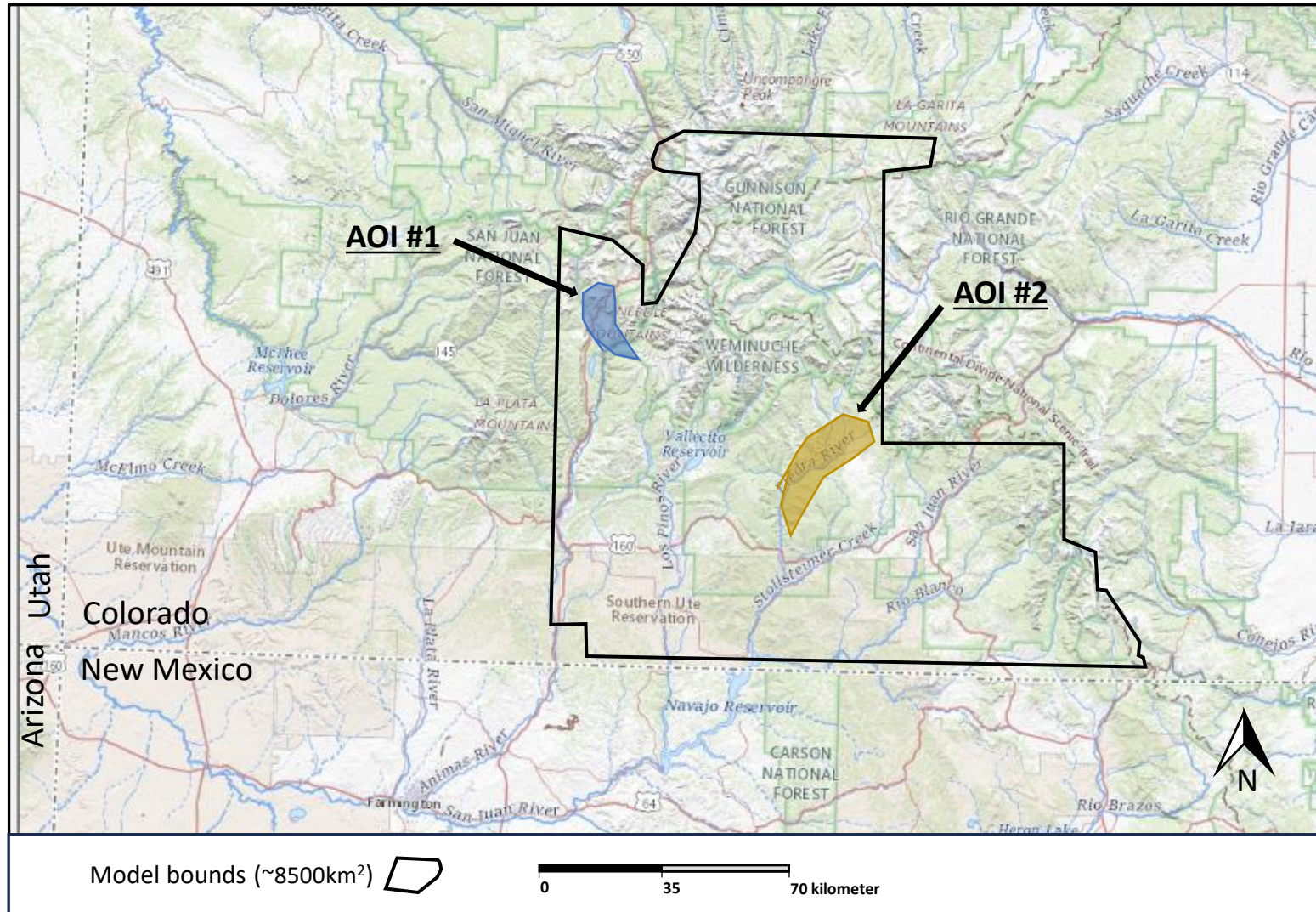
Look at output from

- ☒ [Current Forest Data](#)
- ☒ [Current Meteorologic Data](#)

STEP TWO:

Compare current data with

- ☒ [Potential Forest Scenarios](#)
- ☒ [Potential Future Meteorologic Data](#)



Alpine Area

- ☐ Click here to include area above tree line

SNOTEL

- ☒ Click here to include closest SNOTEL data

AOI Define Tool

-  Grab Tool Here to define spatial domain by clicking mouse. User may define up to two AOI's
- ☐ Or click here for full domain

Time Definition

Define Water Year (WY) or Years from drop down boxes below. User may define up to two time periods.

Time Period One

WY Start WY End

Time Period Two

WY Start WY End

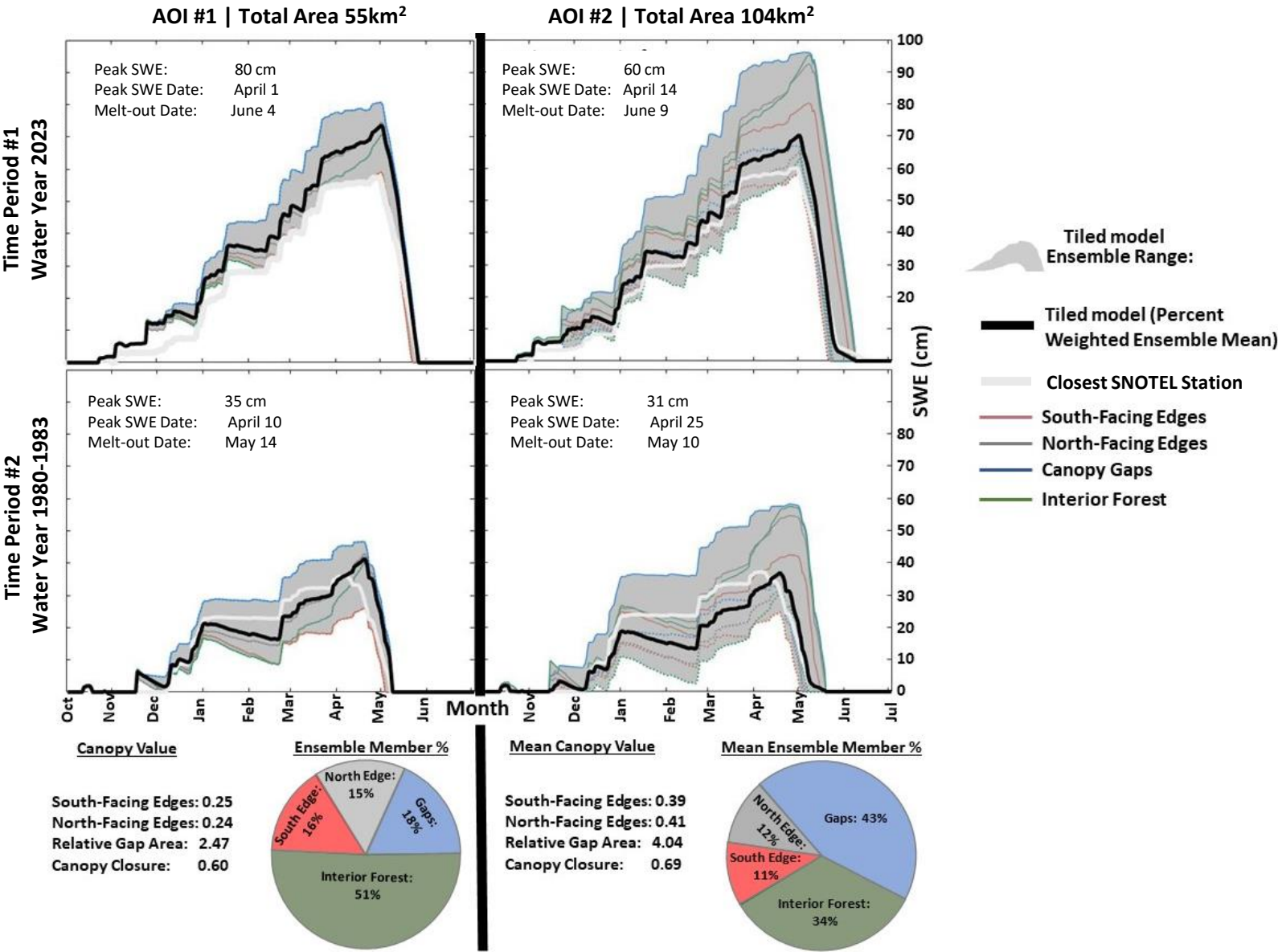
Data Range

Choose one method for graphing the data range of the ensemble members

- ☐ mean
- ☒ min - max
- ☐ 25% - 75% quantile
- ☐ 5% - 95% quantile

Example Output

Screen Pop-up



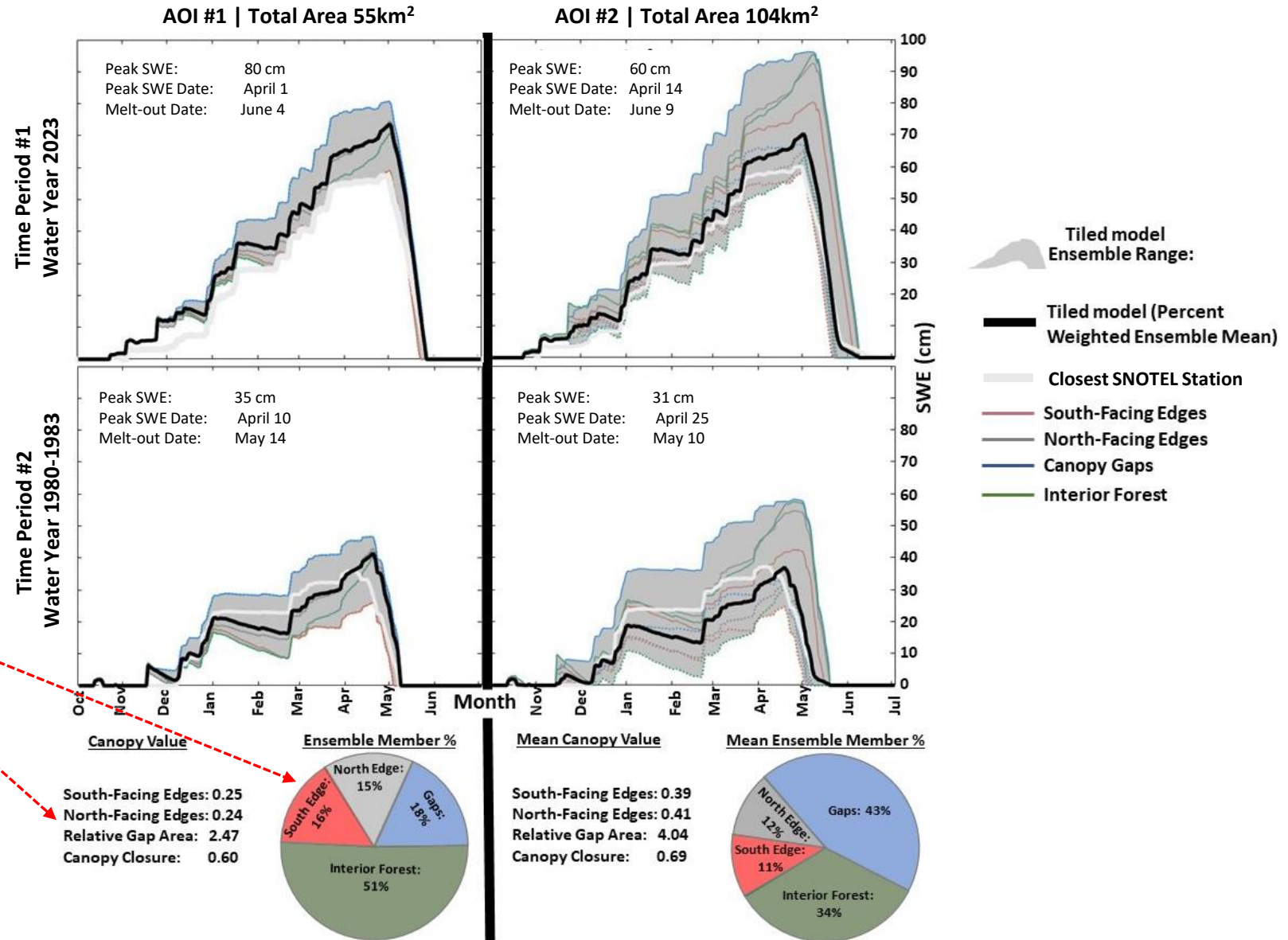
Example Output

STEP TWO:
Compare current data with

☒ Potential Forest Scenarios

☐ Potential Future Meteorologic Data

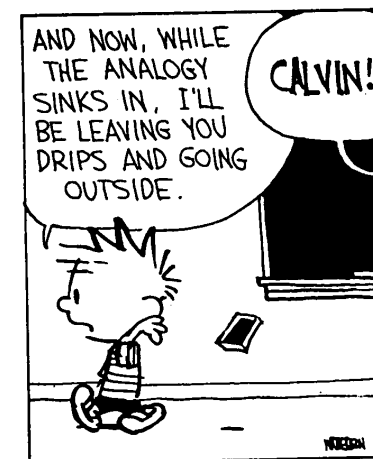
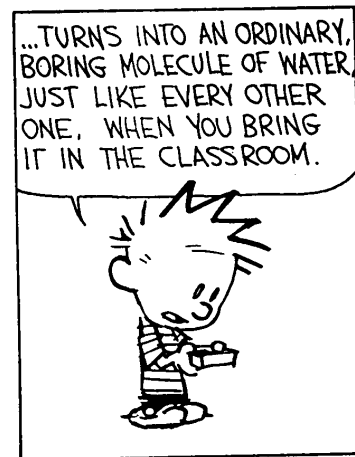
User can then change values and percentages of forest metrics and see how that will change the SWE curve



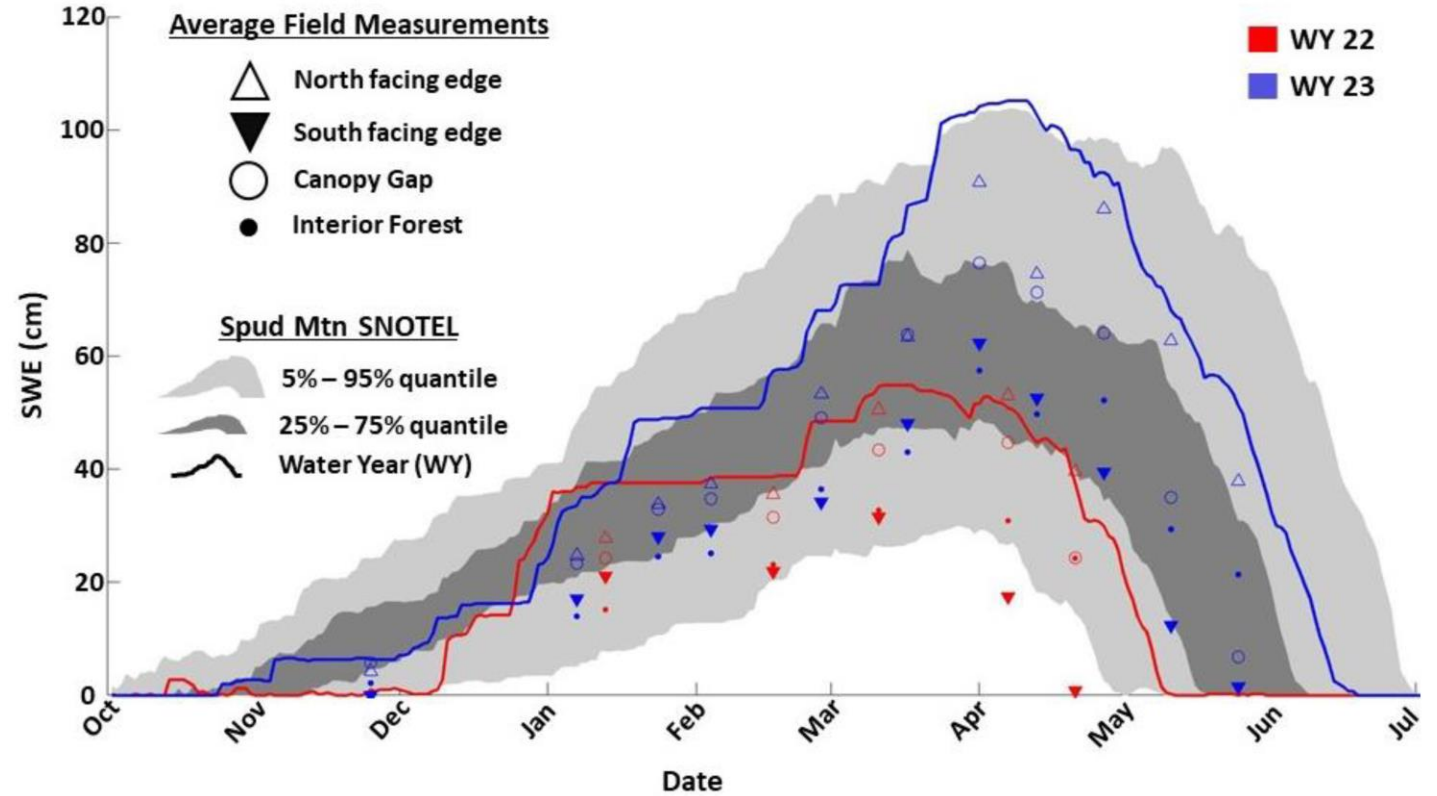
Questions?

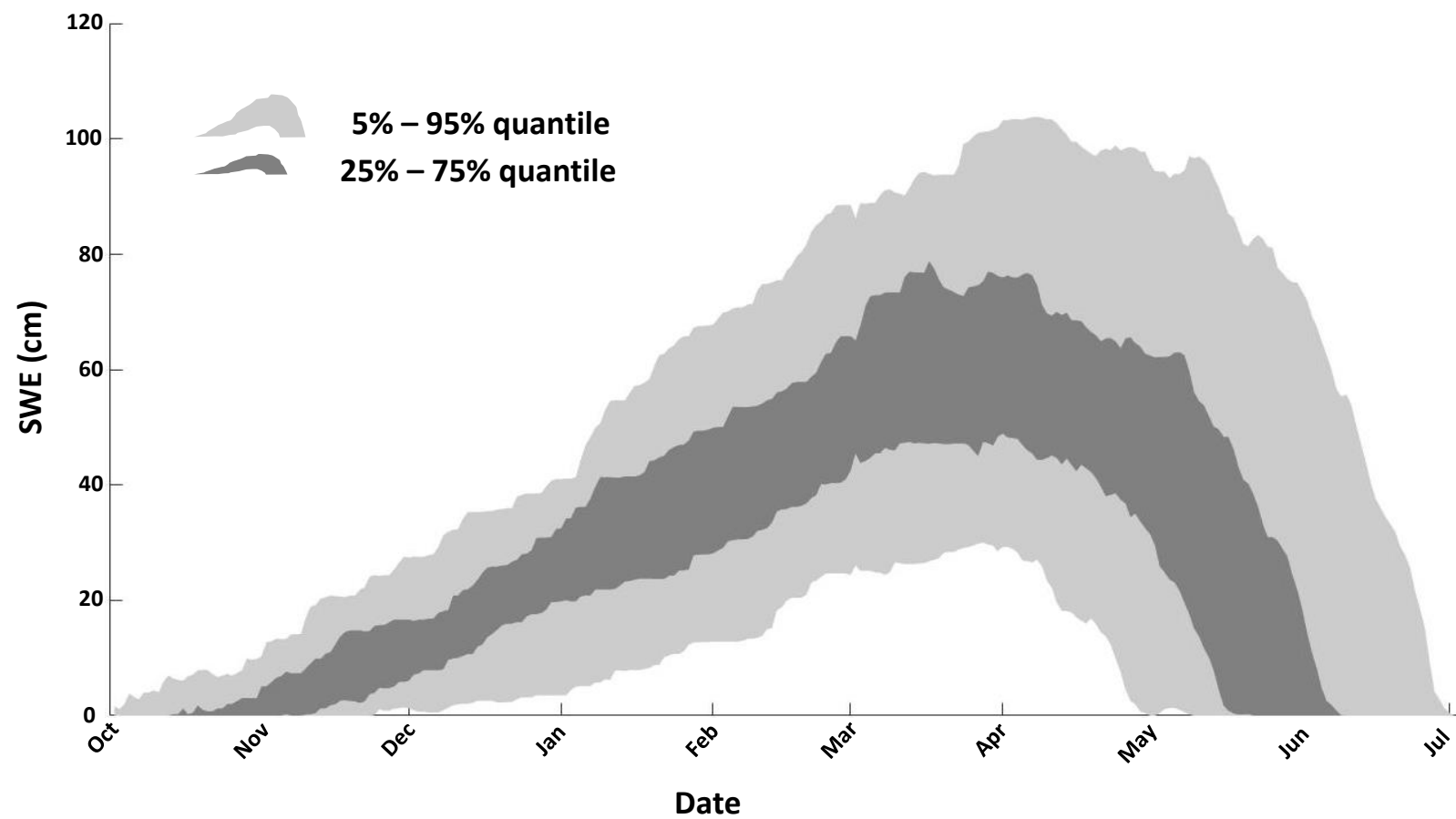


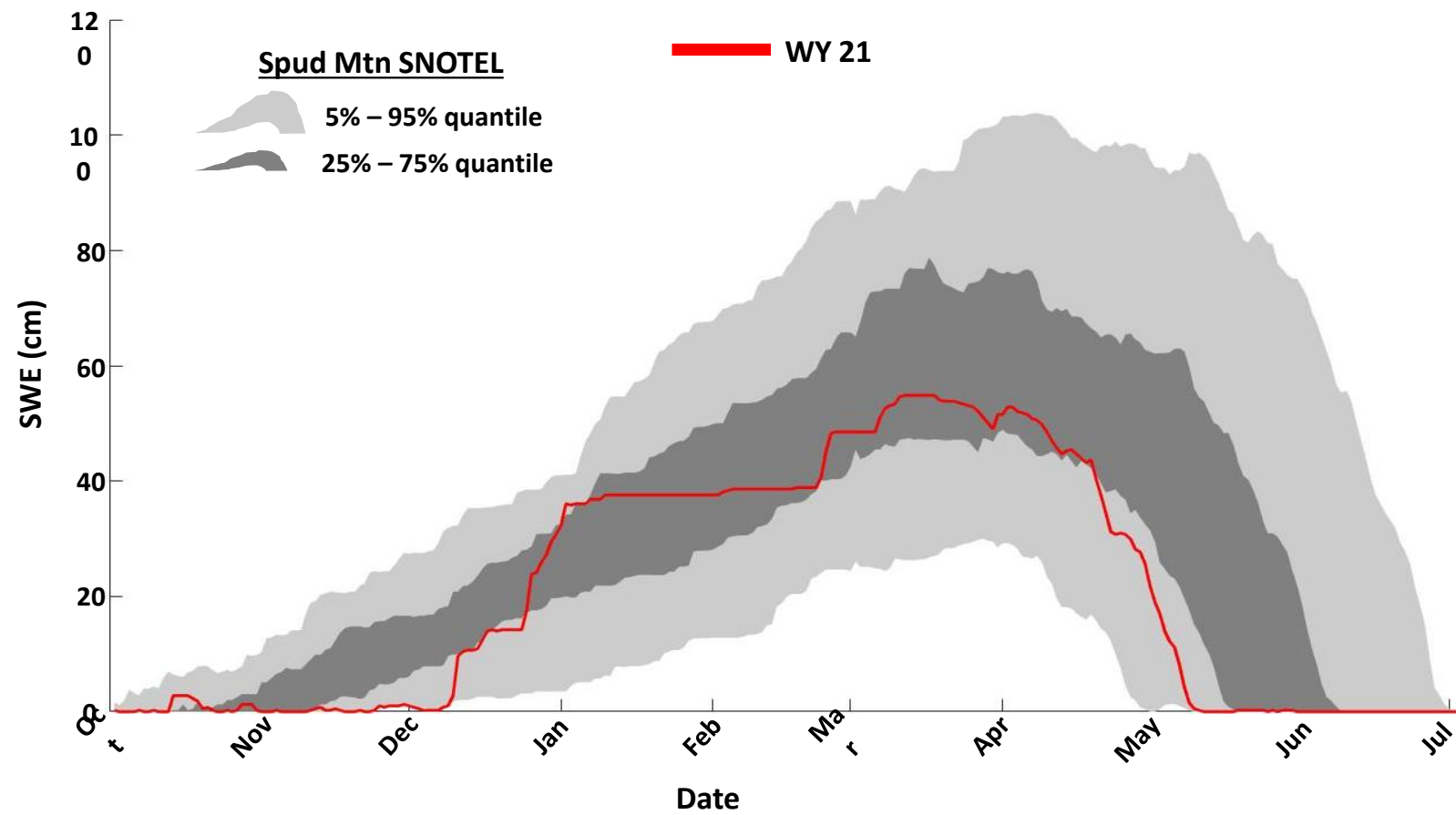
I THINK WE MIGHT ALL LEARN A LESSON FROM HOW THIS UTTERLY UNIQUE AND EXQUISITE CRYSTAL ...

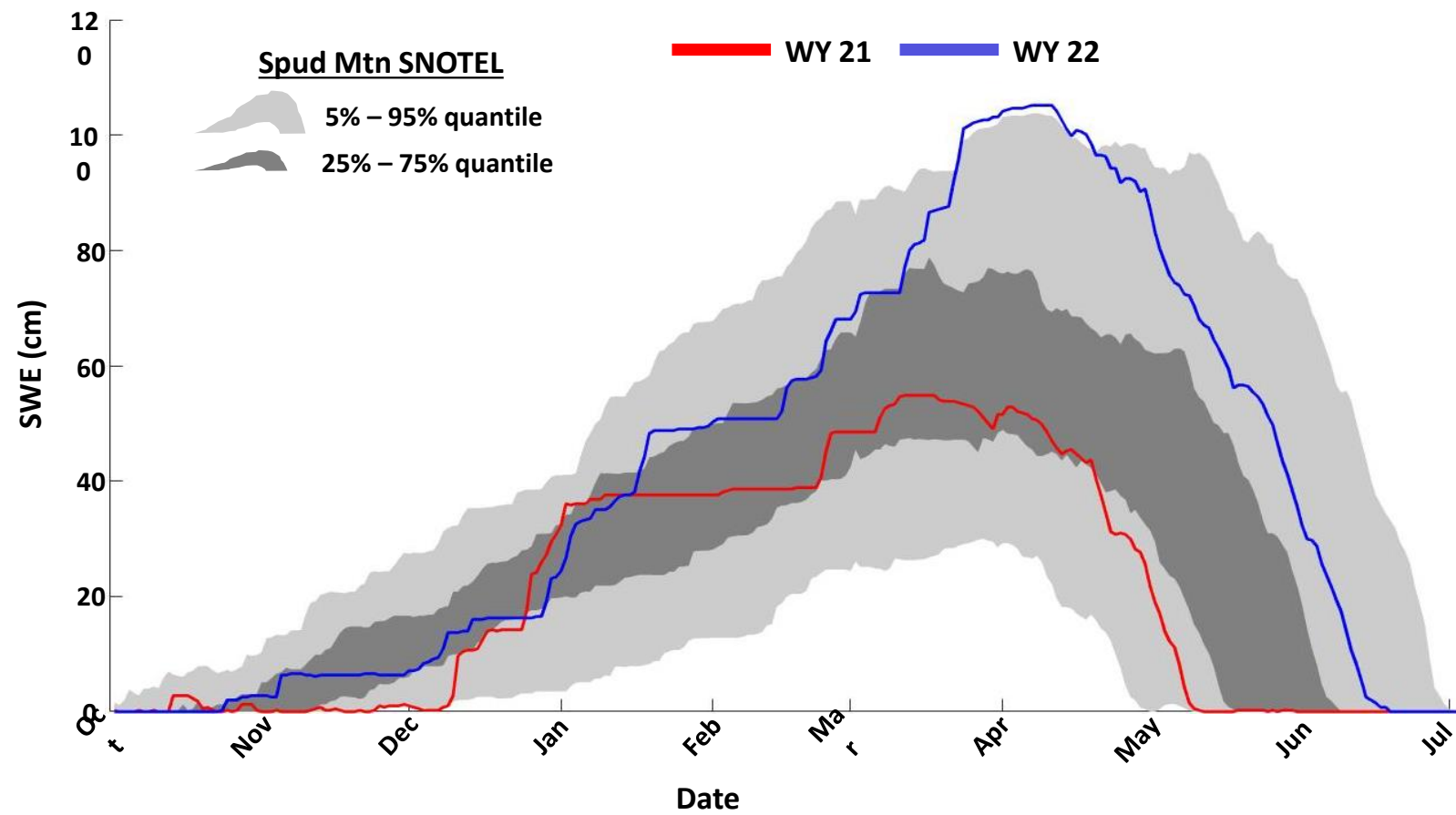


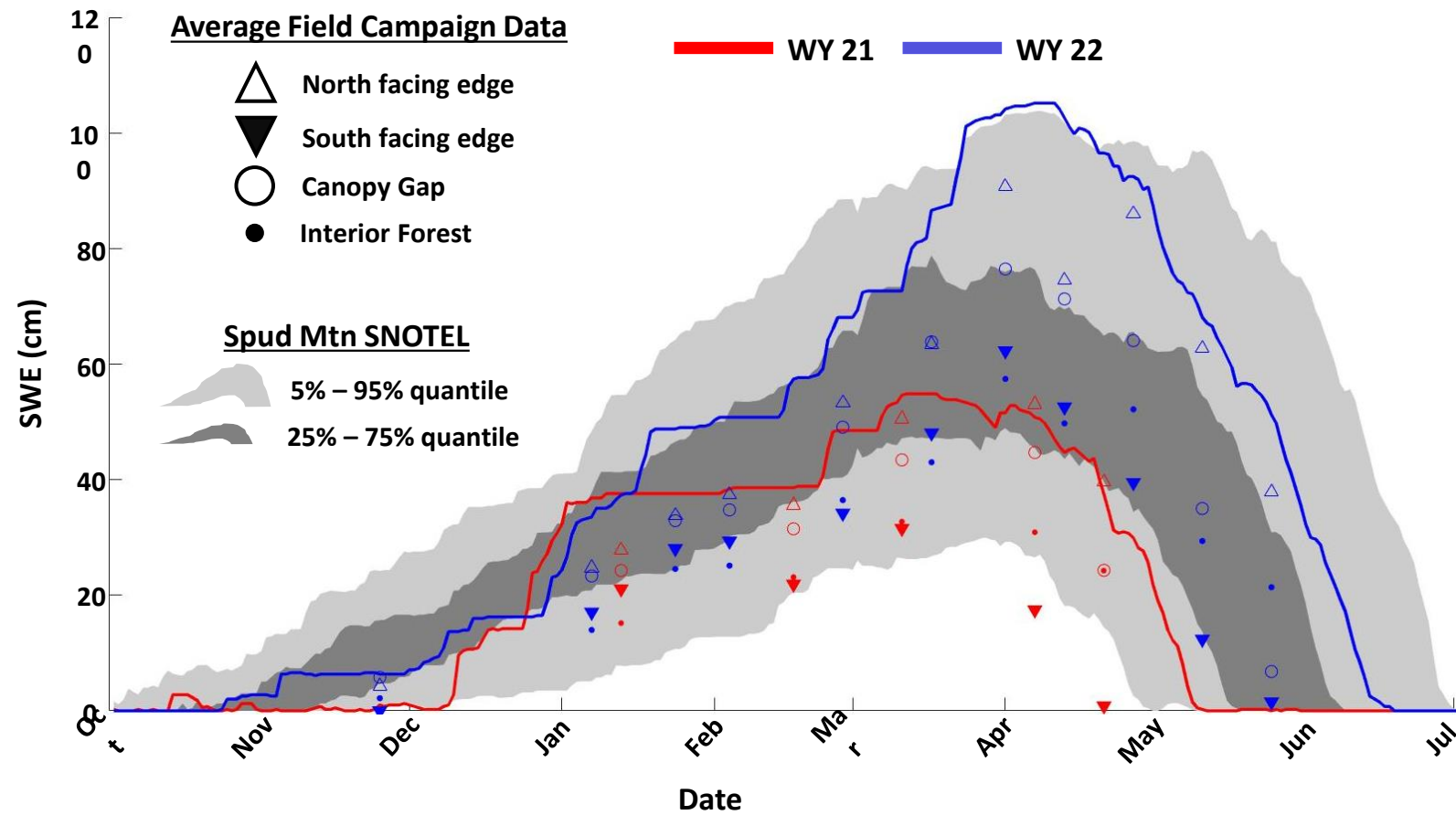
Manual Snow Measurements











Output: ~5,000 manual measurements → Published and Public

